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Chapter 22

GELFAND-P001026

6/IX/82 Gelfand seminar.

(40th year of activity)

Kirillov on the work of Lepowsky and Wilson.

generaliz. (Lepowsky, Wilson
—"—, —"—, Kac, Kazdan)

Lie algebra with six generators.

$e_0 \quad h_0 \quad f_0$ | $\forall$ triple satisfies the standard relations
$e_1 \quad h_1 \quad f_1$ | of matrices of order two with trace 0.

GELFAND-P001027

I. M. G. Cartan algebras No need to learn the roots (everything follows from this)

$e_i, h_i, f_i, \quad i = 1, \dots, n$

$$\begin{cases} [e_i, f_k] = h_i \delta_{ik} \\ [h_i, f_k] = -a_{ik} f_k \\ [h_i, e_k] = a_{ik} e_k \\ [h_i, h_k] = 0 \end{cases} \quad \text{(without summing over } k\text{)} \tag{1}$$

(a_{ik}) — Cartan matrix

a_{ii} can be adjusted by normalizing f_i

Assume: $a_{ii} = 2$.

Important algebras: 1) $a_{ij} < 0$ for $i \neq j$

2) $a_{ij} \in \mathbb{Z}$

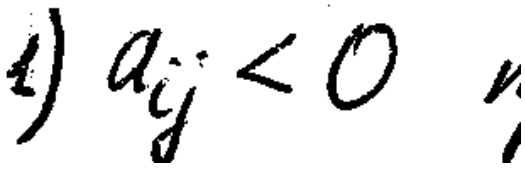


Diagram and typed description. The manuscript has a strict inequality $<$.

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As a rule there are no further relations (so that no linear combination of e_i, h_i, f_i is forced to be 0)

Sometimes one can add relations, and the theorem is that they are unique.

$$\begin{cases} e_i^{-a_{ik}+1} e_k = 0 \\ f_i^{-a_{ik}+1} f_k = 0 \end{cases} \tag{2}$$

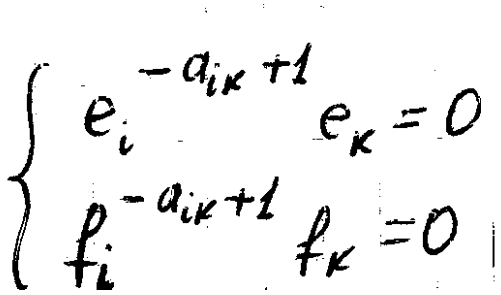


Diagram and typed description. The powers of e_i and f_i in (2) are written without the notation ad.

Original source details: GELFAND-P001027

Sometimes the algebra is finite-dimensional,

$n = 1$: e_1, f_1, h_1

$$[e_1, f_1] = h_1$$
$$[h_1, e_1] = 2e_1$$
$$[h_1, f_1] = -2f_1$$

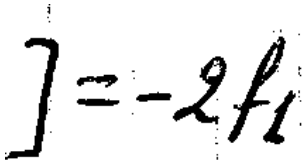


Diagram and typed description. The last subscript in $-2f_1$ has the foot of the numeral 1.

Original source details: GELFAND-P001027

this gives sl(2)



Uncertain reading. A short entry after sl(2) is heavily crossed out.

Original source details: GELFAND-P001027

Finite-dimensionality $\Leftrightarrow$ the matrix $(a_{ij}) > 0$ (strictly)

Affine $\Leftrightarrow$ $(a_{ij}) \geq 0$ and rank$(a_{ij}) = n - 1$.

$n = 2$

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$
$$\text{sl}(3) \qquad \text{so}(5) \qquad G_2$$

This replaces Dynkin diagrams.

Relations (2) for sl(3):

$$[e_1[e_1e_2]] = 0$$
$$[e_2[e_2e_1]] = 0$$

GELFAND-P001028

[source number: 4]

Affine algebras:

$$\begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix} \qquad A_1^1$$



Diagram and typed description. The lower-right entry of the first matrix is written as -2.

Original source details: GELFAND-P001028

$$\begin{pmatrix} 2 & -4 \\ -1 & 2 \end{pmatrix}$$



Uncertain reading. An entry to the right of the second matrix is crossed out.

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Heisenberg algebra:

$$\underset{\text{symplectic}}{E} \oplus R, \qquad [x,y] = \langle x,y\rangle 1, \quad 1 \in R,$$

$\langle x,y\rangle$ — symplectic product.

— · —

Kirillov.

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \qquad \begin{matrix} f_1 & h_1 & e_1 \\ f_2 & h_2 & e_2 \end{matrix}$$

Nontrivial relation

$$[e_1[e_1[e_1,e_2]]] = 0$$

and analogous relations with the replacements $\begin{matrix} 1 \leftrightarrow 2 \\ e \leftrightarrow f \end{matrix}$.

I. M. G. Root vectors — monomials;



Uncertain reading. An entry before “monomials” is crossed out.

Original source details: GELFAND-P001028

weights are the numbers of occurrences of e_j



Uncertain reading. The first strokes of the crossed-out word may begin «вхо...», as in “occurrences”.

Original source details: GELFAND-P001028

K.

Grading: weights (pairs of numbers)



Uncertain reading. An entry after “Grading” is crossed out.

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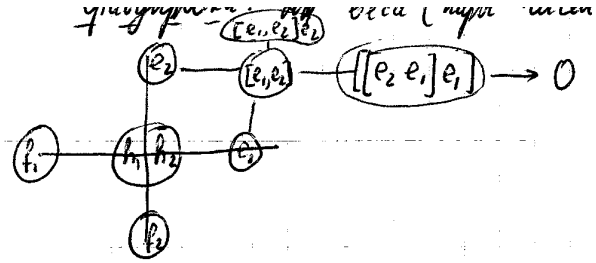
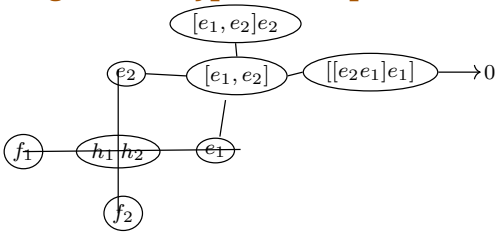


Diagram and typed description.



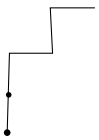
The upper oval reads $[e_1, e_2]e_2$. Any outer brackets remain uncertain.

Original source details: GELFAND-P001028

I. M. G.



Diagram and typed description.



Original source details: GELFAND-P001028

monomials: paths like $e_1e_1e_2e_1\ldots$

Question: how can the paths be made continuous?

K. Second realization. Functions on the circle with values in matrices of order two. $A(t)$ (polynomials in t and t^{-1}; defined for $|t| = 1$).

$$[A, B] = AB - BA + \frac{\mathbf{1}}{2\pi i}\int \mathrm{tr}(A\,dB), \qquad \mathbf{1} \text{ — a new generator} \tag{1}$$



Uncertain reading. An entry above e_1 is crossed out.

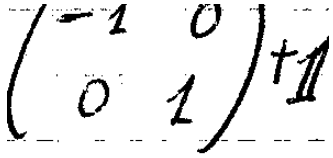
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$$e_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$
$$e_2 = \begin{pmatrix} 0 & 0 \\ t & 0 \end{pmatrix}, \quad h_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \mathbf{1} \quad f_2 = \begin{pmatrix} 0 & t^{-1} \\ 0 & 0 \end{pmatrix} \tag{2}$$



Original source details: GELFAND-P001028

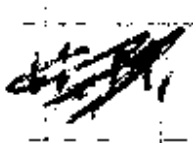
Uncertain reading. The entry between the matrix for h_1 and the comma is heavily crossed out.



Original source details: GELFAND-P001028

Diagram and typed description. The matrix for h_2 in (2) is followed by $+\mathbf{1}$.

They satisfy all the relations.



Original source details: GELFAND-P001028

Uncertain reading. An entry at the beginning of the line before “They” is crossed out.

I. M. G. Pick out the Heisenberg subalgebra, the type of its representations, etc.

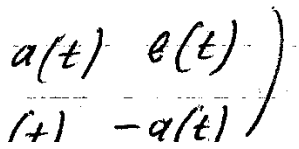


Original source details: GELFAND-P001028

Uncertain reading. The word before “its representations” most likely reads “type”.

K. This is convenient for writing a general element.

$$\begin{pmatrix} a(t) & e(t) \\ c(t) & -a(t) \end{pmatrix} + \alpha\mathbf{1}. \tag{3}$$



Original source details: GELFAND-P001028

Uncertain reading. The upper-right letter resembles e, with a short loop. A reading of b remains possible.

GELFAND-P001029

There are many representations; they have not all been described (and pointless) [tentative reading of the abbreviation]?



Uncertain reading. (and pointless) [tentative reading of the abbreviation]

Original source details: GELFAND-P001029

The highest-weight representations are described.

One of them is singled out and is remarkable:

cyclic vector — highest-weight vector v_0

$$\begin{cases} e_1 v_0 = 0, \\ e_2 v_0 = 0, \\ h_1 v_0 = 0, \\ h_2 v_0 = v_0 \end{cases} \tag{4}$$

The most economical highest-weight representation

The space is realized as the space of [Crossed out: illegible.]?



Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P001029

polynomials in $x_1, x_3, x_5, \ldots, x_{2n+1}, \ldots$

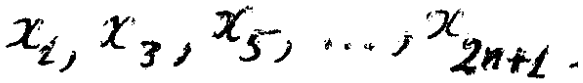


Diagram and typed description. The first indices in the list: $1, 3, 5$.

Original source details: GELFAND-P001029

A Heisenberg (infinite-dimensional) subalgebra is singled out:

$$B_{2k+1} = \begin{pmatrix} 0 & t^k \\ t^{k+1} & 0 \end{pmatrix}, \quad C_{2k+1} = \begin{pmatrix} 0 & -t^k \\ t^{k+1} & 0 \end{pmatrix}, \quad H_{2k} = \begin{pmatrix} t^k & 0 \\ 0 & -t^k \end{pmatrix}. \tag{5}$$

— · —

I. M. G.: This algebra arises naturally in the p-adic case.
Consider the case in which we quotient by $\{\mathbf{1}\}$
K. There are different (nonconjugate) commutative subalgebras.
1) diagonal
2) $\begin{pmatrix} 0 & f(t) \\ tf(t) & 0 \end{pmatrix}$
One cannot put it in diagonal form continuously (but one can for each fixed value).
I. M. G. Quadratic forms: a related problem; Mishchenko; Gelfand — the number is 2^n,
but the canonical forms are undescribed.
K. with **1** these algebras are now noncommutative

$$[B_{2k+1}, B_{2\ell+1}] = \delta(k+\ell)(2\ell+1)\mathbf{1}$$

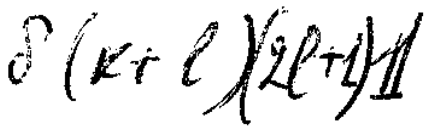


Diagram and typed description. The manuscript reads $\delta(k+\ell)(2\ell+1)\mathbf{1}$.

Original source details: GELFAND-P001029

(Heisenberg algebra)
Take its standard representation.

$B_1, B_3, \ldots$	— multiplication by $x_1, x_2, \ldots$	$B_k = x_k(2k+1)$
$B_{-1}, B_{-3}, \ldots$	— differentiation	$B_{-k} = \partial/\partial x_k$

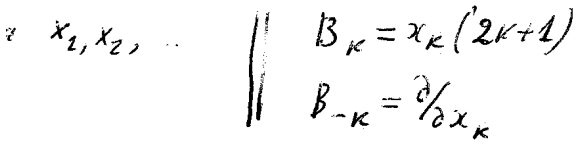


Diagram and typed description. The notation in the two manuscript columns.

Original source details: GELFAND-P001029

The restriction of the representation to the Heisenberg algebra remains irreducible.
Generating function for the action

$$e^{\sum x_k} e^{4\sum \partial/\partial x_k} \stackrel{df}{=} L = \sum L_m, \qquad L_m \text{ — homogeneous component of weight } m.$$

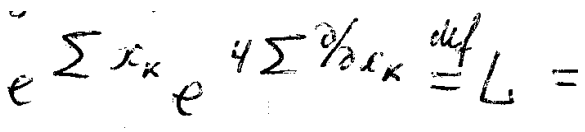


Diagram and typed description. The two exponential arguments in the manuscript.

Original source details: GELFAND-P001029

$(\forall x_k$ has its own weight$)$
$\downarrow$
$(-k)$



Uncertain reading. (not written out) [tentative reading]

Original source details: GELFAND-P001029

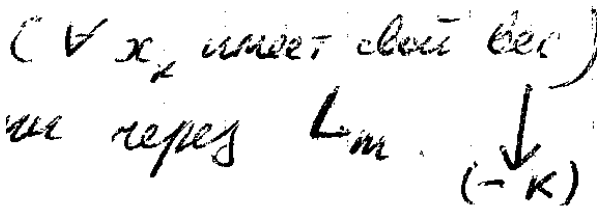


Diagram and typed description. Arrow labelled $(-k)$ below "weight".

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C_{2k+1} and H_{2k} are expressed by simple formulas in terms of L_m.

GELFAND-P001030

[source number: 5]

Zamolodchikov (Sasha).

On the string.

It arises from the theory of strong interactions (we neglect something).

There is a mechanical system with coordinates $x^\mu(\sigma)$, $\mu = 0, 1, \ldots, D-1$. σ — a continuous index; $\sigma \in (0, 2\pi)$ (a circle).

$$\dot{x}^\mu(\sigma) = \frac{\partial}{\partial\tau} x^\mu(\sigma) \qquad \text{(generalized velocities)},$$

$$\dot{x}^\mu(\sigma, \tau) = \frac{\partial}{\partial\tau} x^\mu(\sigma, \tau)$$

$x^\mu(\sigma, \tau)$ The trajectory is a surface (two-dimensional).

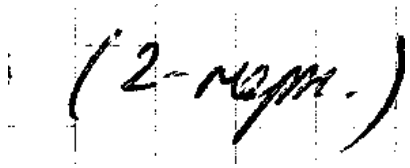


Diagram and typed description. The manuscript abbreviation for "two-dimensional."

Original source details: GELFAND-P001030

Action:

$$A[x^\mu] = \int d\tau \int d\sigma\, \mathcal{L}(x^\mu(\sigma,\tau), \dot{x}^\mu(\sigma,\tau)) \tag{1}$$

x^μ is regarded as a vector in a space with metric

$$\operatorname{diag}\underset{x^0}{(1}, \underset{x^1}{-1}, \ldots, \underset{x^{D-1}}{\underset{\ldots}{-1}})$$

Choose area as the action:

$$A[x^\mu] = \int d\sigma\, d\tau \sqrt{(\dot{x}_\mu)^2 (x'_\nu)^2 - (\dot{x}_\mu x'^\mu)^2} \tag{2}$$

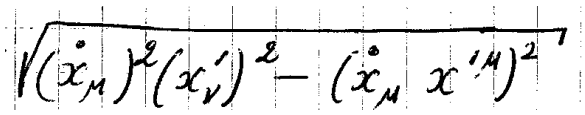


Diagram and typed description. The expression under the square root in (2).

Original source details: GELFAND-P001030

Here $x'_\mu = \dfrac{\partial}{\partial\sigma} x_\mu(\sigma,\tau)$.

$$\text{Invariance:} \quad \begin{matrix} \sigma \longrightarrow f(\sigma,\tau) \\ \tau \longrightarrow g(\sigma,\tau) \end{matrix} \quad // \quad \begin{matrix} \text{area} \\ \text{is unchanged} \end{matrix} \tag{3}$$

This is a relativistically invariant model.

Mass is proportional to length; a point interacts with its nearest neighbors. As it contracts, it reaches relativistic speeds.

Momenta:

$$p_\mu(s) = \frac{\partial \mathcal{L}}{\partial \dot{x}_\mu(\sigma)} \tag{4}$$

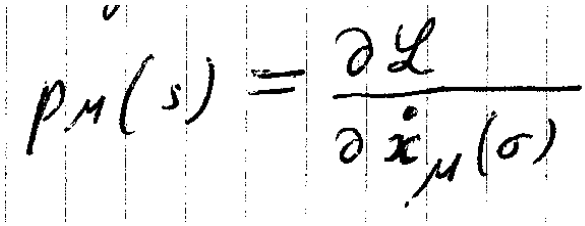
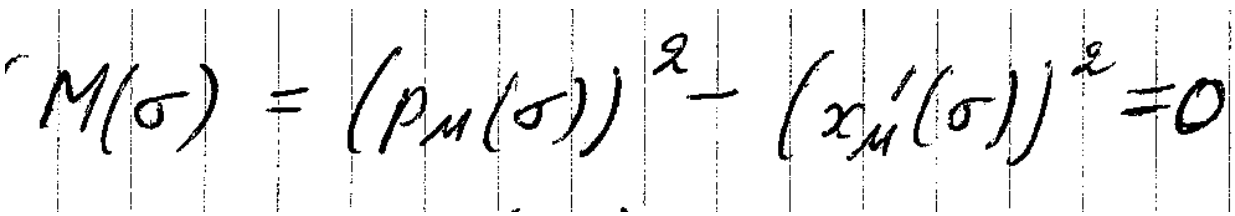


Diagram and typed description. The left side uses s and the right side uses σ.

Original source details: GELFAND-P001030

The Lagrangian is degenerate $\Rightarrow$ (4) cannot be solved for the velocities; there are relations between coordinates and momenta:

$$\begin{cases} M(\sigma) = (p_\mu(\sigma))^2 - (x'_\mu(\sigma))^2 = 0 \\ K(\sigma) = x'_\mu(\sigma) p^\mu(\sigma) = 0 \end{cases} \qquad \text{(constraints)} \tag{5}$$



Original source details: GELFAND-P001030

Diagram and typed description. The minus sign in the first constraint.

$$H = \sum p\dot{x} - L \qquad \Rightarrow \qquad H \equiv 0 \tag{6}$$

Worked on by: Barbashov, Chernikov (classical theory); Nambu — connections with dual models.

$(5), (6) \Rightarrow$ one must use Dirac's formalism (see Dirac's book).

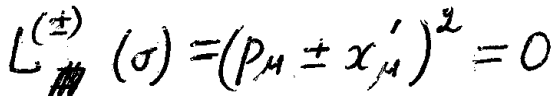
GELFAND-P001031

$$L^{(\pm)}(\sigma) = (p_\mu \pm x'_\mu)^2 = 0 \tag{7}$$



Original source details: GELFAND-P001031

Uncertain reading. The cancelled subscript under $L^{(\pm)}$ is illegible.



Original source details: GELFAND-P001031

Diagram and typed description. The square in (7).

Standard commutation relations defining the Poisson bracket:

$$\{p_\mu(\sigma), x^\nu(\sigma')\} = \delta^\nu_\mu \delta(\sigma - \sigma') \tag{8}$$

Consider the Fourier expansion of x^ν and p_μ:

$$\begin{cases} x_\mu(\sigma) = \sum\limits_{n=-\infty}^{\infty} e^{in\sigma} x^{(n)}_\mu \\ p_\mu(\sigma) = \sum e^{-in\sigma} p^{(n)}_\mu \end{cases} \tag{9}$$

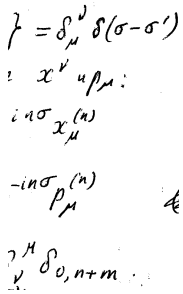


Original source details: GELFAND-P001031

Uncertain reading. Illegible cancelled entry to the right of the second line of (9).

Then

$$\{p^\mu_n, x_{m\nu}\} = \delta^\mu_\nu \delta_{0,n+m}. \tag{10}$$



Original source details: GELFAND-P001031

Diagram and typed description. The exponents $in\sigma, -in\sigma$ and the index $0, n+m$.

Expand $L^{(\pm)}(\sigma)$ from (7):

$$L^{(\pm)}(\sigma) = \sum L_n^{(\pm)} e^{in\sigma} \tag{11}$$

One can show:

$$\{L_n^{(+)}, L_m^{(-)}\} = 0, \qquad \{L_n^{(+)}, L_m^{(+)}\} = (n-m)L_{n+m}^{(+)}, \qquad n, m \in \mathbb{Z}, \tag{12}$$

(and the same upon replacing $+$ by $-$)
i.e. we obtain the algebra of vector fields on the circle.

$$\dot{p}_\mu = \{p_\mu, \widetilde{H}\}, \qquad \dot{x}_\mu = \{x_\mu, \widetilde{H}\} \tag{13}$$

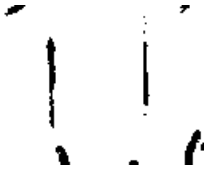


Diagram and typed description. | |

Original source details: GELFAND-P001031

$$\widetilde{H} = \alpha_+(\sigma)L^{(+)} + \alpha_-(\sigma)L^{(-)} \tag{14}$$

Changing $\alpha_\pm(\sigma)$ corresponds to gauge transformations. One can take

$$\widetilde{H} = \alpha_+(\sigma,\tau)L^{(+)} + \alpha_-(\sigma,\tau)L^{(-)} \tag{14'}$$

We must solve the equations (13) with initial data $p_\mu(\sigma,0)$, $x_\mu(\sigma,0)$, satisfying the condition:

$$L^{(\pm)}(p(\sigma,0), x(\sigma,0)) = 0 \tag{15}$$

($\Rightarrow$ they will remain satisfied)
Quantize. We obtain

$$[p_n^\mu, x_{m\nu}] = i\delta_\nu^\mu \delta_{0,n+m}, \qquad p_n^\mu, x_{m\nu} \text{ — Hermitian operators.} \tag{16}$$

A representation of these relations — ?
Introduce the operators

$$a_n^\mu = p_n^\mu + ix_{+n}^\mu, \qquad a_{-n} = a_n^+. \tag{17}$$

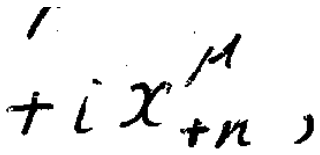


Diagram and typed description. The subscript $+n$ on x in (17).

Original source details: GELFAND-P001031

$$[a_n^\mu, a_n^{+\nu}] = \delta_{n,m}\delta_{\mu\nu}, \qquad n, m \geq 0. \tag{18}$$

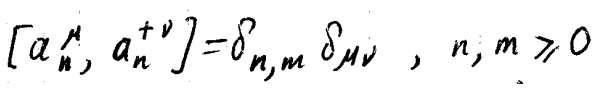


Diagram and typed description. Both operators on the left carry n. The right side has $\delta_{n,m}$.

Original source details: GELFAND-P001031

(the others commute).

GELFAND-P001032

[source number: 6]

Fock representation:

$$a_n|0\rangle = 0, \quad \forall n \geq 0. \tag{19}$$

$|0\rangle$ — the vector called the vacuum state.
All the other states should be obtained by acting with a_n^+ on $|0\rangle$.

$$[L_n^{(+)}, L_m^{(+)}] = (n-m)L_{n+m}^{(+)} + \frac{D}{12}(n^3 - n)\delta_{0,n+m}, \tag{20}$$

where

$$L_n^{(+)} = \sum_{k=-\infty}^{\infty} a_k a_{n-k} \tag{21}$$

(constraint operators)



Uncertain reading. Illegible cancelled entry before “constraint operators.”

Original source details: GELFAND-P001032

(22) Problem (quantum): find a state φ, such that $L_n^{(\pm)}|\varphi\rangle = 0$.
States of the system (stationary):

$$\begin{cases} H(p,q)|\varphi\rangle = E|\varphi\rangle \\ L^i(p,q)|\varphi\rangle = 0 \end{cases} \tag{23}$$

In (20) one must normal-order, with a_n^+ to the left of a_n.

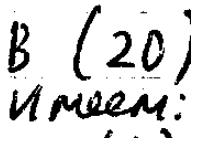


Diagram and typed description. “We have:” is written under the start of the line.

Original source details: GELFAND-P001032

We have: $L_n^{(+)}|0\rangle = 0$, $n \geq 0$, but for $n < 0$ this fails, and there is no state $|\varphi\rangle$, annihilated by all the $L_n^{(+)}$ (because of the last term in (20)).



Uncertain reading. The crossed-out word appears to read “Idea.”

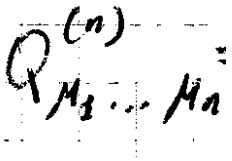
Original source details: GELFAND-P001032

The result is that we satisfy $1/2$ of all the constraints, and that suffices since the matrix elements between all physical states will vanish.

(but only because there is no better option)

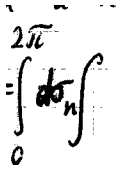
In the string problem there are ∞ operators commuting with all the constraints. These are the only ones that need quantizing! They have the form:

$$Q^{(n)}_{\mu_1 \ldots \mu_n} = \int_0^{2\pi} d\sigma_n \int d\sigma_1 \, d\sigma_2 \ldots d\sigma_{n-1} \, y_{\mu_1}(\sigma_1) \ldots y_{\mu_n}(\sigma_n) \qquad (24)$$
$$\sigma_n < \sigma_1 < \sigma_2 < \ldots < \sigma_n + 2\pi$$



Original source details: GELFAND-P001032

Diagram and typed description. The first subscript on Q in (24) is μ_1.



Original source details: GELFAND-P001032

Diagram and typed description. The differential $d\sigma_n$ is written over a correction.

(normal ordering is understood) in accordance with (26) below.

Here

$$y_\mu(\sigma) = p_\mu(\sigma) + i x'_\mu(\sigma)$$

Then

$$[Q, L_n] = 0 \qquad (25)$$

$$y_\mu = \sum_{n=0}^{\infty} a^n_\mu e^{in\sigma} + (a^n_\mu)^+ e^{-in\sigma} \qquad (26)$$

It is interesting to ask what commutation relations the $Q_{\mu_1 \ldots \mu_n}$ satisfy with one another.

Total momentum operator:

$$\int_0^{2\pi} p_\mu(\sigma) \, d\sigma = P_\mu \qquad (27)$$

GELFAND-P001033

$\{Q\}$ — a complete set of operators commuting with P

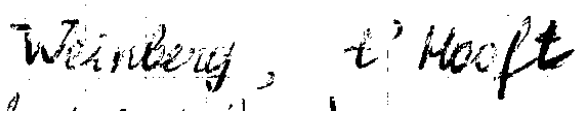
$$\{Q^{(n)}, Q^{(m)}\} = \sum Q^{(n+m-2)}$$

(classically)

13/IX–82 Gelfand Seminar.

A. S. Schwarz

Physicists: Witten, Wilson, Weinberg, t' Hooft



Original source details: GELFAND-P001033

Diagram and typed description. Names written in Latin script: Weinberg, t' Hooft.

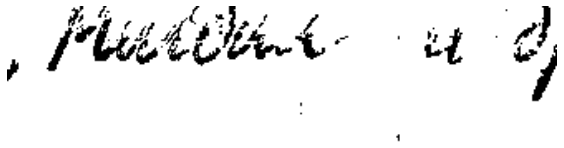
(+ Schwarz; Polyakov according to A. Migdal)

Witten — how topological invariants arise in supersymmetric models.

What is interesting to study and to hear about?

1) Gauge field theory.

String representation; Polyakov, Migdal [final stroke unclear]? and others.



Uncertain reading. Migdal [final stroke unclear]

Original source details: GELFAND-P001033

2) $SU(3), SU(2)$ and others — grand unification.

Schwarz worked on the theory of characteristic classes for supermanifolds.

3) Supersymmetry and supergravity.

— · —

Threads or strings in grand unification.

Gauge field: $A_\mu(x)$ with values in the Lie algebra of a compact Lie group

$$G_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \qquad \text{field strength} \tag{1}$$

Localization of internal symmetry.

$$L = L(\phi) = \langle \partial_\mu \phi, \partial^\mu \phi \rangle - U(\phi), \qquad \phi(x) \in \mathbb{R}^n, \quad S = \int L\, d^4x. \tag{2}$$

$G \subset SO(n)$, $\quad G$ acts on the field values $\Rightarrow$ internal symmetry.

(3) Required L — G-invariant

Sufficient: U — G-invariant.

Example: $U(\vec{\phi}) = \lambda(\vec{\phi}^2 - a^2)^2 \Rightarrow SO(3)$ model

n can be $126, 145$ (for $SO(10)$)

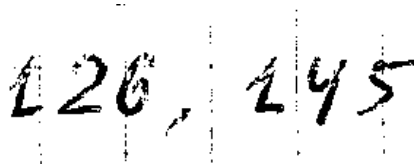


Diagram and typed description. The manuscript gives $126, 145$.

Original source details: GELFAND-P001033

Localize: add to ϕ a gauge field taking values in the Lie algebra of the group G of internal symmetry. We then treat the Lagrangian as follows:

$$\partial_\mu \to \partial_\mu + A_\mu$$

$$L \to \widetilde{L} \to \widetilde{L} + \text{const}\, \langle G_{\mu\nu}, G^{\mu\nu} \rangle \tag{4}$$

(barring cosmic miracles)

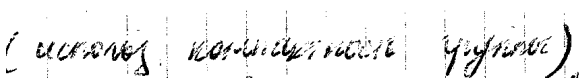
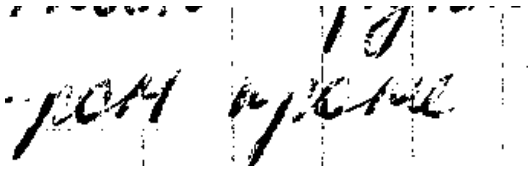


Diagram and typed description. The parenthetical remark below (4).

Original source details: GELFAND-P001033

We obtain a new Lagrangian in which the former [reading of the abbreviation]?



Uncertain reading. former [reading of the abbreviation]

Original source details: GELFAND-P001033

internal symmetries are localized symmetries.

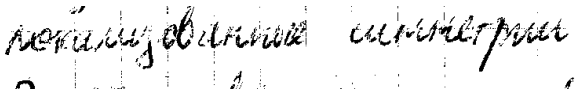
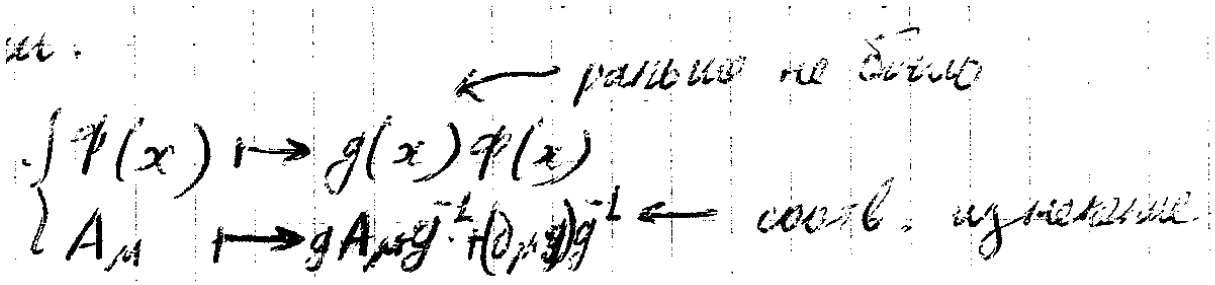


Diagram and typed description. The line ends with “localized symmetries.”

Original source details: GELFAND-P001033

We obtain invariance under

$$\begin{cases} \phi(x) \mapsto g(x)\phi(x) & \longleftarrow \text{absent before} \\ A_\mu \mapsto gA_\mu g^{-1} + (\partial_\mu g)g^{-1} & \longleftarrow \text{change accordingly} \end{cases} \tag{5}$$



Original source details: GELFAND-P001033

Diagram and typed description. In (5), a $+$ precedes the derivative of g.

GELFAND-P001034

[source number: 7]

t’ Hooft: if L is renormalizable, after localization we again obtain a renormalizable Lagrangian.

Classical analogue of the ground state: lowest energy. We obtain:

$$\phi(x) \equiv \phi, \qquad \text{where } \phi \text{ attains } \min U(\phi) \tag{6}$$

In example (3,5) one can take any $\vec{\phi}$ with $|\vec{\phi}| = a$; vacuum degeneracy

(vacuum — a two-dimensional sphere)

We obtain a set of vacua $\mathcal{R}$.



Uncertain reading. Illegible crossed-out entry before “nearby.”

Original source details: GELFAND-P001034

Nearby we expand in the deviations, retaining the quadratic part, and so on.

To agree with nature, one should take:

$$G = SU(3) \times SU(2) \times U(1) \tag{7}$$

+ add certain fermions and scalar fields

(this agrees brilliantly Illegible crossed-out entry.[?] with physics)



Uncertain reading. Illegible crossed-out entry.

Original source details: GELFAND-P001034

Furthermore, G is not simple $\Rightarrow$ extra parameters arise

(8) Idea: embed $G \subset \widetilde{G}$, so as to obtain the experimental results at low energies. — That is the idea of grand unification.

Electromagnetic subgroup $U(1) \subset SU(2) \times U(1)$

Glashow, Weinberg, Salam (before that, Gell-Mann).



Diagram and typed description. “Fritzsch” is written under “Gell-Mann” (manuscript spelling «Фритч»).

Original source details: GELFAND-P001034

Equation (7) is believed by 99% percent of physicists.

(8) is more doubtful.

Candidates for $\widetilde{G}$:



Uncertain reading. Illegible cancelled entry before $SU(5)$.

Original source details: GELFAND-P001034

$SU(5)$, $SO(10)$

still too small

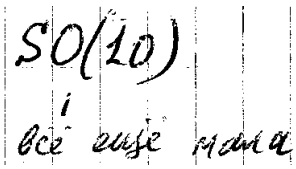
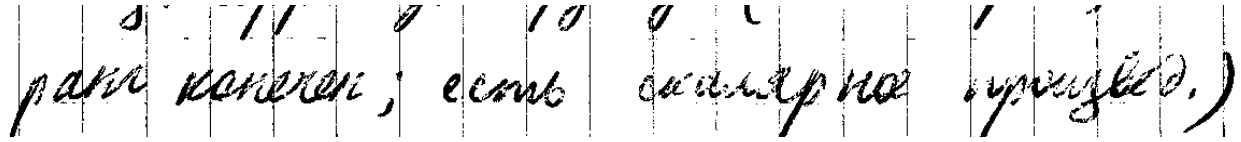


Diagram and typed description. Note under $SO(10)$.

Original source details: GELFAND-P001034

Tyupkin, Schwarz: E_8

I. M. G.'s idea: use an affine group (∞-dimensional, but in no way worse than finite-dimensional ones; the rank is finite; there is a scalar product).



Original source details: GELFAND-P001034

Diagram and typed description. At the end of the parenthesis: "there is a scalar product."

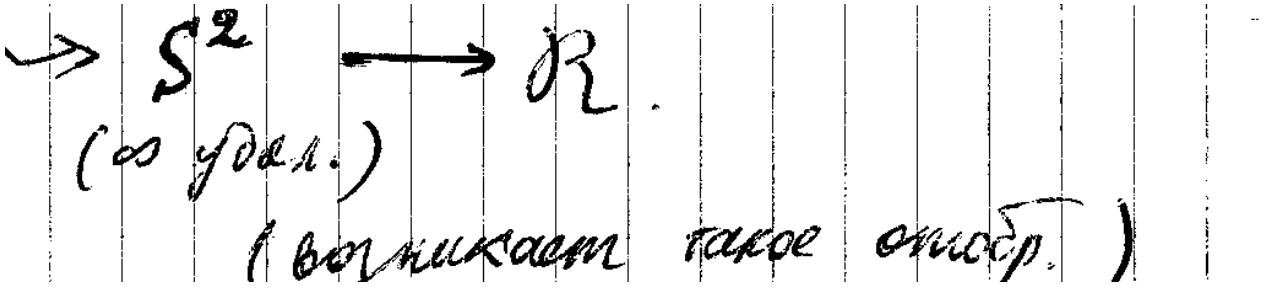
Sch. min possible rank = rank of $G = 4$.

I. M. G. Affine groups are defined using functions of one variable (cf. strings!)

— · —

Sch.: fields of finite energy must tend at ∞ to vacuum values

$$\phi(\vec{x}) \text{ on } \mathbb{R}^3 \quad \rightsquigarrow \quad \underbrace{S^2 \text{ (at infinity)} \to \mathcal{R}}_{\text{(such a map arises)}} \tag{9}$$



Original source details: GELFAND-P001034

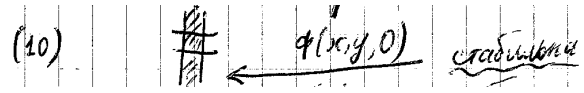
Diagram and typed description. Annotations below S^2 and the map into $\mathcal{R}$.

$\Rightarrow$ there are topological classes of fields; $\pi_2(\mathcal{R})$ — an element of this group arises.

$\Rightarrow$ magnetic monopoles.

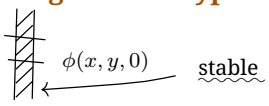
There are topologically nontrivial threads with finite energy density.

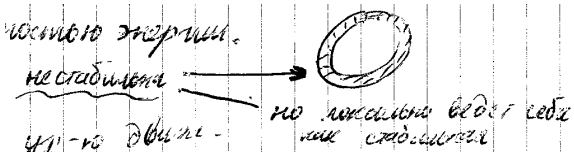
(10)



Original source details: GELFAND-P001034

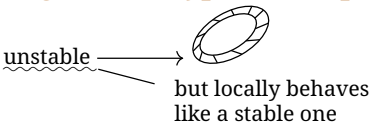
Diagram and typed description.



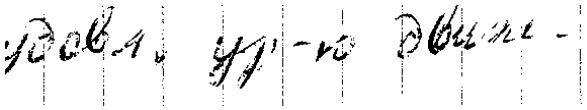


Original source details: GELFAND-P001034

Diagram and typed description.



One can require the thread to satisfy the equation of mo-



Original source details: GELFAND-P001034

Diagram and typed description. The word breaks off at “mo-” at the end of the frame.

GELFAND-P001035

Wonders with a thread: there is a notion of topological nontriviality.

(11) Thread: the field $\phi(x,y,t)$ (i.e. it depends on only two spatial variables)



Original source details: GELFAND-P001035

Uncertain reading. Illegible cancelled entry before the abbreviated “functional.”

The energy functional per unit of the coordinate z must be finite.

$$\int \mathcal{E}(x,y)\,dx\,dy < \infty$$



Original source details: GELFAND-P001035

Diagram and typed description. The manuscript abbreviates “functional” as «ф-л».

($\mathcal{E}$ — density, gives the energy)



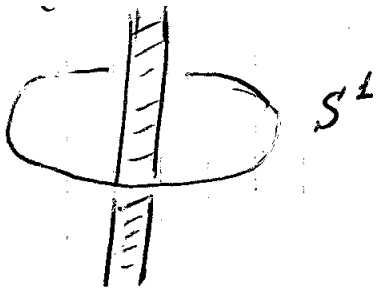
Original source details: GELFAND-P001035

Diagram and typed description. The gloss reads “density, gives the energy.”

In (10) the region where the field is nontrivial is hatched.

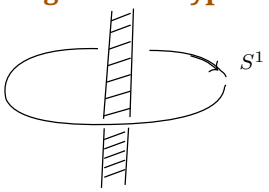
(as $e^{-\alpha x^2}$ is concentrated in a region of size $1/\sqrt{\alpha}$).

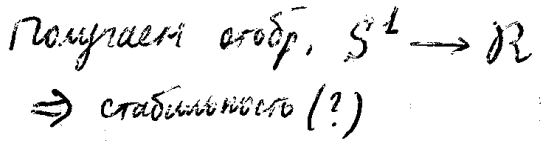
(12)



Original source details: GELFAND-P001035

Diagram and typed description.





Original source details: GELFAND-P001035

Diagram and typed description. We obtain a map $S^1 \to \mathcal{R}$.
$\Rightarrow$ stability (?)

(13) On going around a thread, a particle may change type (become another particle)
Example: it may change its electric charge.

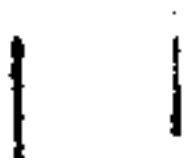


Diagram and typed description. | |

Original source details: GELFAND-P001035

$\forall$ vacuum $\phi \in \mathcal{R}$ $\exists$ a group of unbroken symmetries H_ϕ (preserving ϕ)
Assumption:

$$H_\phi = K \times U(1) \subset \widetilde{G} \tag{14}$$

(here $U(1)$ — the electromagnetic group).
Let $\widetilde{G}$ contain an element $\alpha \in \widetilde{G}$, which reverses $U(1)$:

$$\alpha u \alpha^{-1} = u^{-1} \qquad \text{for } u \in U(1) \tag{15}$$

(16) Claim. Then $\exists$ a topologically nontrivial thread (and even a solution) around which a particle changes its charge.

— · —

On parity.



Uncertain reading. The crossed-out mark is illegible.

Original source details: GELFAND-P001035

Hypothesis: mirror matter exists.

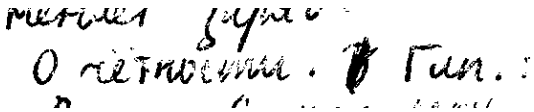


Diagram and typed description. “On parity,” then the abbreviation for “Hypothesis.”

Original source details: GELFAND-P001035

Inside the Sun there is a planet made of mirror matter. It interacts weakly but exerts a gravitational influence $\Rightarrow$ this explains certain experiments.
(17) Claim. 1) $\exists$ grand unifications that include mirror particles
2) Illegible cancelled entry after “2).”[?]



Uncertain reading. Illegible cancelled entry after “2).”

Original source details: GELFAND-P001035

∃ threads around which particles turn into mirror particles.

The interaction of our particles with mirror particles is weak (interaction through heavy bosons; if we neglect it — the Lagrangian splits).



Original source details: GELFAND-P001035

Diagram and typed description. The start of “neglect” is written over earlier strokes.

These are not antiparticles. There are mirror photons, mirror charge, and so on.

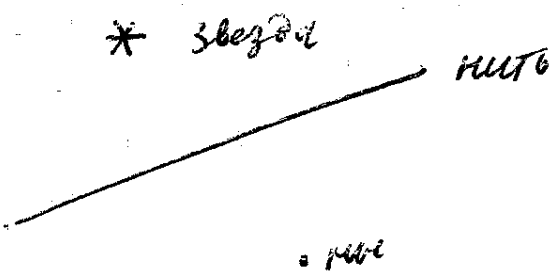
The coupling is very weak.

Thread of such unifications. Effect.



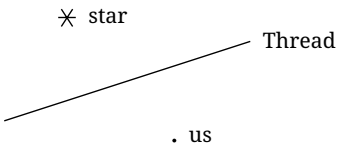
Original source details: GELFAND-P001035

Diagram and typed description. The manuscript underlines “Thread of such unifications.”



Original source details: GELFAND-P001035

Diagram and typed description.



We stop seeing the star because its photons



Original source details: GELFAND-P001035

Uncertain reading. Illegible cancelled entry before the abbreviated “become.”

become mirror photons.

The dynamics is semiclassical, which applies because we are far from the thread.

The threads are thought to have survived from the early universe.



Original source details: GELFAND-P001035

Diagram and typed description. The last sentence begins “It is thought.”

GELFAND-P001036

[source number: 8]

It is unclear whether there are quantum-mechanically stable threads (classically they are stable).

As the thread moves, the stars should be replaced (by mirror ones).

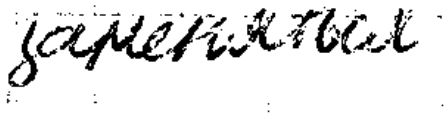


Diagram and typed description. The verb means “to be replaced.”

Original source details: GELFAND-P001036

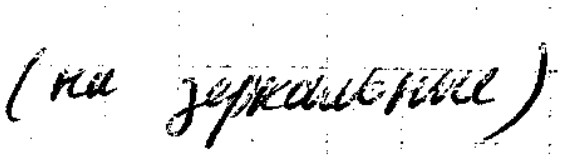
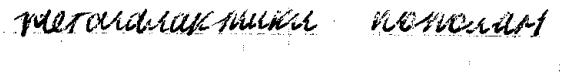


Diagram and typed description. In parentheses: “by mirror ones.”

Original source details: GELFAND-P001036

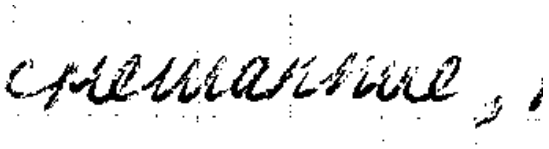
Khlopov: metagalaxies in half [tentative reading]?



Uncertain reading. metagalaxies in half [tentative reading]

Original source details: GELFAND-P001036

and nothing will change; the only chance — quasars; they are special [tentative reading]?,



Uncertain reading. special [tentative reading]

Original source details: GELFAND-P001036

but they have a characteristic rhythm of variation.

There is a preprint: one quasar over 100 years sharply changed its luminosity.



Diagram and typed description. The underlined word reads “preprint.”

Original source details: GELFAND-P001036

Khlopov’s remark: it is explained, if a thread passed.

GELFAND-P001037

20/IX-82. Gelfand Seminar.

Sinai.

Topics (in physics)

1. Renormalization group.

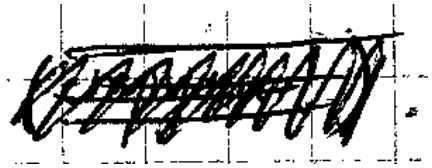
2. Debye screening as the principal mechanism of phase transitions in systems with continuous symmetry.

3. Gauge models.

4. Gravity, supersymmetry, supergravity.

A. A. Migdal.

5. Lattice models.



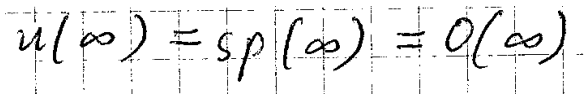
Original source details: GELFAND-P001037

Uncertain reading. Illegible cancelled expression in parentheses.

6. Strings.

7. Large N (N — rank of the gauge group):

$$u(\infty) = \mathrm{sp}(\infty) = O(\infty)$$



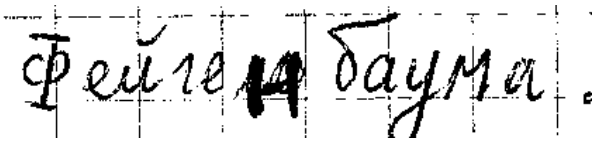
Original source details: GELFAND-P001037

Diagram and typed description. Manuscript notation: $u(\infty), \mathrm{sp}(\infty), O(\infty)$.

8. Quarks and leptons. Unified theories.

9. Anderson localization.

10. Feigenbaum universality.



Original source details: GELFAND-P001037

Diagram and typed description. The n in the handwritten surname “Feigenbaum” is written over earlier strokes.

Sinai on item 10.

In the one-dimensional case: Let

$$\begin{aligned} [-1,1] &\longrightarrow [-1,1] \\ x &\longmapsto f(x;\mu) \end{aligned}$$

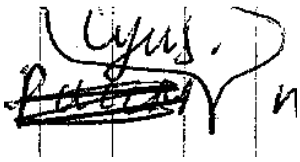
a) $f(-x;\mu) = f(x;\mu)$

b) $f(0;\mu) = 1.$

[At right:] Example: $f(x,\mu) = 1 - \mu x^2$, $1 < \mu < 2$.

GELFAND-P001038

[source number: 9]



Original source details: GELFAND-P001038

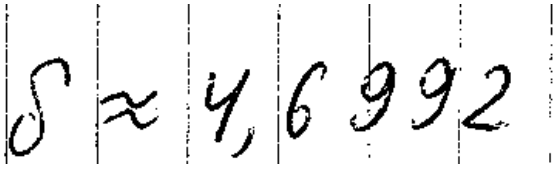
Uncertain reading. The cancelled word may be the abbreviation for “Consider.”

[inserted above the line] There is a sequence of values μ_n:

$$\forall \mu: \quad \mu_n < \mu < \mu_{n+1} \quad \Rightarrow \quad \exists \text{ a stable orbit of period } 2^n. \tag{1}$$

Consider

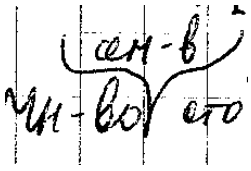
$$\frac{\mu_n - \mu_{n+1}}{\mu_{n-1} - \mu_n} \underset{n\to\infty}{\longrightarrow} \delta^{-1}, \qquad \delta \approx 4{,}6992 \tag{2}$$



Original source details: GELFAND-P001038

Diagram and typed description. The manuscript reads $\delta \approx 4{,}6992$.

The set of [inserted above the line] families of maps for which $\exists$ a sequence μ_n — an open set in the space of maps.

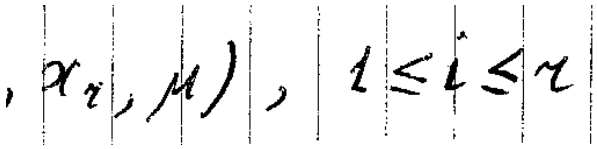


Original source details: GELFAND-P001038

Diagram and typed description. The abbreviation for “families,” with an insertion mark.

There is a generalization for differential equations.

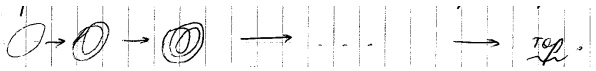
$$\frac{dx_i}{dt} = f(x_1, \ldots, x_r, \mu), \qquad 1 \le i \le r \tag{3}$$



Original source details: GELFAND-P001038

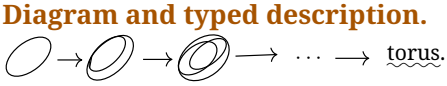
Uncertain reading. The subscript of the last x resembles the r in the upper bound. The strokes are small.

(4) μ_n, $\mu_n < \mu < \mu_{n+1} \Rightarrow$ there is a periodic orbit attracting everything, except a finite number of closed orbits.



Original source details: GELFAND-P001038

Diagram and typed description.



The exact joins in the two middle contours are unclear.

$$\mu_n - \mu_\infty \sim C(f)\delta^{-n}, \tag{5}$$

where δ — the same as above, and the set of such systems is open.

More parameters.

(6) $\mu_1, \mu_2 \Rightarrow$ period tripling occurs (and one also finds a const).

— · —

For the function f there is a complicated functional equation:

$$-\frac{1}{\alpha} f(f(\alpha x)) = f(x) \tag{7}$$

It is solved: with a polynomial (degree 10) and numerically on a computer.

The renormalization group arises from linearizing this equation.

(8) Question: what happens when $\mu > \mu_\infty$ for the vector field?

For maps, it is known that a map with an ergodic invariant measure arises (mixing and so forth $\Rightarrow$ onset of stochasticity).

A. A. Migdal.

Given a field $\varphi(x)$ with values in the algebra $u(N)$, i.e. (φ_{ij}) — an anti-Hermitian matrix,

$$\varphi_{ij}(x) = -\varphi^*_{ji}(x) \qquad \text{or} \qquad \varphi(x) = -\varphi^+(x), \qquad \text{(for definiteness)} \tag{1}$$

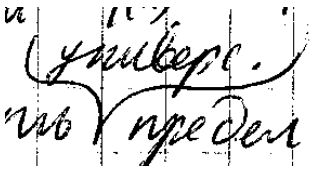
As $N \to \infty$ there is a [inserted above the line] universal limit (non-Gaussian!) and for the limit there are certain equations and



Original source details: GELFAND-P001038

Uncertain reading. Illegible cancelled entry before “observations.”

observations.



Original source details: GELFAND-P001038

Diagram and typed description. The abbreviation for “universal” is inserted above the line.

$$L = \text{tr}\left[\frac{1}{2}(\partial_\mu \varphi)^2 + U(\varphi)\right] N/\lambda, \tag{2}$$

$U(\varphi)$ — a polynomial of degree 4 — for example, φ^4.

(3) • The classical equation $\dfrac{\delta S}{\delta \varphi} = 0$, where

$$S = \int L\, d^4x \tag{4}$$

(the space is Euclidean)

GELFAND-P001039

Consider the equation

$$\frac{\partial \varphi}{\partial \tau} + \frac{\delta S}{\delta \varphi} = J(x, \tau), \tag{5}$$



Original source details: GELFAND-P001039

Diagram and typed description. ↑ random current = Gaussian random noise

where

$$\langle J^{ik}(x, \tau) J^{i'k'}(x', \tau') \rangle = \delta^{ii'} \delta^{kk'} \delta(x - x') \delta(\tau - \tau') \tag{6}$$

(this describes them completely, since they are Gaussian)

(7) Claim. Let $\varphi|_{\tau=0} = 0$. Then the solution $\varphi(x, \tau)$ has the property that if

$$\overline{Q} = \frac{1}{T} \int_0^T Q(\varphi(x, \tau))\, d\tau$$

[At right:] (Q — a functional; e.g. $Q = \mathrm{tr}(\varphi^2)$). invariant under the group for the adjoint action

$$\varphi \longmapsto S^{+} \varphi S$$

then $\overline{Q}$ is given by the formula

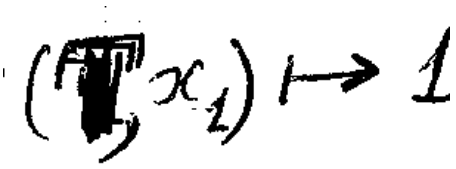
$$\overline{Q} = \int D\varphi \, \exp(-S)\, Q(\varphi(x)) \Big/ \int D\varphi \, e^{-S} \tag{8}$$

Instead of (8) we consider (5).

① Self-averaging. Consider

$$G(1, \ldots, n) = \frac{\mathrm{tr}}{N}(\varphi(1) \ldots \varphi(n)) \tag{9}$$

Here $(T, x_1) \mapsto 1, \ldots$ (for brevity); T is fixed.

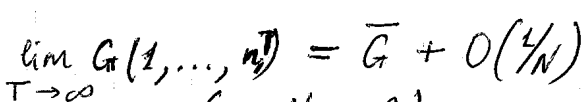


Original source details: GELFAND-P001039

Uncertain reading. The first character in the pair is overwritten and most likely reads T.

Then

$$\lim_{T \to \infty} G_T(1, \ldots, n, T) = \overline{G} + O(1/N), \qquad (\text{as } N \to \infty) \tag{10}$$



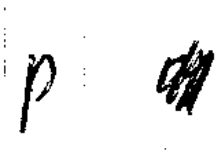
Original source details: GELFAND-P001039

Uncertain reading. A mark beside the base of G resembles T. After n there is an oblique stroke and a raised mark resembling T.

[At right:] here $\overline{G}(1,\dots,n)$ — a function of the points, independent of T and N.
(decrease in fluctuations).
The points coincide: $\frac{1}{N}\operatorname{tr}\varphi^n = \frac{1}{N}\sum_\lambda \lambda^n$ and the variance decreases.
In general there arises a $\sum$ of terms with identical probability distributions.
Thus: traces do not fluctuate.
② At $N=\infty$ a reduction is possible:

$$\varphi(x) \longmapsto e^{ipx}\varphi(0)e^{-ipx}, \qquad p = i\,\partial/\partial x \tag{11}$$

p An illegible mark after p is crossed out.?

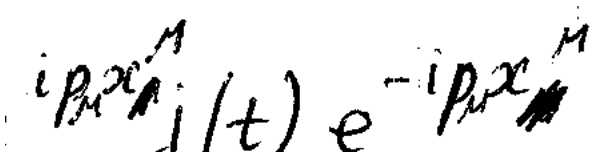


Uncertain reading. An illegible mark after p is crossed out.

Original source details: GELFAND-P001039

can be replaced by a matrix in the Cartan algebra (diagonal)

$$\widetilde{J}(x,t) = e^{ip_\mu x^\mu} j(t) e^{-ip_\mu x^\mu}, \tag{12}$$



Uncertain reading. In both exponents the x subscripts are crossed out and μ is written above. The cancelled subscripts are illegible.

Original source details: GELFAND-P001039

[At right:] p_μ — a diagonal matrix, $p = \operatorname{diag}(p_\mu^1,\dots,p_\mu^N)$
$j(t)$ — Gaussian noise

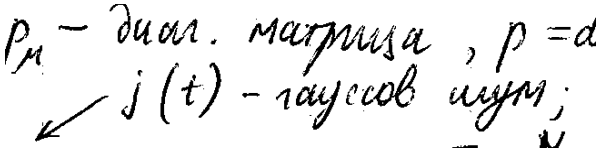


Diagram and typed description. $j(t)$ — Gaussian noise. ↙ to (13)

Original source details: GELFAND-P001039



Uncertain reading. A cancelled entry on the right begins with p.

Original source details: GELFAND-P001039

p^μ — a random quantity (independent of both t, and x)



Uncertain reading. The superscript resembles μ. The lower stroke may be a cancelled index.

Original source details: GELFAND-P001039

$$\langle j^{ik}(t), j^{i'k'}(t')\rangle = \delta(t-t')\delta^{ii'}\delta^{kk'} \tag{13}$$

GELFAND-P001040

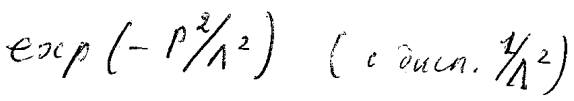
[source number: 10]

p^j — random variables, distributed [inserted above] (according to a Gaussian law) with density $\exp(-p^2/\Lambda^2)$ (with variance $1/\Lambda^2$)



Original source details: GELFAND-P001040

Uncertain reading. The superscript resembles j. It may be i.



Original source details: GELFAND-P001040

Diagram and typed description. The manuscript gives $\exp(-p^2/\Lambda^2)$ and "with variance $1/\Lambda^2$."

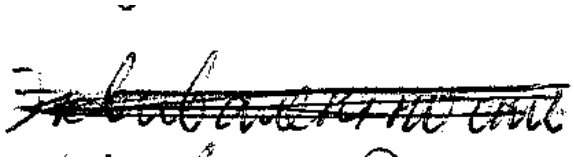
where $\underset{\text{cutoff radius}}{\Lambda} \to \infty$ (after $N \to \infty$).

In the limit as

$$N \to \infty$$
$$\Lambda \to \infty$$

we get the same white noise $J(x,t)$ (see above).

It reduces to an ordinary equation (a random matrix).



Original source details: GELFAND-P001040

Diagram and typed description. ~~Equivalence~~

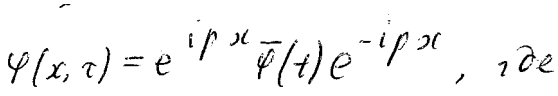
Assertion.

$$\underset{\text{with } J}{G} - \underset{\text{with } \widetilde{J}}{\widetilde{G}} \to 0$$

$$\frac{\partial \widetilde{\varphi}}{\partial \tau} + \frac{\delta S}{\delta \widetilde{\varphi}} = \widetilde{J}(x,\tau) \tag{14}$$

The solution is:

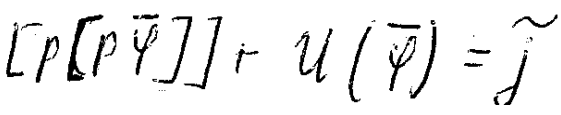
$$\varphi(x,\tau) = e^{ipx}\bar{\varphi}(t)e^{-ipx}, \qquad \text{where} \tag{15}$$



Original source details: GELFAND-P001040

Diagram and typed description. The left side uses τ. The argument of $\bar{\varphi}$ is t.

$$\frac{\partial \bar{\varphi}}{\partial \tau} + [p[p\bar{\varphi}]] + U(\bar{\varphi}) = \widetilde{j} \tag{16}$$



Original source details: GELFAND-P001040

Diagram and typed description. There are no visible commas in the brackets or prime on U.

$$p = \operatorname{diag}(p_1, \ldots, p_N)$$

Thus the dependence on the coordinates has been eliminated.

The integrals pass into the integral:

$$\bar{Q} = \int dp \int d\varphi \, \exp\big(\operatorname{tr}\big(([p,\varphi]^2 + U_\Lambda(\varphi))\big)Q(\varphi) \tag{17}$$

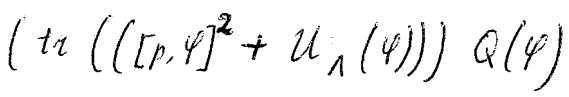
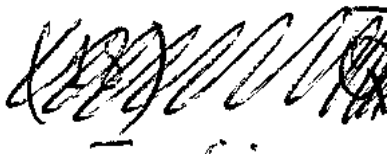


Diagram and typed description. One opening parenthesis after exp, two after tr, and three closing ones after φ in $U_\Lambda(\varphi)$.

Original source details: GELFAND-P001040

More precisely,



Uncertain reading. The cancelled line may have started with (18) and $\bar{Q}$.

Original source details: GELFAND-P001040

$$\begin{array}{l} \bar{Q} \\ \uparrow \\ \text{along trajectories (see above)} \end{array} = \lim_{\Lambda\to\infty}\lim_{N\to\infty} M_p \left\{ \int_{U(N)} d\varphi \, \exp\big(\operatorname{tr}\big(([p,\varphi]^2 + U(\varphi))\big)Q(\varphi) \right\} \tag{18}$$

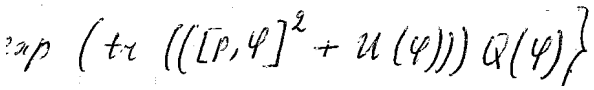


Diagram and typed description. The parentheses follow the same pattern as in (17).

Original source details: GELFAND-P001040

To be calculated:

$$\lim_{N\to\infty} \frac{1}{N^2} \ln \int_{U(N)} \prod_{\mu=1}^{4} dU_\mu \, \exp\Big(-\beta N \sum_{\mu,\nu} \operatorname{tr}(U_\mu U_\nu U_\mu^+ U_\nu^+)\Big) = F(\beta) \tag{19}$$

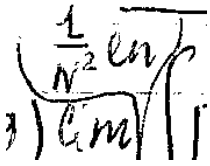


Diagram and typed description. The manuscript inserts $\frac{1}{N^2}$ ln before the integral.

Original source details: GELFAND-P001040

(free energy)

dU_μ — Haar measure

(Partition function of lattice theory [inserted above] (field) as $N \to \infty$ in four-dimensional space)

This is $\sim$ field theory

Evaluated:

$$\int dU \ \exp\bigl(\beta N \operatorname{tr}(Uh + h^+U^+)\bigr) \tag{20}$$

for an arbitrary matrix h

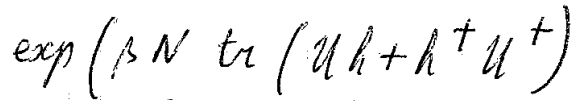


Diagram and typed description. Only one closing parenthesis is visible at the end of the formula.

Original source details: GELFAND-P001040

(a complicated expression)

First. Pirizen, Gross; then Fateev.



Uncertain reading. The first surname appears to read “Pirizen.” Its first vowel is unclear.

Original source details: GELFAND-P001040

GELFAND-P001041

27/IX-82 Gelfand Seminar.

Fuchs.

Representations of the diffeomorphism group of the circle.

(1) Consider the algebra $\{e_i,\ i \in \mathbb{Z}\}$

$$[e_i, e_j] = (j - i)e_{i+j}.$$

(of vector fields on the circle)

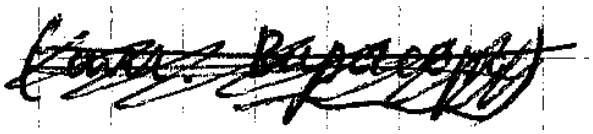


Diagram and typed description. [At right:] ~~(Virasoro alg.)~~

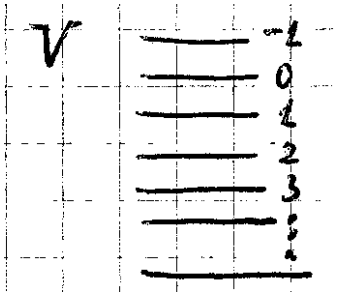
Original source details: GELFAND-P001041

There is a unique central extension (Gelfand–Fuchs):

$$\{e_i, z\}, \qquad [e_i, e_j] = \left[\begin{cases} (j-i)e_{i+j}, & i+j \neq 0, \\ \text{—"—} + \frac{1}{12}(j^3 - j)z, & i+j = 0, \end{cases} \right. \qquad \left. [e_i, z] = 0, \quad \forall i \right] \qquad \begin{matrix} \text{Virasoro} \\ \text{algebra} \end{matrix} \tag{2}$$

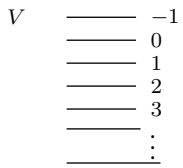
Representations of the Virasoro algebra.

(3) Verma modules,

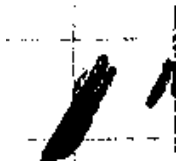


Original source details: GELFAND-P001041

Diagram and typed description.



graded The cancelled mark is illegible.[?]



Original source details: GELFAND-P001041

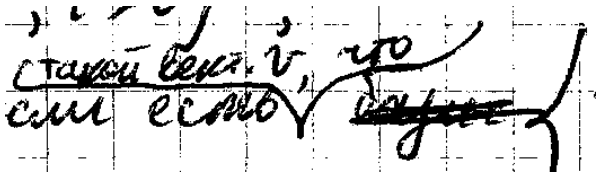
Uncertain reading. The cancelled mark is illegible.

modules bounded below

(graded and bounded on one side)

(4) Subalgebra $\{e_i,\ i > 0\}$;

Verma module: if there is [inserted above] a vector v, such that



Original source details: GELFAND-P001041

Diagram and typed description. ~~a basis~~

$$\{v,\ e_1v,\ e_1^2v,\ e_2v,\ e_1^3v,\ e_2e_1v,\ e_3v,\ \ldots\} \quad —$$

(5) — a basis, and $e_{-j}v = 0$ for $j > 0$. (it suffices that $e_{-1}v = e_{-2}v = 0$)

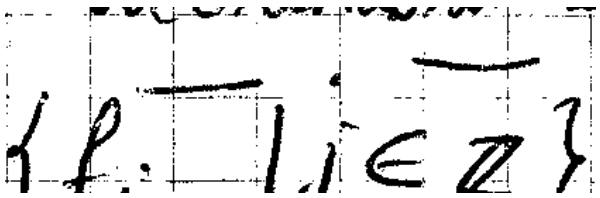
Fix complex numbers h and c.

$$e_0v = hv, \quad zv = cv \tag{6}$$

(additional condition). —

From h, c it is recovered uniquely. Notation $V_{h,c}$.

Construct a module $\{f_j \mid j \in \mathbb{Z}\}$



Uncertain reading. Two strokes appear above the entry, one to the right of f_j and one above $\mathbb{Z}$. Their relation to the entry is unclear.

Original source details: GELFAND-P001041

(auxiliary) by the formulas:

$$e_if_j = [j + \mu - (i+1)\lambda]f_{i+j} \tag{7}$$

(One can picture:

$$e_i = z^{i+1}\frac{d}{dz}, \quad f_j = z^{j+\mu}dz^{-\lambda}) \tag{8}$$

Denote it by $\overline{W}_{\lambda,\mu}$

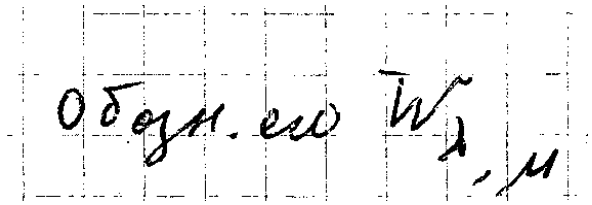


Diagram and typed description. A bar is visible over W.

Original source details: GELFAND-P001041

Consider the sequences.

$$\begin{gathered} \cdots \wedge f_{j_3} \wedge f_{j_2} \wedge f_{j_1} \\ \cdots < j_3 < j_2 < j_1 \\ \cdots \wedge f_{-4} \wedge \underset{-3}{f_2} \wedge \underset{-2}{f_5} \wedge \underset{-1}{f_9} \end{gathered} \tag{9}$$

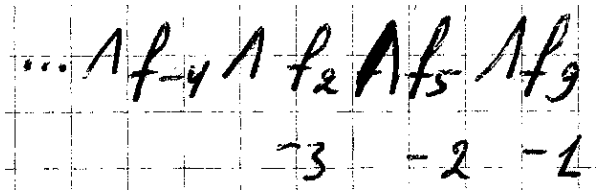


Diagram and typed description. The final factor is f_9, with -1 beneath it.

Original source details: GELFAND-P001041

(the numbers run consecutively up to a certain point; if the subsequent ones are replaced by

consecutive ones, then The short cancelled entry is illegible.?



Uncertain reading. The short cancelled entry is illegible.

Original source details: GELFAND-P001041

it ends at -1

e_i is applied successively from the right; one obtains a sum of infinitely many terms.

If the resulting f_{i+j} is already present, the product is 0.

GELFAND-P001042

[source number: 11]

In fact, one obtains a finite sum.

This defines e_i for $i \neq 0$, and e_0 cannot be defined this way.

$e_0 = [e_{-1}, e_1]$ — define by this formula e_0.

$e_0 = [e_{-2}, e_2]$ — a different coefficient results! — contradiction.

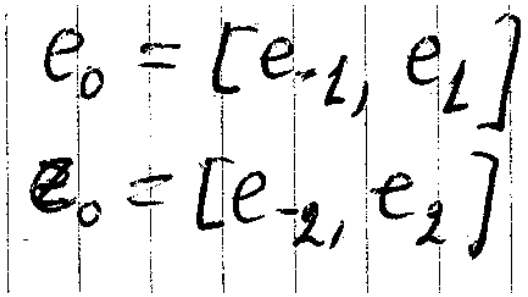


Diagram and typed description. The initial symbol of the second equality has been overwritten and reads as e_0.

Original source details: GELFAND-P001042

But one can determine e_0 and z simultaneously (and uniquely!).

z — multiplication by a number.

(see the work of Feigin and Fuchs; ΦA, 1982, $N = 2$)

e_0 — multiplication by $h + n$, $\quad h = \frac{1}{2}\mu(\mu - 2\lambda + 1)$. (?)

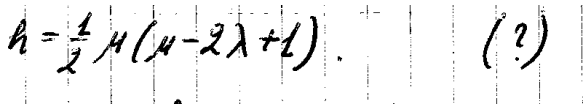


Diagram and typed description. The manuscript has $+1$ and "(?)" at right.

Original source details: GELFAND-P001042

(10) Highest-weight vector:

$$\cdots f_{-3} \wedge f_{-2} \wedge f_{-1}$$
(all consecutive)

If irreducible, then it is a Verma module.

Kac. (Lect. Notes Phys., 1979):

Which Verma modules $\overline{V}_{h,c}$ are irreducible?

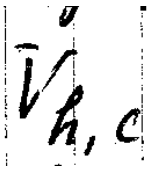


Diagram and typed description. A bar appears over V in the question.

Original source details: GELFAND-P001042

$$\overline{\Phi}_{k_1,k_2}(h,c) = \left(h - \frac{k_1^2-1}{24}(c-13) + \frac{k_1k_2-1}{2}\right)\left(h - \frac{k_2^2-1}{24}(c-13) + \frac{k_1k_2-1}{2}\right) + \\ + \frac{(k_1^2-k_2^2)^2}{16} \tag{11}$$



Diagram and typed description. A bar over Φ, and a minus sign before the first fraction.

Original source details: GELFAND-P001042

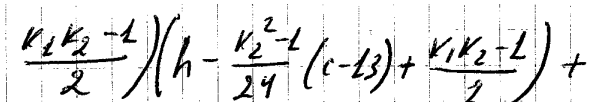
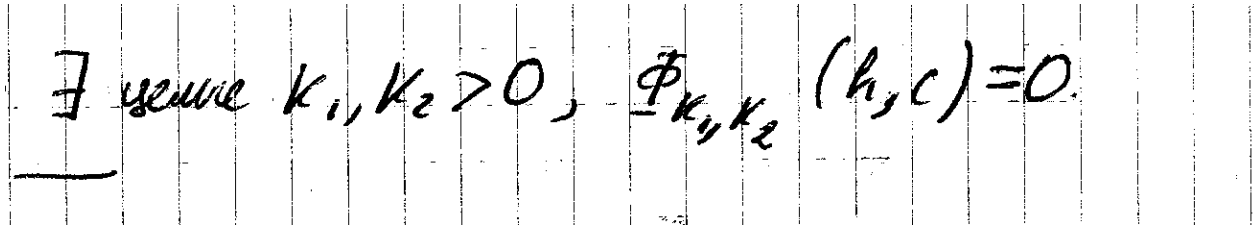


Diagram and typed description. The sign before the last 1 crosses a vertical grid line. It reads as a minus.

Original source details: GELFAND-P001042

(12) Thm. $V_{h,c}$ is reducible $(\Leftrightarrow)$ $\exists$ integers $k_1, k_2 > 0$, $\overline{\Phi}_{k_1,k_2}(h,c) = 0$.



Original source details: GELFAND-P001042

Uncertain reading. A line is visible below Φ. The short top stroke may be part of the letterform.

— . —

$$c = 0 \quad \Rightarrow \quad h = \frac{m^2 - 1}{24}$$

— . —

2nd construction.

(13) Consider polynomials in the variables

$$\begin{array}{rcccc} & x_1, & x_2, & x_3, & \dots \\ \text{of weights} & 1, & 2, & 3, & \dots \end{array}$$

There are two transformations: multiplication by x_i and $\partial/\partial x_i$.

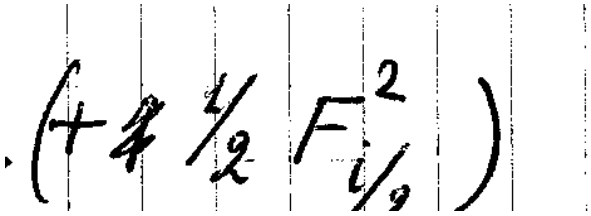
Set

$$F_k = \sum_i x_i \frac{\partial}{\partial x_{i+k}}. \tag{14}$$

Then

$$e_i = \alpha F_i + \sum_{\substack{p,q \\ p\neq 0,\ q\neq 0 \\ p+q=i \\ p>q}} F_p F_q \left(+\tfrac{1}{2} F_{i/2}^2\right), \quad \alpha \text{ — parameter.} \tag{15}$$

(if i is even)



Uncertain reading. Between $+$ and $1/2$ there is a cancelled or overwritten mark, possibly a 2.

Original source details: GELFAND-P001042

Here

$$\alpha = \mu - i(\lambda + \tfrac{1}{2}) \tag{16}$$

GELFAND-P001043

Zamolodchikov. (Sasha).

(1) Let there be a surface $\vec{x}(\xi^1, \xi^2)$, ξ^j — parameters.

We want to study sums over surfaces of different shapes.

Metric:

$$G_{ab} = \partial_a \vec{x} \cdot \partial_b \vec{x}, \qquad a, b = 1, 2. \tag{2}$$

The surface area has the form

$$A = \int \sqrt{\det G_{ab}}\, d\xi \tag{3}$$

The expression A is gauge invariant under $\xi^a \to f^a(\xi)$.

By a choice of gauge one can achieve:

$$G_{ab} = \rho \delta_{ab} \tag{4}$$

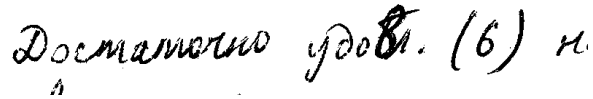
Equation of motion with Lagrangian A:

$$\partial_a \partial^a \vec{x} = 0, \qquad \text{(Laplace equation)} \tag{5}$$

$$\begin{cases} \partial_1 \vec{x} \cdot \partial_2 \vec{x} = 0, \\ (\partial_1 \vec{x})^2 = (\partial_2 \vec{x})^2 \end{cases} \tag{6}$$

(from (4)).

It suffices to satisfy (6) at the boundary $\Rightarrow$ the solution gives a minimal surface.



Original source details: GELFAND-P001043

Uncertain reading. The ending of the abbreviation for "satisfy" has been overwritten. The original strokes are indistinct.

We want to evaluate, in some sense, the sum

$$\sum_{\text{surfaces}} e^{-\mu_0^2 A(\text{surface})} \tag{7}$$

Measure — ?

Consider first

$$\sum_{\text{paths}} e^{-m L_{\text{path}}} \tag{8}$$

L_{path} — length of the path (in the usual sense; in d-dimensional Euclidean space)

For the analogy, introduce a parametrization $\vec{x}(t)$, $t \in (1, 2)$. Then

$$L = \int_1^2 \sqrt{(\dot{\vec{x}})^2}\, dt, \qquad \dot{\vec{x}} = \frac{d\vec{x}}{dt} \tag{9}$$

Minimizing (9), we get straight lines:

$$\ddot{\vec{x}} = 0 \qquad \text{(10) — analogue of (5)} \tag{10}$$

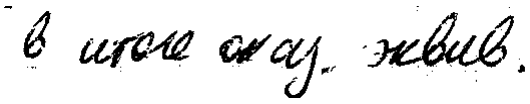
Freedom (gauge) under substitutions $t \mapsto f(t)$

$$\begin{cases} \dot{\vec{x}} = \vec{v} \\ \vec{v}^2 = 1 \end{cases} \qquad \text{(11) — analogue of (6)} \tag{11}$$

GELFAND-P001044

[source number: 12]

There are several ways to define the integral. They all turn out equivalent in the end.



Original source details: GELFAND-P001044

Uncertain reading. The final abbreviation reads as “оказ.” and may read “окаж.”. Both suggest “turn out.”

One:

(12) method replace R^d by a lattice.

$$x^\mu = 0, \pm 1, \pm 2, \dots$$

Length = number of steps.

The sum converges and can be evaluated by elementary means (explicitly!)

Another

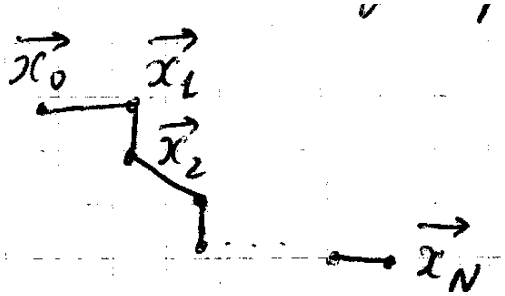
(13) method Replace the curve by a polygonal path of N links of length Δ.

Let Δ be fixed. The result is a finite-dimensional integral.

(a system of hinged rods).

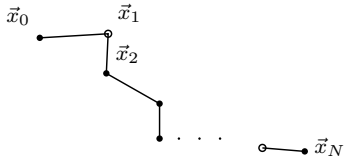
Then sum over N.

(14)



Original source details: GELFAND-P001044

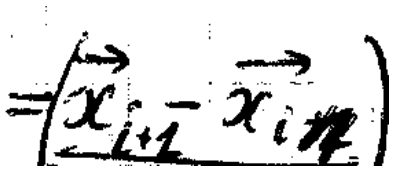
Diagram and typed description.



$$|\vec{x}_i - \vec{x}_{i+1}| = \Delta$$

Consider unit vectors

$$\vec{n}_i = \left(\frac{\vec{x}_{i+1} - \vec{x}_i}{\Delta} \right).$$



Original source details: GELFAND-P001044

Uncertain reading. After the index of the second $\vec{x}$, a symbol or group of symbols is crossed out, possibly $+1$.

$$\vec{x}_1 = \vec{x}_0 + n_1\Delta,$$
$$\vec{x}_2 = \vec{x}_0 + (\vec{n}_1 + \vec{n}_2)\Delta \qquad \sum \vec{n}_i = \frac{x_1 - x_2}{\Delta}$$
$$\cdots\cdots\cdots$$

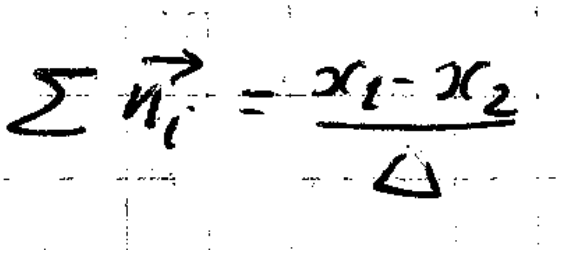


Diagram and typed description. The numerator reads $x_1 - x_2$ with no arrows.

Original source details: GELFAND-P001044

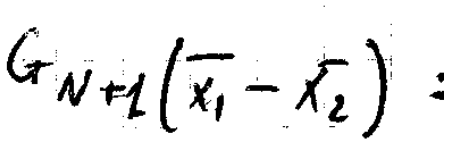
$\sum \vec{n}_i$ is given.

$\int$ over unit vectors. We are interested in the dependence on x_0 and x_N.

$$G_N(\vec{R}) = \int \prod_{i=1}^{N} d\vec{n}_i \, \delta\left(\sum \vec{n}_i - \frac{\vec{R}}{\Delta}\right), \qquad \vec{R} = \vec{x}_N - \vec{x}_0 \tag{15}$$

Recurrence relation:

$$G_{N+1}(\bar{x}_1 - \bar{x}_2) = \cdots \qquad \text{in terms of } G_N \text{ write out.}$$



Uncertain reading. There are short strokes over x_1, x_2 with no discernible arrowheads.

Original source details: GELFAND-P001044

$$G(x_1 - x_2) = \sum_{\text{paths}} e^{-N\Delta m}, \qquad G_N(x_1 - x_2) = \sum_{\substack{\text{paths} \\ \text{with } N \text{ links}}} \tag{16}$$

Consider the Fourier components

$$G(\vec{p}) = \int G(\vec{x}) e^{i\vec{p}\vec{x}} d\vec{x} \tag{17}$$

$$\begin{aligned} G_N(\vec{p}) &= \int G_N(\vec{R}) e^{i\vec{p}\vec{R}} d\vec{x} \\ &= \iint \prod d\vec{n}_i \, \delta\left(\sum \vec{n}_i - \frac{\vec{R}}{\Delta}\right) e^{i\vec{p}\vec{R}} d\vec{R} \\ &= \left(\int d\vec{n}\, e^{i\vec{p}\vec{n}\Delta}\right)^N = (C(\vec{p}, \Delta))^N \end{aligned} \tag{18}$$

Eliminate the δ-function. Independent integrations remain.

This is a Bessel function.

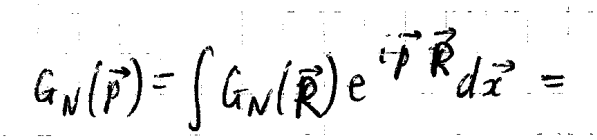


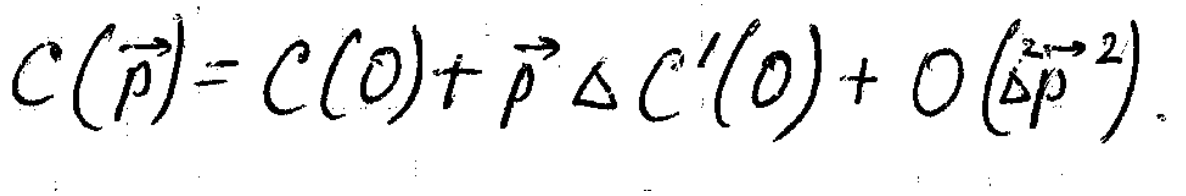
Diagram and typed description. The differential in the first integral in (18) is $d\vec{x}$.

Original source details: GELFAND-P001044

Total: geometric progression.

$$G(\vec{p}) = \sum G_N(\vec{p}) e^{-mN\Delta} = \sum [C(\vec{p}, \Delta) e^{-m\Delta}]^N \tag{19}$$

$$C(\vec{p}) = C(0) + \vec{p}\Delta C'(0) + O(\Delta^2 \vec{p}^2). \qquad (\text{take } \vec{p}\Delta \ll 1)$$



Original source details: GELFAND-P001044

Diagram and typed description. The manuscript has the term $\vec{p}\Delta C'(0)$.

GELFAND-P001045

Hence

$$G(\vec{p}) = \sum_N e^{(a+b\vec{p}-m)N\Delta} \tag{20}$$

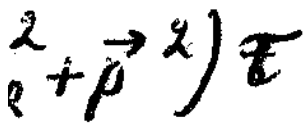
Set

$$m'^2 = m_R^2 = \frac{m-a}{b} \qquad (a, b \text{ — absolute constants})$$

(renormalized mass)

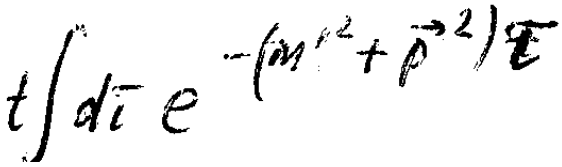
The sum ultimately passes to an integral:

$$\begin{aligned} G(\vec{p}) &= \text{const} \int d\tau\, e^{-(m_R^2 + \vec{p}^2)\tau} \\ &= \text{const} \int d\tau\, e^{-(m'^2 + \vec{p}^2)\tau} \end{aligned} \tag{21}$$



Uncertain reading. The mark after the parenthesis in the first exponent reads as τ over earlier strokes.

Original source details: GELFAND-P001045



Uncertain reading. The symbol in $d\tau$ has a detached top stroke. The τ in the exponent is overwritten, and its earlier strokes are indistinct.

Original source details: GELFAND-P001045

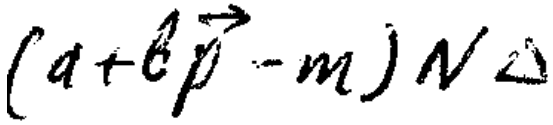
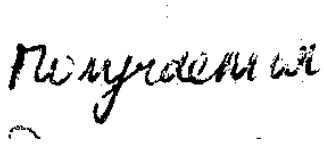


Diagram and typed description. Equation (20) has $b\vec{p}$. There is no square over $\vec{p}$.

Original source details: GELFAND-P001045

The result obtained is independent of the discretization, [crossed out: ~~one must?~~] one must carry out the appropriate mass renormalization.



Original source details: GELFAND-P001045

Uncertain reading. The first word reads “Получаемый” (“obtained”). The middle strokes are close together.



Original source details: GELFAND-P001045

Uncertain reading. The crossed-out word may read “надо” (“one must”).

3rd method. Treat the rods as springs. The length has a Gaussian distribution.

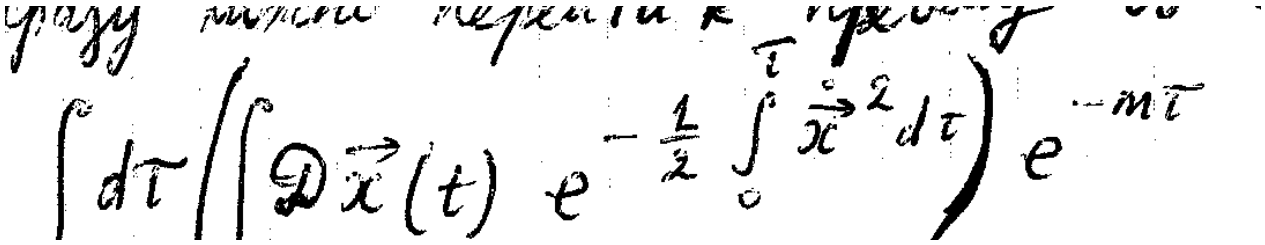
$$\sum_N e^{-Nm\Delta} \int \prod_i d\vec{x}_i \, e^{-\frac{1}{2\Delta}\sum_i^N (\vec{x}_i - \vec{x}_{i+1})^2} = G(\vec{R}) \tag{22}$$

(leads to the same answer).

Here we can pass to the limit straightaway, before evaluation.

$$\int d\tau \left(\int \mathcal{D}\vec{x}(t) e^{-\frac{1}{2}\int_0^\tau \dot{\vec{x}}^2 d\tau} \right) e^{-m\tau} \tag{23}$$

(integral with respect to Wiener measure).

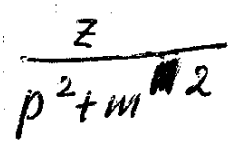


Original source details: GELFAND-P001045

Diagram and typed description. Equation (23) has $\vec{x}(t)$ and two differentials $d\tau$.

Answer:

$$\int G(\vec{R}) e^{i\vec{p}\vec{R}} d\vec{R} = \frac{z}{p^2 + m^{[?]\,2}} \qquad (z \text{ depends on normalization}) \tag{24}$$



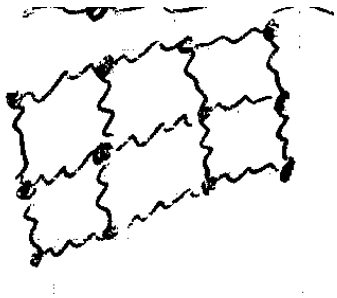
Original source details: GELFAND-P001045

Uncertain reading. Between m and the superscript 2 there is a heavily cancelled mark whose reading is uncertain.

$G(\vec{R})$ — inverse Fourier transform (Hankel function).

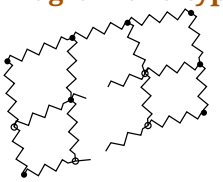
The case of surfaces.

(25)



Original source details: GELFAND-P001045

Diagram and typed description.



The gaps in the middle and lower rows are shown. The exact connections are unclear.

Connect the points with springs.

A regular lattice will not do:



Original source details: GELFAND-P001045

Diagram and typed description.



The drawing shows the visible gaps. The exact closure of the outline is unclear.

For the lattice:

$$\int \mathcal{D}\vec{x}(\xi^1,\xi^2)e^{-\frac{1}{2}\int \partial_a\vec{x}\cdot\partial^a\vec{x}\,d^2\xi} \tag{26}$$

(for a two-dimensional surface in 26-dimensional space, one obtains the correct answer)

Suppose there is an intrinsic geometry described by $g_{ab}(\xi)$. The integral will take the form:

GELFAND-P001046

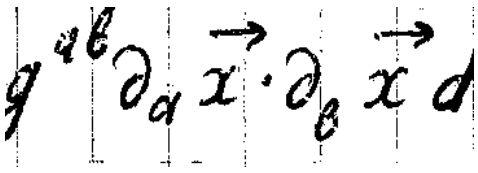
[source number: 13]

$$G=\int \mathcal{D}\vec{x}(\xi)\exp\left\{-\frac{1}{2}\int\sqrt{g}\,g^{ab}\partial_a\vec{x}\cdot\partial_b\vec{x}\,d^2\xi+\lambda\int\sqrt{g}\,d^2\xi\right\} \tag{27}$$

with the addition of λ, which we will choose.

Answer (correct):

$$\begin{aligned} G=&\int\mathcal{D}g_{ab}\int\mathcal{D}\vec{x}(\xi)\\ &\times\exp\left\{-\frac{1}{2}\int\sqrt{g}\,g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\,d^2\xi+\lambda\int\sqrt{g}\,d^2\xi\right\}\end{aligned} \tag{28}$$



Original source details: GELFAND-P001046

Diagram and typed description. A dot separates the derivatives in (27). In (28) they are juxtaposed.

By minimizing, we obtain the equations.

4/X-82 Gelfand Seminar.

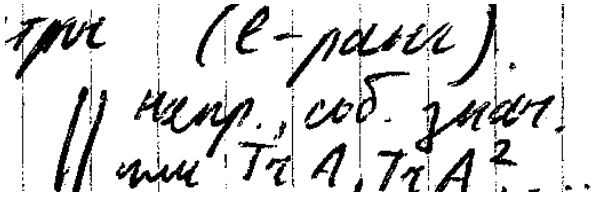
Kirillov on the work of Ginzburg.

G — a Lie group, $\mathfrak{g}$ — a Lie algebra, consider the orbits of the adjoint representation;

there are orbits in general position, and there are nilpotent ones (corresponding to nilpotent elements of the algebra $\mathfrak{g}$);

Ω_c — an orbit, $c = (c_1, \ldots, c_\ell)$ — its parameters (ℓ — the rank).

Example: $SL(2, \mathbb{R})$; $\mathfrak{g} \cong \mathbb{R}^3$; the orbits are hyperboloids.



Original source details: GELFAND-P001046

Diagram and typed description. [At right ||] e.g., eigenvalues or, Tr A, Tr $A^2, \ldots$

Desingularization according to Springer.

G — a semisimple group, $N \subset G$, N — the set of unipotent elements (with all eigenvalues 1).

N — an algebraic set with singularities.

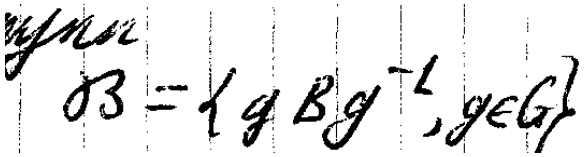


Original source details: GELFAND-P001046

Uncertain reading. A short entry before "singularities" is heavily crossed out. It may be "o."

B_0 — a Borel subgroup in G; $\mathcal{B}$ — the collection of all Borel subgroups (they are all conjugate), so that

$$\mathcal{B} = \{gBg^{-1},\ g \in G\}.$$



Original source details: GELFAND-P001046

Diagram and typed description. The formula has B without a subscript, while the preceding line has B_0.

A desingularization of the algebraic set X (with singularities) — a set Y (without singularities) of the same dimension, $Y \longrightarrow X$ is one-to-one almost everywhere (except in smaller dimension).



Original source details: GELFAND-P001046

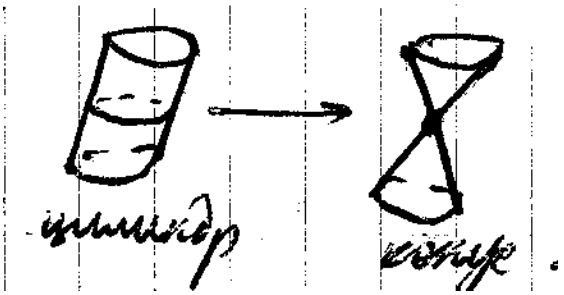
Uncertain reading. The word before "Desingularization" is crossed out and may read "Всп.".



Original source details: GELFAND-P001046

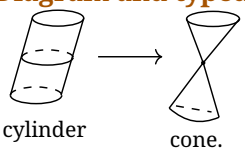
Uncertain reading. The word corresponding to "same" is written over a correction. The earlier strokes are indistinct.

Example:



Original source details: GELFAND-P001046

Diagram and typed description.



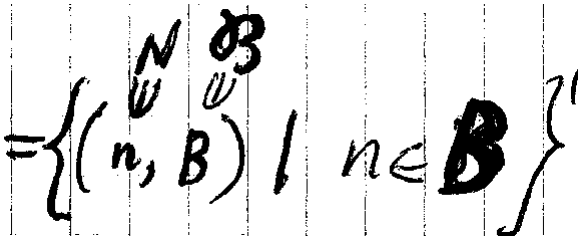
$$(x, y, z) \longmapsto (xz, yz, z).$$



Original source details: GELFAND-P001046

Uncertain reading. An indistinct symbol is crossed out after $\mathcal{M}$.

$$\mathcal{M} = \left\{ \left(\begin{matrix} N & \mathcal{B} \\ \cup & \cup \\ n, & B \end{matrix} \right) \middle| \, n \in B \right\} \qquad \text{(flags)}.$$



Original source details: GELFAND-P001046

Diagram and typed description. The membership signs above n and B are written vertically.

Then $\mathcal{M}$ — a desingularization of N.



Original source details: GELFAND-P001046

Uncertain reading. A symbol after "Then" is crossed out and illegible.

For $SL(2,\mathbb{R})$ this is precisely what occurs in the example.

On Ginzburg again.

$$f \in C^\infty(\mathfrak{g}) \longmapsto \psi(c) = \int_{\Omega_c} f$$

(the measure is chosen consistently on orbits in general position).

$$\int_{\Omega_0} f = P\psi(c)|_{c=0}\,, \qquad P \text{ — a differential operator}$$

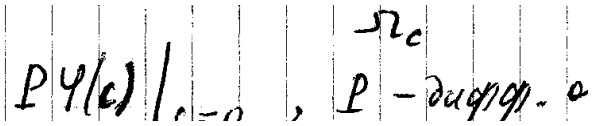


Diagram and typed description. The operator is written P in both places.

Original source details: GELFAND-P001046

GELFAND-P001047

I. M. G. Integrals over hyperbolas are given. Find the value of the function at 0.

(more generally — a hyperboloid).



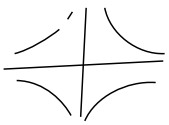


Diagram and typed description. The short bend at the right end of the horizontal line is unclear.

Original source details: GELFAND-P001047

$$xy = c; \qquad \text{distance from } 0 \text{ of order } \sqrt{c}.$$

For $|x| \geqslant 1$ distance $\sim c$.



Uncertain reading. The $>$ sign has a separate slanted stroke beneath it. The intended sign is likely $\geqslant$.

Original source details: GELFAND-P001047

$\Rightarrow$ we evaluate the integral over the lines (limit of the integral over the hyperbolas) and so on.

See the expansion of $\delta(xy - c)$ in powers of c in the first volume of Generalized Functions.

Zamolodchikov.

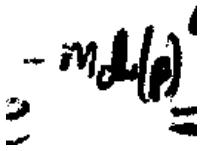
We want to find the sum

$$\sum_S e^{-\mu_0^2 A(S)} \tag{1}$$

S — surface

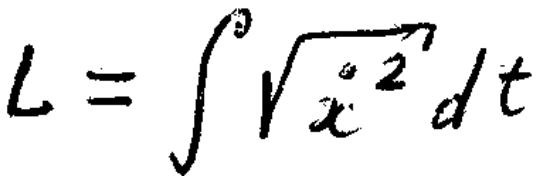
The integral over the set of all curves (see above) had the form

$$\sum_P e^{-m_0 L(P)} = \int_0^\infty d\tau\, e^{-m^2\tau} \int_{\substack{\vec{x}(0)=\vec{0}\\ \vec{x}(\tau)=\vec{R}}} \mathcal{D}\vec{x}(t)\, e^{-\frac{1}{2}\int_0^\tau \dot{\vec{x}}^2(t)\,dt},$$
$$L = \int \sqrt{\dot{x}^2}\, dt. \tag{2}$$



Original source details: GELFAND-P001047

Diagram and typed description. A subscript zero is visible between m and L, touching the stroke of L.



Original source details: GELFAND-P001047

Diagram and typed description. In the formula for L, $\dot{x}$ has no arrow.

($m \neq m_0$ — renormalization)

And it is equal to:

$$(\text{const}) \int_0^\infty d\tau \left(e^{-m^2\tau - \frac{\vec{R}^2}{2\tau}} \right) \frac{1}{\tau^{d/2}}, \qquad d \text{ — dimension of the space.} \tag{2'}$$

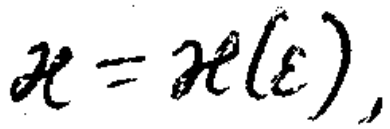
The relation between m and m_0 depends on the method of regularization [approximation — fine structure of the space].

$$\sum_P e^{-m_0 L} = \sum_{\substack{L=0 \\ \text{(integral)}}}^{\infty} f(L) e^{-m_0 L} \tag{3}$$

where

$$f(L) \sim L^\beta e^{\varkappa L} \qquad \text{when } L \gg \varepsilon \qquad (\varepsilon \text{ — lattice spacing})$$

$$\varkappa = \varkappa(\varepsilon), \qquad \beta \text{ is independent of } \varepsilon.$$



Uncertain reading. The looped symbol is rendered as $\varkappa$. A cursive x is also possible.

Original source details: GELFAND-P001047

Hence

$$(3) = \text{const} \sum (L)^\beta e^{-(m_0 - \varkappa)L} \tag{4}$$

$$m^2 = m_0 - \varkappa.$$

2 dangers:

a) concentration on straight lines

b) it diverges

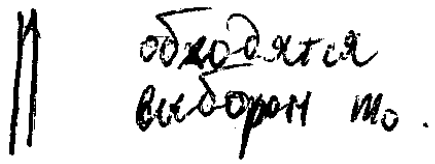


Diagram and typed description. [At right of both items ||] are avoided by the choice of m_0.

Original source details: GELFAND-P001047

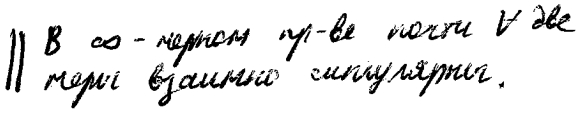


Diagram and typed description. [Farther right ||] In an infinite-dimensional space almost any two measures are mutually singular.

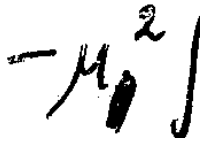
Original source details: GELFAND-P001047

GELFAND-P001048

[source number: 14]

Let us pass to surfaces.
$\vec{x}(\xi^1, \xi^2)$; $g_{ab}(\xi^1, \xi^2)$ — the metric on the surface.

$$\sum_S e^{-\mu_0^2 A(S)} = \int d\mu(\vec{x}, g) \exp\Big\{-\frac{1}{2}\int \sqrt{g}\, g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\, d^2\xi - \\ - \mu_0^2 \int \sqrt{g}\, d^2\xi\Big\}, \qquad g = \det g_{ab}. \tag{5}$$



Uncertain reading. The subscript of the second μ is heavily overwritten. It is probably 0.

Original source details: GELFAND-P001048

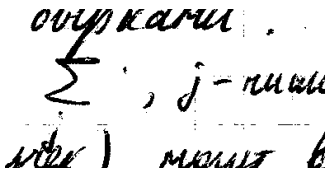
(Polyakov formula) Here we sum over surfaces [inserted above: with] given topological properties.



Diagram and typed description. The preposition “c” (“with”) is inserted between “surfaces” and “given.”

Original source details: GELFAND-P001048

[In fact one has to take account of surfaces with handles and holes.
It is unclear how the contributions compare. One needs: $\sum$, j — the number of handles.
Foamy surfaces (with many handles) can make an overwhelming contribution.



Uncertain reading. A small mark is visible under $\sum$, without a discernible j there. The text to the right calls j the number of handles.

Original source details: GELFAND-P001048

Here we choose a parameter domain in $\mathbb{R}^2$. One introduces [inserted above: elements of] distance in the function space:

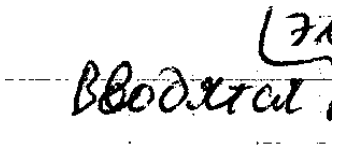


Diagram and typed description. The start of the phrase reads “Вводятся” (“are introduced”).

Original source details: GELFAND-P001048



Diagram and typed description. The word “elements” is written above the line.

Original source details: GELFAND-P001048

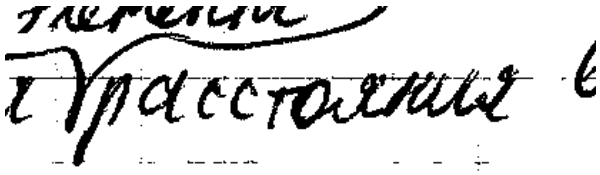


Diagram and typed description. The word “расстояния” (“distance” or “distances”) is written on the line.

Original source details: GELFAND-P001048

$$\|\delta\vec{x}(\xi^1,\xi^2)\|^2 = \int \sqrt{g}\,(\delta\vec{x}(\xi))^2\,d^2\xi$$
$$\|\delta g\|^2 = \int \sqrt{g}\,\left(g^{aa'}g^{bb'} + c\,g^{ab}g^{a'b'}\right)\delta g_{ab}\,\delta g_{a'b'}\,d^2\xi \tag{6}$$

Gauge invariance under the transformations:

$$\begin{cases}\xi^a \to \xi^a + \varepsilon^a(\xi),\\ g_{ab} \to g_{ab} + \nabla_a\varepsilon_b + \nabla_b\varepsilon_a,\end{cases} \qquad \nabla_a\varepsilon_b = \partial_a\varepsilon_b + \Gamma^c_{ab}\varepsilon_c. \tag{7}$$

nothing changes.

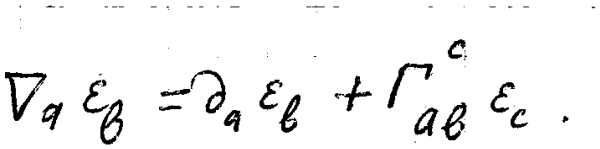


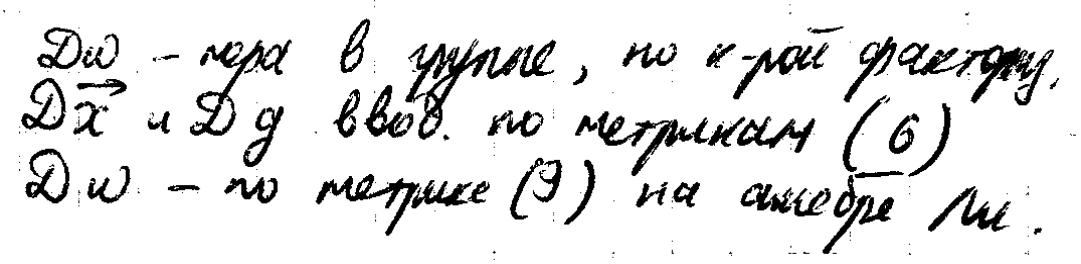
Diagram and typed description. A $+$ appears before Γ^c_{ab} in (7).

Original source details: GELFAND-P001048

One must integrate over the space of orbits.

$$d\mu(x,g) = \frac{\mathcal{D}\vec{x}\,\mathcal{D}g}{\mathcal{D}\omega} \tag{8}$$

$$\|\varepsilon\|^2 = \int \sqrt{g}\,g^{ab}\varepsilon_a\varepsilon_b\,d^2\xi \tag{9}$$



Original source details: GELFAND-P001048

Diagram and typed description. [At right of (8) and (9)] $\mathcal{D}\omega$ — the measure on the group one quotients by,
$\mathcal{D}\vec{x}$ and $\mathcal{D}g$ are introduced using the metrics (6)
$\mathcal{D}\omega$ — using the metric (9) on the Lie algebra.

Now the integral [inserted above: (5)] is well defined.



Diagram and typed description. The reference “(5)” is inserted above the line.

Original source details: GELFAND-P001048

$\int \mathcal{D}\vec{x}$ is Gaussian and reduces to the det of the operator.
Polyakov proposes choosing representatives as follows:

$$g_{ab} = \delta_{ab}\,\rho(\xi^1, \xi^2) \qquad \text{(conformally flat form).} \tag{10}$$

(choice of gauge). The conformal group remains (finite-dimensional).



Diagram and typed description. The opening strokes of “finite-dimensional” have been overwritten.

Original source details: GELFAND-P001048

det are the same. One obtains

$$\int d\mu(\rho)\,[\det(\Delta_g)]^{(26-d)/2}, \qquad \Delta_g \text{ — Laplacian of the metric} \tag{11}$$

(ρ — the coefficient in (10)).

GELFAND-P001049

One obtains:

$$(11) = \int d\mu(\varphi)\, e^{\frac{d-26}{48\pi}\int \frac{1}{2}(\partial_a\varphi)^2 + m^2 e^{\varphi})\, d^2\xi} \tag{12}$$



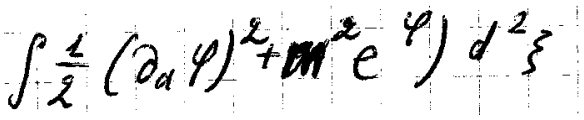
Diagram and typed description. The reference “(11)” to the left of the equals sign is written over earlier strokes.

Original source details: GELFAND-P001049



Uncertain reading. The base of the power before e^{φ} is heavily overwritten and reads as m. The earlier strokes are indistinct.

Original source details: GELFAND-P001049



Uncertain reading. A closing parenthesis follows e^{φ}. An opening parenthesis for the whole sum is indistinct.

Original source details: GELFAND-P001049

$d\mu(\varphi)$ — the measure from the metric

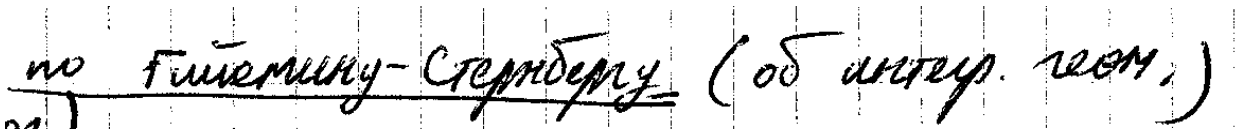
$$\|\delta\varphi\|^2 = \int e^{\varphi}(\delta\varphi)^2\, d^2\xi \tag{13}$$

GELFAND-P001050

11/X-82 Gelfand Seminar

V. Ginzburg.

after Guillemin–Sternberg (on integral geometry)

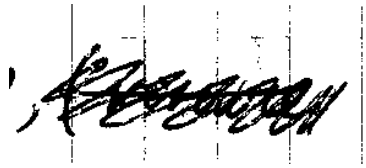


Original source details: GELFAND-P001050

Diagram and typed description. The right-hand heading reads “after Guillemin–Sternberg.”

X — a set (complex)

$\Sigma \subset T^*X$, Σ — a coisotropic subset,



Uncertain reading. The word after “subset” is crossed out and may read “which” or “conic.”

Original source details: GELFAND-P001050

has the following properties:

1. Σ — is conic;
2. $\Sigma \to X$ is a fibration.

On Σ there is a 0-foliation (kernel of the symplectic form), which is always integrable.

Locally arranged as follows: $q_1, \ldots, q_n, p_1, \ldots, p_k$

0-foliation

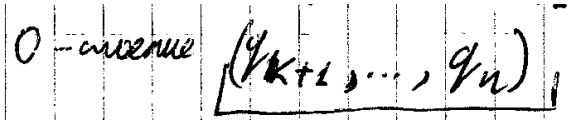


Diagram and typed description. $\underbrace{(q_{k+1}, \ldots, q_n)}$

Original source details: GELFAND-P001050



Uncertain reading. There is a mark before “Suppose,” possibly “Мы” (“we”) or “Или” (“or”).

Original source details: GELFAND-P001050

Suppose it is a fibration (all leaves are closed, etc.)

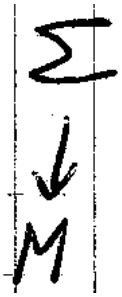


Diagram and typed description. $\begin{array}{c}\Sigma\\ \downarrow\\ M\end{array}$

Original source details: GELFAND-P001050

and M — the quotient space.

From Σ construct a Lagrangian subset $\Lambda \subset T^*X \times T^*X$:

$$\Lambda = \{(\xi_1, \xi_2) : \quad \xi_1, \xi_2 \in \Sigma, \quad \xi_1 \sim \xi_2\}$$

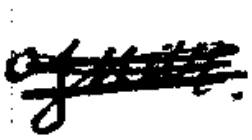
Properties:

1) $\Lambda^t = \Lambda$

2) $\Lambda \circ \Lambda = \Lambda$ (projector)

GELFAND-P001051

Apparently the converse: if Λ is Lagrangian and satisfies 1), 2), then it is obtained from Σ.



Uncertain reading. A word after "and" is heavily crossed out. A loop is visible beneath the cancellation.

Original source details: GELFAND-P001051

Fourier integral operators are constructed from Λ: denote their set by $\mathcal{E}_\Lambda$.

It is a ring by property 2).



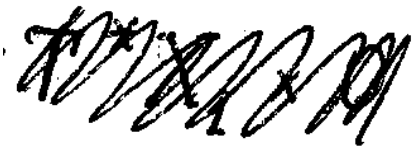
Uncertain reading. A short entry before "property" is crossed out.

Original source details: GELFAND-P001051

This is an analogue of the Radon transform.

$X = X_1 \times X_2$; Σ — directions normal to X_1 (along X_2) in the cotangent bundle.

$$\Lambda = \text{diag}(T^*X_1 \times T^*X_1) \times 0$$
$$\subset T^*X_1 \times T^*X_1 \times T^*X_2 \times T^*X_2.$$



Uncertain reading. A formula before diag is crossed out. Individual signs resembling T^*, X_1, $\times$, and 0 are visible.

Original source details: GELFAND-P001051

Operators: in X_1 all pseudodifferential.

in X_2 smoothing.

Shmelyov.

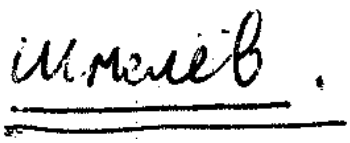


Diagram and typed description. The twice-underlined surname reads "Shmelyov."

Original source details: GELFAND-P001051

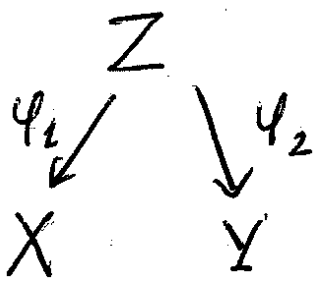


Diagram and typed description.

$$\begin{array}{ccc} & Z & \\ {\scriptstyle\psi_1}\swarrow & & \searrow{\scriptstyle\psi_2} \\ X & & Y \end{array}$$ fibrations

Original source details: GELFAND-P001051

Requirement: $\psi_2^* : N^*Z \longrightarrow T^*Y$ an embedding;

Fröhlich

1) Magnetic monopoles

2) Gauge theories $G \overset{\text{e.g.}}{=} SU(2)$

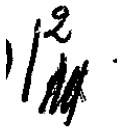
Lagrangian: φ : isospin $= 1$ (3 components)

$$\begin{aligned} E &= \int \{(D_A\varphi)^2 + F^2\} \\ &= \int \left\{ \left|(\partial_\mu - eA_\mu\wedge)\varphi(x)\right|^2 + \operatorname{tr}(F_{\mu\nu}\cdot F^{\mu\nu})(x) \right\} dx, \end{aligned} \tag{1}$$



Diagram and typed description. A sign resembling $\wedge$ follows A_μ.

Original source details: GELFAND-P001051



Uncertain reading. A mark beneath the squared norm is heavily crossed out.

Original source details: GELFAND-P001051



Uncertain reading. Below the dot between the two Fs are overwritten strokes and a wavy line. The earlier marks are indistinct.

Original source details: GELFAND-P001051

$x \in \mathbb{R}^3$



Uncertain reading. An equality to the right of $x \in \mathbb{R}^3$ is crossed out. It appears to begin $E = E$.

Original source details: GELFAND-P001051

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

Monopole: a solution of the Euler–Lagrange equations such that the degree of $\varphi|_{S^2} \longrightarrow S^2$ at ∞ equals 1. ‖ (stationary solutions)

$\exists$ solutions attaining min or loc. min — explicitly, using twistor techniques.



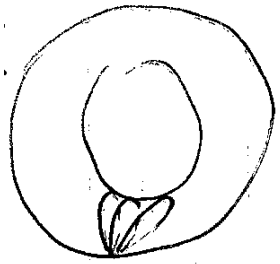
Uncertain reading. A short entry before "twistor" is crossed out.

Original source details: GELFAND-P001051

Question: other stationary points?

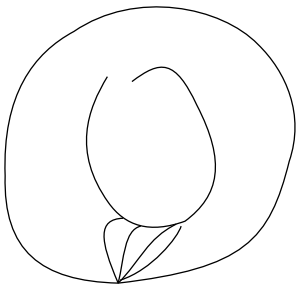
Answer: solutions have been found that are not minima

Idea:



Original source details: GELFAND-P001051

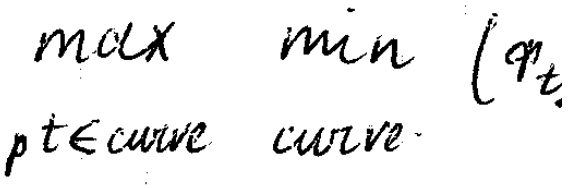
Diagram and typed description.



The inner loop has a gap. The exact connections of the thin lines at the bottom are indistinct.

one-parameter family:

$$\max_{[\rho?]\, t \in \text{curve}} \min_{\text{curve.}} (\varphi_t, F_t)$$



Uncertain reading. Before $t \in$ curve is a sign resembling ρ or p. Below min the manuscript has "curve.".

Original source details: GELFAND-P001051

C. Taubes

The new solution resembles a pair

monopole / antimonopole.

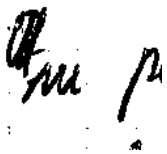
GELFAND-P001052

[source number: 16]

There is a manifold $\underset{\text{(compact)}}{M^4}$, $\pi_1(M^4) = \{\text{id}\}$; $H^2(M^4) = \{0\}$.

M^4 with a conformal structure.

$\exists\,[A_\mu?]$ a solution of the equation $F = \pm F^*$?

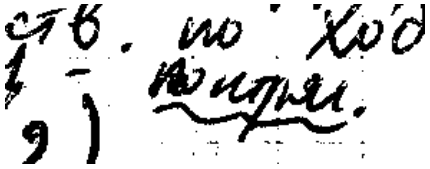


Uncertain reading. The symbol after $\exists$ is overwritten and may be A_μ.

Original source details: GELFAND-P001052

$*$ — Hodge duality;

with characteristic class 1 — [e.g.?]



Uncertain reading. The underlined word after "class 1" may read "напр." ("e.g."). Its opening has been overwritten.

Original source details: GELFAND-P001052

F is the curvature of the connection A; the group — $SU(2)$.

Answer: a solution exists!

Space of parameters labelling solutions $= \underset{\text{manifold } B}{5}$ (as on the sphere!)

(Taubes, Comm. Math. Phys, 1981).

Instanton parameters

B — a manifold with boundary, $\partial B \underset{\text{Donaldson}}{\simeq} M^4$ (the original manifold)

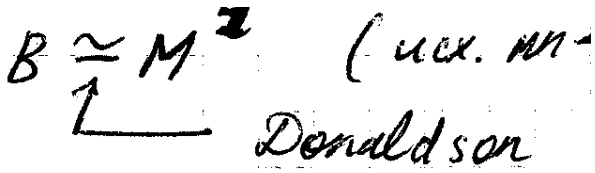
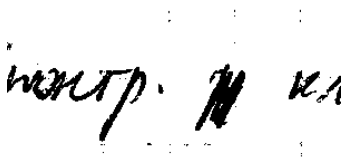


Diagram and typed description. The name "Donaldson" is connected to $\simeq$ by an angled arrow.

Original source details: GELFAND-P001052

M^4 — a boundary $\Longleftrightarrow$ (Pontr. [?] class $= 0$)

$\Longleftrightarrow$ (signature $= 0$ for the form in H^2).



Uncertain reading. A sign between "Pontr." and "class" is heavily overwritten. It may be p.

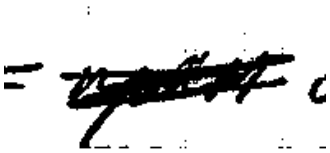
Original source details: GELFAND-P001052

Donaldson: if the form on H^2 is positive definite, then the manifold = generalized projective plane $CP^2 \Rightarrow$ the form reduces to a sum of squares (over $\mathbb{Z}$).



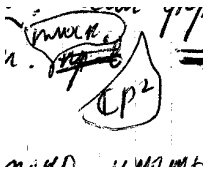
Original source details: GELFAND-P001052

Uncertain reading. A short entry between "positive" and "definite" is crossed out.



Original source details: GELFAND-P001052

Uncertain reading. The word immediately after the equals sign is crossed out.



Original source details: GELFAND-P001052

Uncertain reading. An entry after "projective" is crossed out. "Plane" is added above and CP^2 below. Both additions are outlined.

B must have singularities — cones over CP^2.

On the work of Witten.

M — n-dimensional manifold, B_p — the pth Betti number.

f — a function on M with isolated critical points.

M_p — the number of critical points of index p (the Hessian has p negative eigenvalues).

$$M_p \geqslant B_p \qquad \text{(Morse inequality)}$$

the simplest one

$d : \Omega_p \longrightarrow \Omega_{p+1}$ (exterior differential)



Original source details: GELFAND-P001052

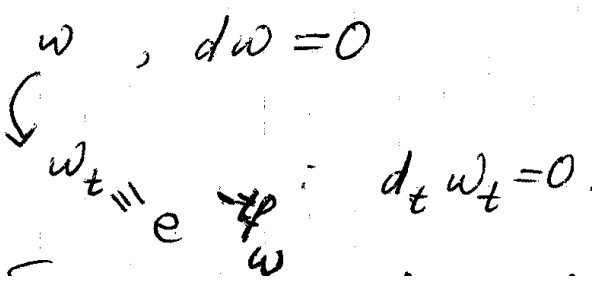
Uncertain reading. A mark before "exterior differential" is crossed out.

Introduce a metric on M; $*$ — the corresponding operator.

Choose in H^p a basis of harmonic p-forms (B_p in number)

Alternatively: set

$$d_t = e^{-tf} d\, e^{tf}$$



Original source details: GELFAND-P001052

Diagram and typed description.

$\omega, \quad d\omega = 0$

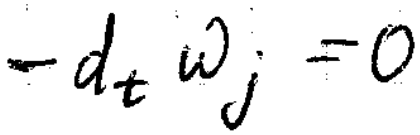
$\omega_t = e^{-tf}\omega; \quad d_t\omega_t = 0$

The exponent of e has been overwritten and reads as $-tf$.

Choose a basis in H^p, satisfying the conditions:

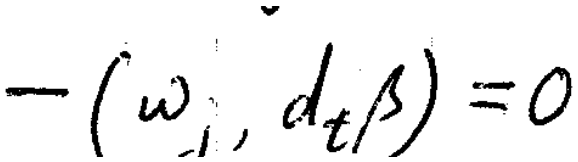
$$\left\{\begin{array}{c} -d_t\omega_j = 0 \\ -(\omega_j, d_t\beta) = 0, \quad \forall\beta \in \Omega_{p-1} \end{array}\right\} \Rightarrow (d_t^* d_t + d_t d_t^*)\omega_j = 0 \tag{1}$$

(closed in the new sense).



Original source details: GELFAND-P001052

Uncertain reading. The short stroke before $d_t\omega_j$ may be a minus or an item dash.



Original source details: GELFAND-P001052

Uncertain reading. A separate horizontal stroke also precedes the second line.

GELFAND-P001053

d_t-cohomology is the same as d-cohomology, since there is a homotopy.

$$d_t^2 = 0 \quad \text{(obvious)}$$

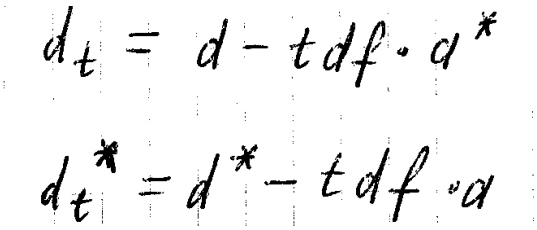
Let us compute the operator $\Box_t = d_t^* d_t + d_t d_t^*$ and find its asymptotics as $t \to \infty$.



Original source details: GELFAND-P001053

Diagram and typed description. The manuscript reads “Let us compute the operator.” A diagonal stroke is visible inside the box sign.

$$\begin{cases} d_t = d - tdf \cdot a^*, \\ d_t^* = d^* - tdf \cdot a \end{cases} \qquad \begin{array}{c} df\, a^*\omega \equiv df \wedge \omega \\ \text{(physics notation)} \end{array} \tag{2}$$

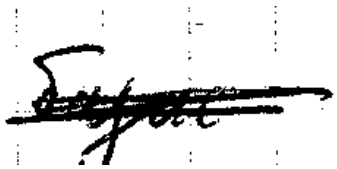


Original source details: GELFAND-P001053

Diagram and typed description. Both tdf terms in (2) carry minus signs.

$$\{a_j, a_i^*\} = \delta_{ij}. \tag{3}$$

a_i — ~~basis~~ correspond to the basis of 1-forms; (operators of exterior multiplication).



Original source details: GELFAND-P001053

Diagram and typed description. After a_i, "basis" is crossed out.

Choose so that one has (3).



Original source details: GELFAND-P001053

Uncertain reading. The entry before "one has" is heavily crossed out. It may contain a_i.

Kirillov:

$$a = (a^1, a^2, \ldots, a^n) \text{ — a vector}, \qquad df = (\partial_1 f, \partial_2 f, \ldots, \partial_n f)$$

a^i — interior multiplication (annihilation operator) — strike out dx^i, where it occurs, where it does not occur, annihilate.



Original source details: GELFAND-P001053

Uncertain reading. A word before "annihilation" is crossed out. It may be an abbreviation for "creation."

a^{i*} — exterior multiplication.

Indices can be lowered and raised using the metric.

a^j — interior multiplication by $\dfrac{\partial}{\partial x^j}$ (substitution in the first argument)

$df \cdot a$ — interior multiplication by the field obtained from df by raising the index

$$a_i^* = dx_i \wedge \qquad \text{(exterior multiplication)}$$

Fröhlich.

$$a_i = \partial/\partial x_i \llcorner \qquad \text{(interior multiplication)}$$

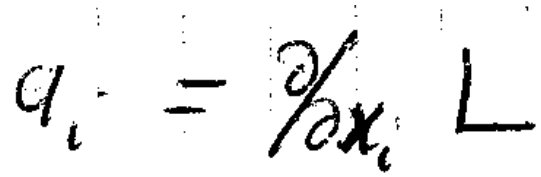


Diagram and typed description. An angle-shaped sign follows $\partial/\partial x_i$.

Original source details: GELFAND-P001053

Meaning:

One-particle space (fermions)

$$\{1, 2, \ldots, n\} \to \ell_2(\{1, 2, \ldots, n\}).$$

(ℓ_2 on a finite set).

$$m \text{ particles (fermions)} \Rightarrow f_{i_1 \ldots i_m}$$

(antisymmetric δ-functions — basis)

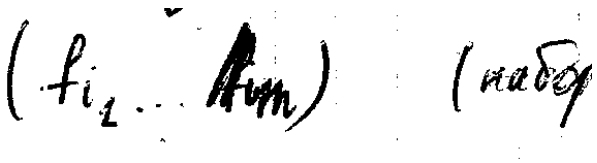
A way of writing elements of the Grassmann algebra.

(see Berezin: The Method of Second Quantization)

$$\sum_{i_1 \ldots i_m} a_{i_1}^* \ldots a_{i_m}^* f_{i_1 \ldots i_m} \Omega_0 \tag{4}$$

$$\updownarrow$$

$$(f_{i_1} \ldots f_{i_m}) \quad \text{(list of coefficients)}$$



Uncertain reading. The last base is overwritten and may read f_{i_m} or h_{i_m}.

Original source details: GELFAND-P001053

GELFAND-P001054

[source number: 17]

It is postulated:

$$\{a_i, a_j^*\} = \delta_{ij} \tag{5}$$

Exterior multiplication by df — a fermion creation operator.

$$a_i \Omega_0 = 0, \quad \forall i \tag{6}$$

$$\square_t = \square + t^2 (df)^2 + t \sum [a_i^*, a_j] \partial_i \partial_j f \tag{7}$$

In a neighborhood of the critical point

$$f(x) = \sum_{i=1}^{n-p} \lambda_i (x^i - x_0^i)^2 - \sum_{j=n-p+1}^{n} \lambda_j (x^j - x_0^j)^2 + O(|x - x_0|^3) \tag{8}$$

— . —

I. M. G. Idea: instead of changes in cycles — changes in forms!

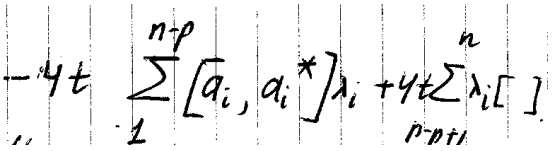
Connection between PDOs and Morse changes.

F. the principal term of $\square_t$ equals:

$$\square + 4t^2 \sum_{j=1}^{n} \lambda_j^2 (x^j - x_0^j)^2 + \overbrace{t \sum_{i<j} [a_i^*, a_j]}^{\text{const}} c_{ij} \tag{9}$$

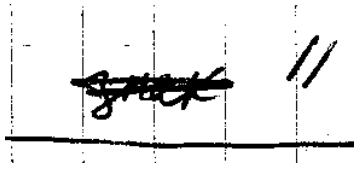
[Above the last term]

$$-4t \sum_{1}^{n-p} [a_i, a_i^*] \lambda_i + 4t \sum_{n-p+1}^{n} \lambda_i [\,].$$



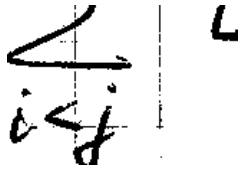
Original source details: GELFAND-P001054

Diagram and typed description. The square brackets in the second sum are empty.



Original source details: GELFAND-P001054

Uncertain reading. A short word above the bracket is crossed out, possibly “sign.” Two slanted strokes follow.



Original source details: GELFAND-P001054

Diagram and typed description. The index beneath the sum in (9) reads $i < j$.

$$\Big|\approx \quad \textit{Annihilates}: \quad \underline{dx_{n-p+1}\wedge\cdots\wedge dx_n} \quad (p\text{-form}) \tag{10}$$

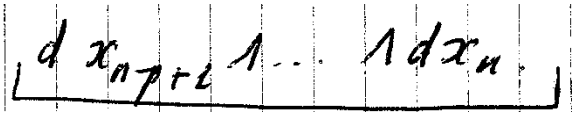


Diagram and typed description. A lower bracket surrounds the p-form.

Original source details: GELFAND-P001054

Let us look at eigenvalues other than the 0-th, the small ones.

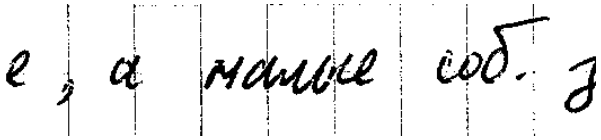
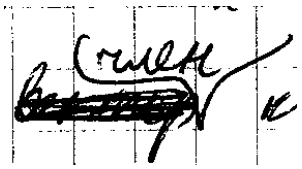


Diagram and typed description. The manuscript reads “small eigenvalues.”

Original source details: GELFAND-P001054

$\forall$ critical point gives an approximation to an eigenfunction.

The second [inserted above: term] cancels the 1-st exactly on the first vector (10).



Uncertain reading. After “second,” a word, possibly “vector,” is crossed out and “term” is inserted.

Original source details: GELFAND-P001054

The cancellation follows from supersymmetry considerations.

d_t plays the role of a supercharge; Yukawa model — the cancellation is known.

$$a = \frac{\delta}{\delta\varphi(x)} + ta^*(x)\varphi^n(x) \leftrightarrow d + tdf\cdot a^* \tag{11}$$

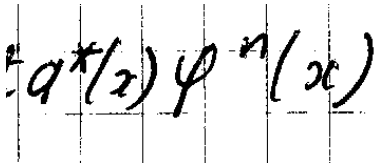


Diagram and typed description. In (11) the exponent of φ is n.

Original source details: GELFAND-P001054

$\underline{d=2}\Rightarrow$ fermions $=$ forms on the set of Bose-field configurations.

(since the Poincaré group is simple)

$\underline{d>2}\Rightarrow$ (?) gauge fields are needed for the interpretation. —

— there is no (such) interpretation yet.

V. Enss — asymptotic completeness of systems of particles with Coulomb interactions.

(three-particle) (worked on it for 15 years)

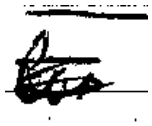
— a classical problem of mathematical physics.

GELFAND-P001055

18/X̄-82 Gelfand Seminar

Fröhlich. Statistical mechanics of surfaces.

Arises in the following settings:



Uncertain reading. A short entry after “в” (“in”) is crossed out and may read “во.”

Original source details: GELFAND-P001055

(a) Crystal growth

(b) Interfaces separating different phases

(c) (lattice) gauge theories

(d) dual resonance models (strings)

Simplified model for (a), (b):

$$\underset{\text{finite subset}}{\Lambda} \subset \mathbb{Z}^2; \qquad h : p \subset \mathbb{Z}^2 \to h_p \in \mathbb{Z}. \tag{1}$$

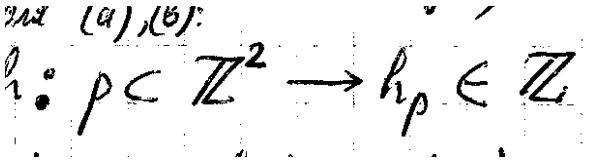


Diagram and typed description. The symbol following p is $\subset$.

Original source details: GELFAND-P001055

(solid-on-solid model)

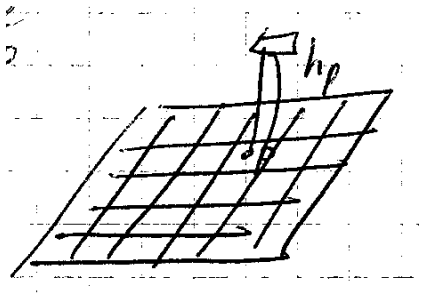
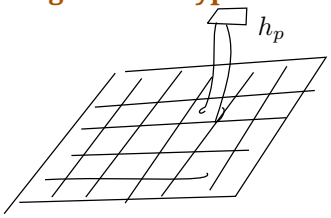


Diagram and typed description.



Loops appear at the lower ends of the vertical lines. It is unclear whether one is the letter p.

Original source details: GELFAND-P001055

p — a little square

h_p is interpreted as the height of the surface above the plane

$$w(S) \equiv e^{-\beta\,\mathrm{ares}(S)}, \quad \beta > 0 \tag{2}$$

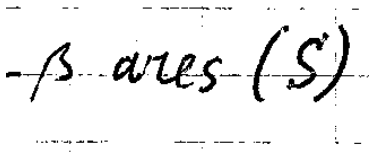
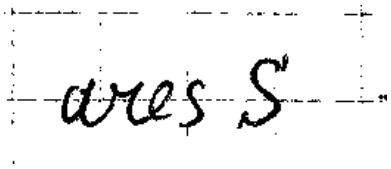


Diagram and typed description. The exponent has $\mathrm{ares}(S)$.

Original source details: GELFAND-P001055

ares S — the area of what is obtained, if supplemented with vertical walls



Original source details: GELFAND-P001055

Diagram and typed description. The explanation also has ares.

(build it out of solid cubes)

$$= e^{-\beta|\Lambda|} \prod_{\langle p,p'\rangle} e^{-\beta|h_p - h_{p'}|}, \tag{3}$$

$h_p = 0$ when $p \notin \Lambda$.

(boundary condition)

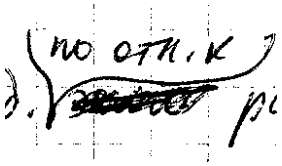
GELFAND-P001056

$$\mu(S) \equiv Z_\Lambda^{-1} w(S), \qquad Z_\Lambda = \sum_S{}' w(S). \tag{4}$$

— probability distribution.

Behavior of typical surfaces — ?

Results: let E — the expectation [inserted above the line: with respect to] the probability distribution μ as $\Lambda \nearrow Z^2$



Original source details: GELFAND-P001056

Uncertain reading. The word crossed out after “expectation” is difficult to read. It may be «этого» (“this”).

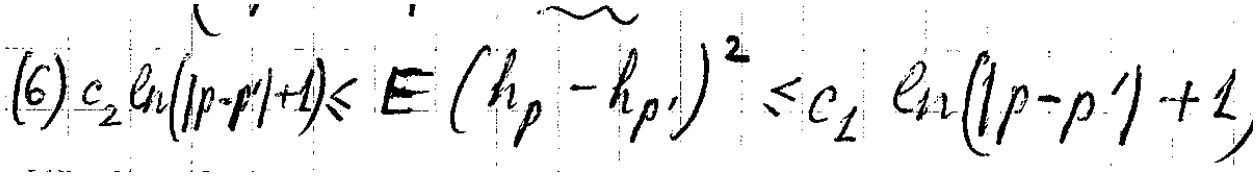
(a) for large β Eh_p^2 is bounded

$$\text{Prob}\{h : |h_p| = n\} \sim e^{-\text{const}\cdot n} \tag{5}$$

(a special case of Dobrushin’s general theorem)

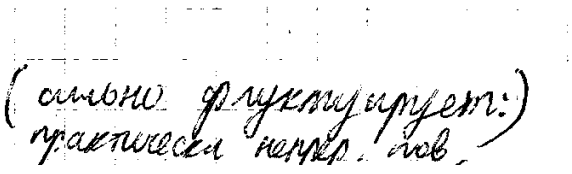
(b) β is small

$$c_2 \ln(|p-p'|+1) \leqslant E(h_p - h_{p'})^2 \leqslant c_1 \ln(|p-p'|+1) \tag{6}$$



Original source details: GELFAND-P001056

Diagram and typed description. The coefficient on the left is c_2, and the one on the right is c_1.

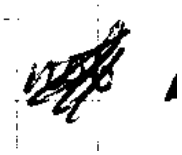


Original source details: GELFAND-P001056

Diagram and typed description. (fluctuates strongly.) Practically continuous behavior.

Enhancement of symm.

We shrink the little squares ⇒ a continuous symmetry group appears.



Original source details: GELFAND-P001056

Uncertain reading. A short entry is crossed out between “shrink” and “the little squares.” It may be «кв».

(too complicated to explain)

$$\mu(S) \sim \prod_{\langle p,p'\rangle} e^{-\beta|h_p - h_{p'}|} \tag{7}$$

$h_p \mapsto h_p + \text{const}$ symmetry (continuous)

Broken by the boundary conditions.

The same questions have yet to be addressed at small β

$$Z_{\beta,\gamma} \equiv \sum_{S:\partial S=\gamma} w_\beta(S), \qquad w_\beta(S) = e^{-\beta|S|} \tag{8}$$

$$\#\{S : \partial S = \gamma,\ |S| = \text{area } S = n\} \sim C_\gamma^n n^{\delta(\gamma)}, \quad n \text{ is large.} \tag{9}$$

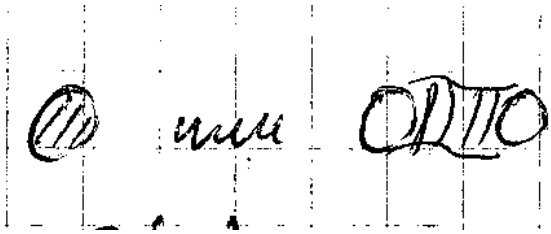
(from general combinatorial theorems)

Found in papers on the Peierls method.

$$\ln C_\gamma \equiv \lim_{n\to\infty} \frac{1}{n} \ln \#\{S \mid \partial S = \gamma,\ |S| = n\} \tag{10}$$

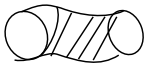
(11) Lemma C_γ does not depend on γ

Conjecture. Depends on the topological properties of γ:

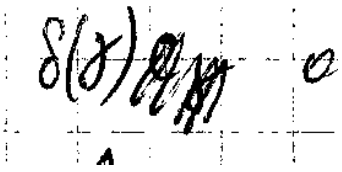


Original source details: GELFAND-P001056

Diagram and typed description.

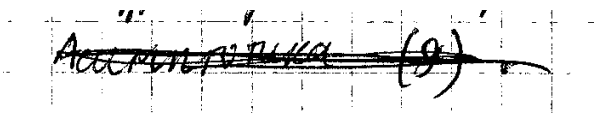
 or 

$\delta(\gamma)$ determines the critical phenomena. $\delta(\gamma)$, apparently depends on the topology.



Original source details: GELFAND-P001056

Uncertain reading. A short entry after the first $\delta(\gamma)$ is crossed out. It may be «для» (“for”).



Original source details: GELFAND-P001056

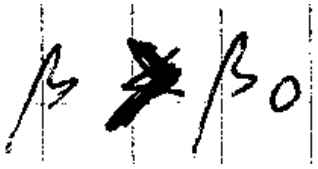
Diagram and typed description. ~~Asymptotics (9).~~

GELFAND-P001057

[source number: 27]

A difficult question: what if we take surfaces with $\forall$ topology.
How many handles does a typical surface have?

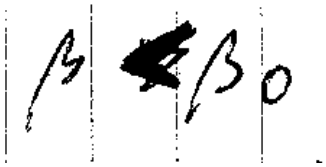
$$Z_{\beta,\gamma} < \infty \quad \text{when } \beta > \beta_0 \tag{12}$$



Uncertain reading. The relation sign has been overwritten. Possible readings are $\beta > \beta_0$ and $\beta \geqslant \beta_0$.

Original source details: GELFAND-P001057

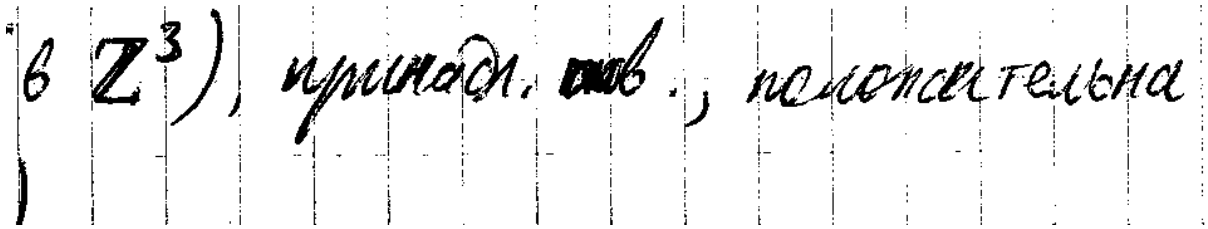
$$Z_{\beta,\gamma} = +\infty \quad \text{when } \beta < \beta_0 \qquad \text{(like a gas)} \tag{13}$$



Uncertain reading. The relation sign has been overwritten. Possible readings are $\beta < \beta_0$ and $\beta \leqslant \beta_0$.

Original source details: GELFAND-P001057

(13): The density of the squares (in Z^3), belonging to the surface, is positive
(like a Peano curve)



Original source details: GELFAND-P001057

Diagram and typed description. The abbreviation «пов.» (“surface”) is written over other strokes.

Conjecture, when $\beta_0 < \beta < \beta_R$ logarithmic fluctuations,
$\beta > \beta_R$ — roughening (coarsening)
— the shape becomes fixed, almost smooth (almost const)
Why bother?
1° NaCl is described by this model
2° This theory is equivalent to gauge theory $SU(N)$ in the limit $N \Rightarrow 0$ (!) (degenerate case)



Diagram and typed description. A double arrow follows N.

Original source details: GELFAND-P001057

3° It is a fun problem.



Original source details: GELFAND-P001057

Uncertain reading. A short entry is crossed out after “fun.” It may begin with “pr.”

$$w_N(S) \underset{N\to 0}{\longrightarrow} w(S)$$

(Muscovites like to expand in $1/N$ when $N \to \infty$).

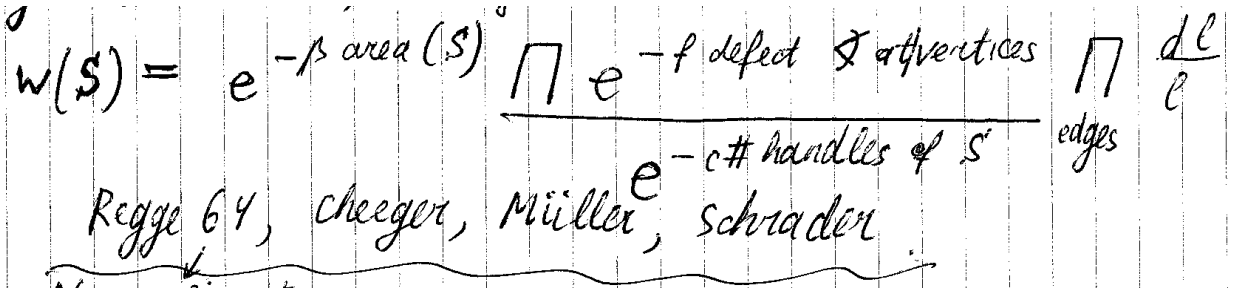
References. Phys. Lett. Amos Maritan, C. Omero

Generalizations→

Continuous case. Surfaces that do not lie on a lattice.

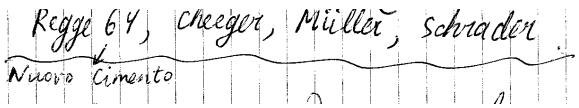
Piecewise linear surfaces made of Δs.

$$w(S) = e^{-\beta \,\text{area}(S)} \frac{\prod e^{-f \,\text{defect}\,\angle \,\text{at vertices}}}{e^{-c\,\#\,\text{handles of } S}} \prod_{\text{edges}} \frac{d\ell}{\ell} \tag{14}$$



Original source details: GELFAND-P001057

Uncertain reading. The line above the lower exponential is typeset as a fraction bar. Its placement is visible in the manuscript.



Original source details: GELFAND-P001057

Diagram and typed description.
Regge 64, Cheeger, Müller, Schrader

Nuovo Cimento

Question: convergence when approaching a continuous surface.

$$R = \sum_f \text{area of } (d-2) \text{ simplex, } f \times \text{defect } \angle$$

$$\text{Theor.} \quad R \longrightarrow \int \sqrt{g(x)} R(x) d^d x. \tag{15}$$

$(14) \longrightarrow$ Polyakov's string theory

Exercise: Prove this!

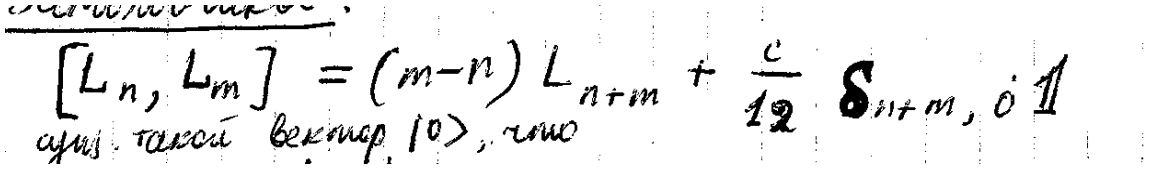
GELFAND-P001058

1/XI-82 Gelfand Seminar.

Zamolodchikov.

Suppose

$$[L_n, L_m] = (m-n)L_{n+m} + \frac{c}{12}\delta_{n+m,0}\mathbf{1} \qquad \text{(Virasoro algebra)}$$



Original source details: GELFAND-P001058

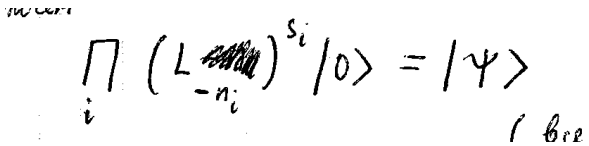
Uncertain reading. The central term visibly contains $\frac{c}{12}\delta_{n+m,0}\mathbf{1}$.

and there is a vector $|0\rangle$, such that

$$L_n|0\rangle = 0, \quad n > 0,$$
$$L_0|0\rangle = \Delta|0\rangle.$$

and

$$\prod_i (L_{-n_i})^{s_i}|0\rangle = |\psi\rangle$$



Uncertain reading. An index after L is crossed out. It may be n_i or m_i. The index $-n_i$ is inserted below.

Original source details: GELFAND-P001058

(all vectors are obtained in this way)

Suppose $\varphi(z)$ — an operator acting in this space, and

$$[L_n, \varphi(z)] = z^{n+1}\frac{d\varphi(z)}{dz} + (n+1)\Delta z^n \varphi(z)$$

It is uniquely determined by this formula.

First question: what can be said about it?

— how to calculate

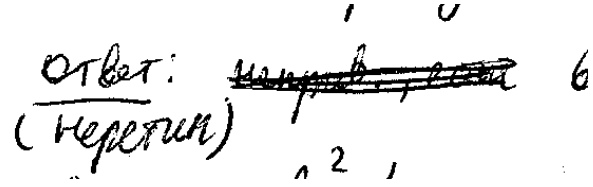
$$\langle 0|\varphi(z_1)\dots\varphi(z_n)|0\rangle \quad ?$$

$L_{-n}^{+} = L_n$, $\langle 0 \mid 0\rangle = 1 \Rightarrow$ recover the inner product.

Second question: what relations on the states can be written down?

Or: is this module irreducible?

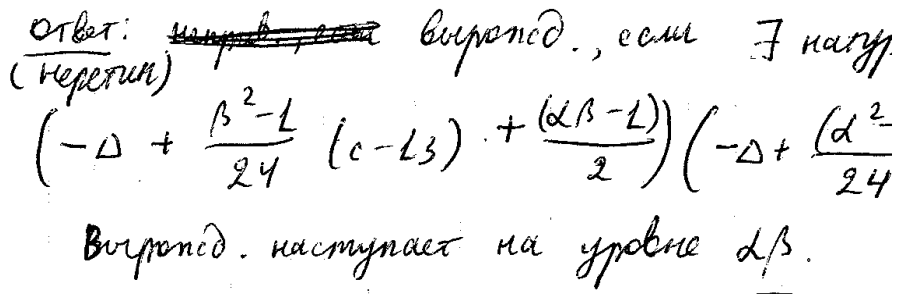
Answer: (Neretin) degenerate if $\exists$ natural numbers α, β, such that



Uncertain reading. The crossed-out words before “degenerate” appear to read “irreducible if.”

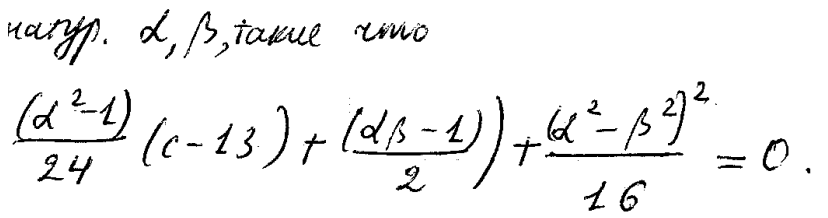
Original source details: GELFAND-P001058

$$\left(-\Delta+\frac{\beta^2-1}{24}(c-13)+\frac{(\alpha\beta-1)}{2}\right)\left(-\Delta+\frac{(\alpha^2-1)}{24}(c-13)+\frac{(\alpha\beta-1)}{2}\right)+\frac{(\alpha^2-\beta^2)^2}{16}=0.$$



Original source details: GELFAND-P001058

Diagram and typed description. The first factor contains β^2-1.



Original source details: GELFAND-P001058

Diagram and typed description. The second factor contains α^2-1.

Degeneracy occurs at level $\alpha\beta$.

$\vec{x}(\xi_1,\xi_2)$ — surface, $g_{ab}(\xi_1,\xi_2)$ — metric, $a,b=1,2$.

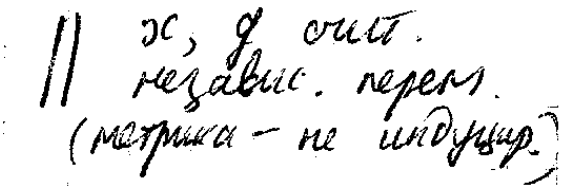


Diagram and typed description. // x, g are treated as independent variables (the metric is not induced)

Original source details: GELFAND-P001058

Suppose

$$Z \equiv \int \mathcal{D}\vec{x}\,\mathcal{D}g_{ab}\,\exp\left\{-\tfrac{1}{2}\int \sqrt{g}\,g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\,d^2\xi\right\} \exp\left\{-\mu_0^2\int\sqrt{g}\,d^2\xi\right\}. \quad (1)$$

Evaluation — ?

Invariant under the transformations

$$\begin{cases}\xi^a \mapsto \xi^a + \varepsilon^a(\xi),\\ g_{ab}(\xi) \mapsto g_{ab} + \nabla_a\varepsilon_b + \nabla_b\varepsilon_a.\end{cases}$$

We divide by this group (the volume element is understood to be factored)
— one must integrate over the orbits.



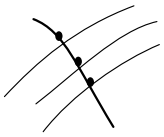


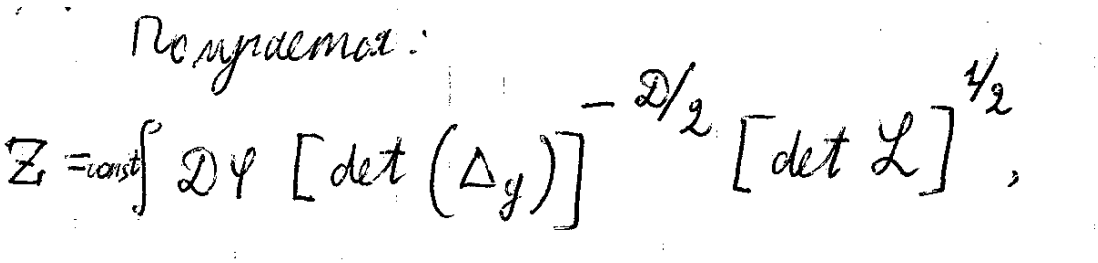
Diagram and typed description. The strokes thicken at the three intersections. It is unclear whether these are arrowheads.

Original source details: GELFAND-P001058

Representative: $\vec{x}(\xi)$ — arbitrary, $g_{ab}(\xi) = \delta_{ab}\rho(\xi)$

This gives:

$$Z = \text{const}\int \mathcal{D}\varphi\,[\det(\Delta_g)]^{-D/2}\,[\det\mathcal{L}]^{1/2}, \quad (2)$$



Original source details: GELFAND-P001058

Uncertain reading. The symbol after $\mathcal{D}$ has a loop on the right. It is provisionally transcribed as φ. Another possible reading is ψ.

GELFAND-P001059

[source number: 38]

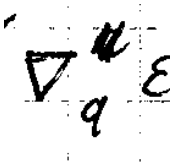
where



Uncertain reading. A short entry in the upper margin is crossed out. Its reading is uncertain.

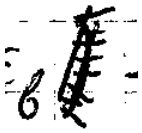
Original source details: GELFAND-P001059

$$(\mathcal{L}\varepsilon)_a = \nabla^b\big(\nabla^a_a\varepsilon_b + \nabla_b\varepsilon_a - g_{ab}\nabla^c\varepsilon_c\big). \tag{3}$$



Original source details: GELFAND-P001059

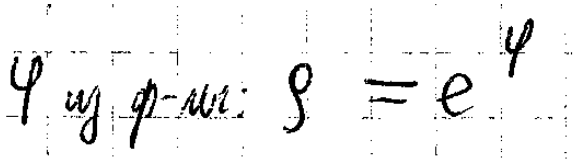
Uncertain reading. The first ∇ in parentheses has a subscript a and a thickened superscript resembling a. The superscript may be cancelled.



Original source details: GELFAND-P001059

Uncertain reading. A vertical entry after the first ε_b is crossed out.

On evaluating this, we obtain:



Original source details: GELFAND-P001059

Diagram and typed description. The same looped sign appears in (4), (6), and $\rho = e^{\varphi}$. It is transcribed as φ. Another possible reading is ψ.

D — dimension of the space, $D < 26$.

$$Z = \int \mathcal{D}\varphi \cdot \exp\left\{-\frac{26-D}{48\pi}\int\left[\frac{1}{2}\partial_a\varphi\,\partial_a\varphi + m_0^2 e^{\varphi}\right]d^2\xi\right\}. \tag{4}$$

φ from the formula: $\rho = e^{\varphi}$.

$$\Delta_g = \frac{1}{\sqrt{g}}\partial_a\big(\sqrt{g}\,g^{ab}\partial_b\big).$$

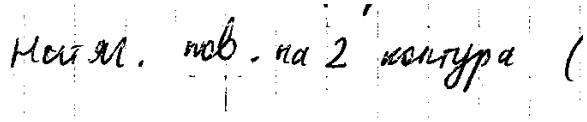
The string problem will be solved if we evaluate this integral.

$D = 26 \Rightarrow$ the answer is the same as for $g_{ab} = \delta_{ab}$.

This prescription is used in dual models. They are exact at $D = 26$.

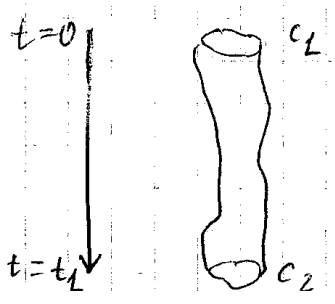
What surfaces should we integrate over?

Surfaces stretched across 2 contours (evolution of a contour).



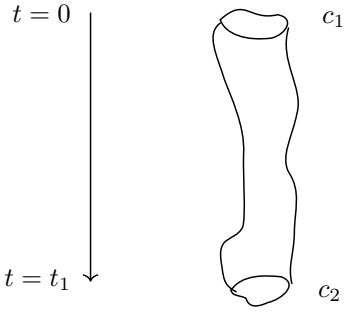
Original source details: GELFAND-P001059

Uncertain reading. The abbreviation at the start of the line appears to read «Натян.», for “stretched.”



Original source details: GELFAND-P001059

Diagram and typed description.



$$Z(c_1, c_2, t_1) \underset{t_1 \to \infty}{=} \sum_n G(c_1, n)\, G(c_2, n)\, e^{-E_n t_1}. \tag{5}$$

The spectrum of stationary states in Liouville theory is the QFT spectrum for the Lagrangian



Original source details: GELFAND-P001059

Uncertain reading. An entry above the words “Liouville theory” is crossed out.

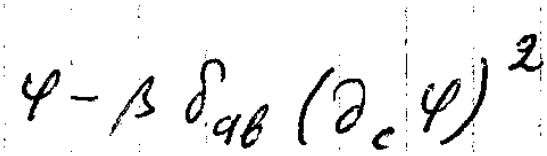
$$\int_M \left[\frac{1}{2}\partial_a\varphi\,\partial_a\varphi + m_0^2 e^{\varphi}\right] d^2\xi. \tag{6}$$

Conjecture: the spectrum is related to the system L_n, described on the previous page.

$$z = \xi_1 + i\xi_2,$$
$$\bar{z} = \xi_1 - i\xi_2.$$

Traceless two-dimensional conserved energy–momentum tensor:

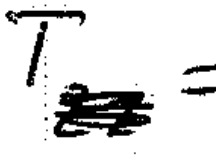
$$T_{ab} = \partial_a\varphi\,\partial_b\varphi - \tfrac{1}{2}\delta_{ab}(\partial_c\varphi)^2 + \alpha\partial_a\partial_b\varphi - \beta\delta_{ab}(\partial_c\varphi)^2.$$



Original source details: GELFAND-P001059

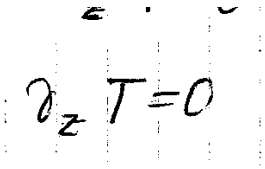
Diagram and typed description. The last term reads $-\beta\delta_{ab}(\partial_c\varphi)^2$.

$$T = \varphi_z^2 + 2\varphi_{zz}, \qquad \partial_{\bar{z}} T = 0,$$
$$\bar{T} = \varphi_{\bar{z}}^2 + 2\varphi_{\bar{z}\bar{z}}, \qquad \partial_z T = 0.$$



Original source details: GELFAND-P001059

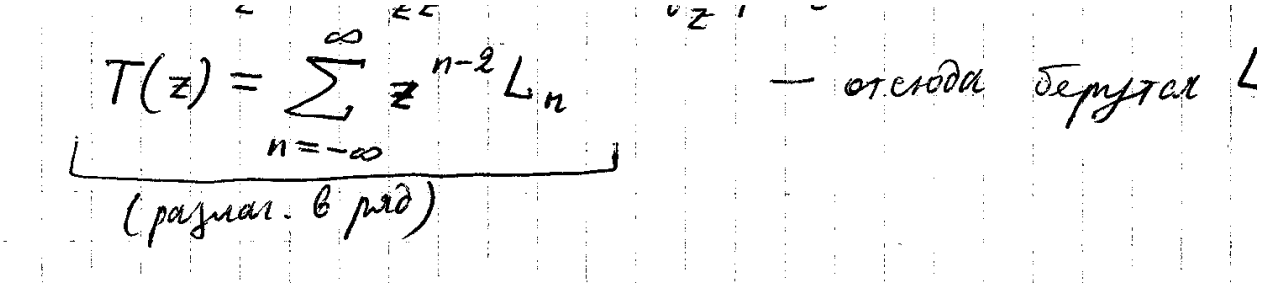
Uncertain reading. The subscript on the first T is crossed out. It may be zz.



Original source details: GELFAND-P001059

Diagram and typed description. The T in the second equality on the right has no additional bar.

$$\underbrace{T(z) = \sum_{n=-\infty}^{\infty} z^{n-2} L_n}_{\text{(expansion in a series)}} \quad \text{— this is where we obtain } L_n!$$



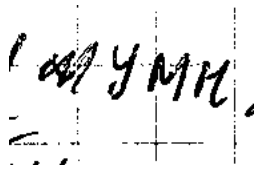
Original source details: GELFAND-P001059

Diagram and typed description. A lower bracket encloses the series and its left-hand side.

GELFAND-P001060

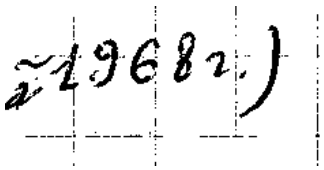
Beilinson A.

I.M.G. Gelfand, Ponomarev (UMN, $\underset{[2?]}{\sim}$ 1968) reduced the description of indecomposable representations of the Lorentz group to a problem in linear algebra.



Original source details: GELFAND-P001060

Uncertain reading. A short mark before "UMN" is crossed out.



Original source details: GELFAND-P001060

Uncertain reading. A mark resembling 2 appears below ∼.

B. model example. — $\mathrm{SL}(2,\mathbb{R})$

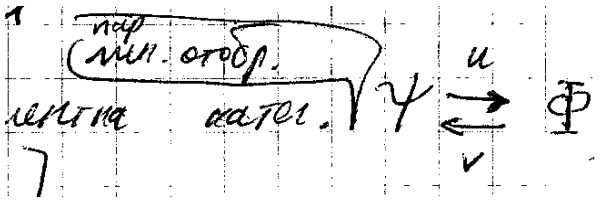
On $\mathbb{P}^1$ we must classify D-modules with singularities at two fixed points.

Theorem 1. the category of D-modules on the disk with a fixed singularity is equivalent to the category



Original source details: GELFAND-P001060

Diagram and typed description.



Original source details: GELFAND-P001060

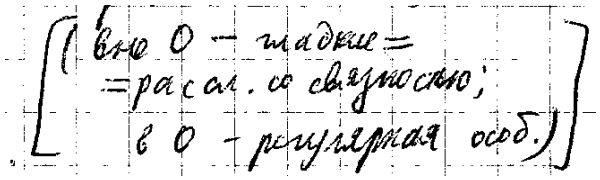
Diagram and typed description.

pairs of linear maps

$$\Psi \underset{v}{\overset{u}{\rightleftarrows}} \Phi$$

"Pairs of" is written above "linear maps."

such that $uv - \mathrm{Id}$ is invertible.

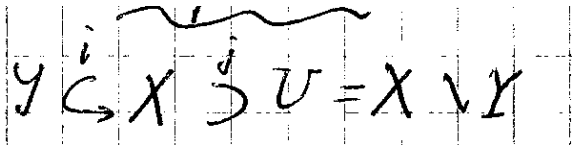


Original source details: GELFAND-P001060

Diagram and typed description.

$$\left[\begin{array}{l}\text{outside } 0 \text{ — smooth } = \\ = \text{bundles with connection;} \\ \text{at } 0 \text{ — regular singularity.}\end{array}\right]$$

Geometric problem.



Original source details: GELFAND-P001060

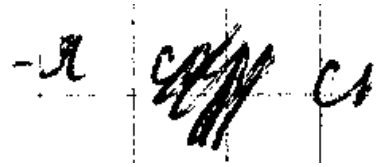
Diagram and typed description.

$$Y \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} U = X \setminus Y$$

A sheaf on X is given as follows:

$$\mathcal{F}_X = (\mathcal{F}_U, \mathcal{F}_Y, \quad \mathcal{F}_Y \longrightarrow i^* j_* \mathcal{F}_U)$$

(gluing function)



Original source details: GELFAND-P001060

Uncertain reading. A short entry between "function" and "gluing" is crossed out. It may begin with «ск».

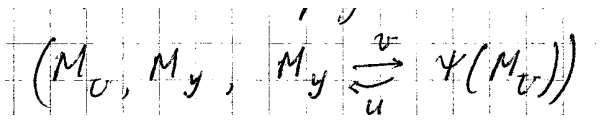
For representations we must solve the analogous problem on the space of flags.



Original source details: GELFAND-P001060

Uncertain reading. The crossed-out formula may read $Y = f^{-1}(0)$.

Theorem. the category of D-modules with regular singularities on X is equivalent to the category of tuples

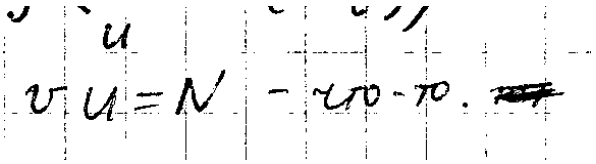


Original source details: GELFAND-P001060

Diagram and typed description.

$$(M_U,\ M_Y,\quad M_Y \underset{u}{\overset{v}{\rightleftarrows}} \Psi(M_U))$$

$vu = N$ — something.



Uncertain reading. A short entry after "something" is crossed out.

Original source details: GELFAND-P001060

GELFAND-P001061

[source number: 48]

15/XI-82 Gelfand Seminar.

I.M.G. Problem. (imprecise)

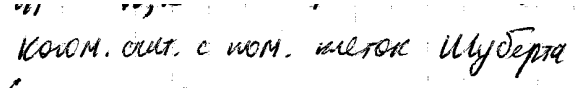
$G_{n,k}$ — Grassmannian, k-dimensional planes in an n-dimensional space.



Uncertain reading. An entry at the start of the line is crossed out. It resembles G with additional marks.

Original source details: GELFAND-P001061

Cohomology is calculated using Schubert cells.



Uncertain reading. The first two abbreviations provisionally read «Когом. счит.», for "cohomology is calculated."

Original source details: GELFAND-P001061

(The cell is specified by the dimensions of intersections with the flag

$$0 = d_0 \leqslant d_1 \leqslant d_2 \leqslant \ldots \leqslant d_n = k,$$

$$\dim \text{ of the cell} = \sum d_j.$$

)

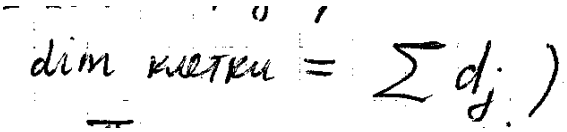


Diagram and typed description. The dimension formula reads $\sum d_j$.

Original source details: GELFAND-P001061

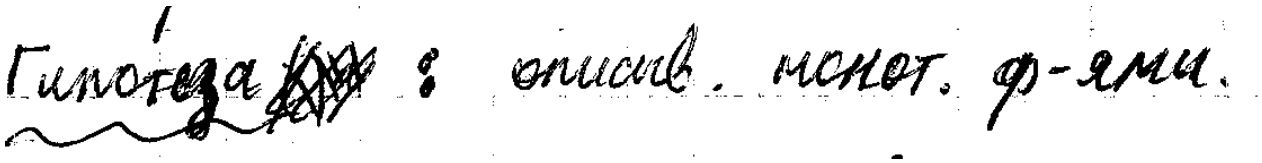
Consider a II_1 factor. In it, consider α-dimensional projections (a continuous analogue of Grassmannians). Is there an analogue of a Schubert cell?

Flag of continuous dimension $[0, 1]$.

Does this give a Grassmannian?

Continuous cohomology!?

Conjecture: described by monotone functions.



Original source details: GELFAND-P001061

Uncertain reading. A short entry after "Conjecture" is crossed out.

An even vaguer question: what topology? and what cohomology?



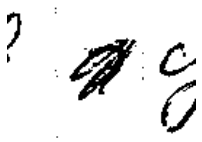
Uncertain reading. A mark before “An even” is crossed out.

Original source details: GELFAND-P001061

Yu. I. Manin.

1) Flag supermanifolds

2) The Radon–Penrose transform in a supermathematical setting.



Uncertain reading. A single mark is crossed out before “supermathematical.”

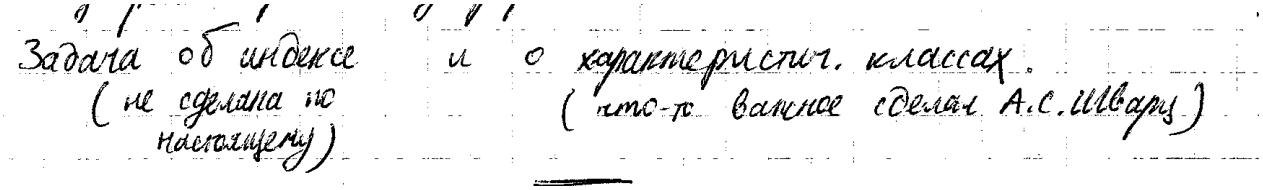
Original source details: GELFAND-P001061

3) Applications to solving nonlinear equations: new solutions of the non-self-dual Yang–Mills equations

(ordinary and supersymmetric)

4) Specific flag manifolds, their deformation, and so on in supersymmetry and supergravity.

5) The problem of the index and of characteristic classes.



Original source details: GELFAND-P001061

Diagram and typed description. [Below “of the index”] (has not really been done)
[Below “of characteristic classes”] (A. S. Schwarz did something important)

The talk will cover 1)–4)

1) Reminder about the self-dual Yang–Mills equations

$T = \mathbb{C}^4$ [antiholomorphic involution $T \oplus T^*$]

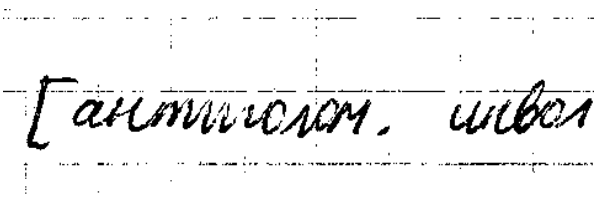


Diagram and typed description. The first abbreviation appears to read «антиголом.», for “antiholomorphic.”

Original source details: GELFAND-P001061

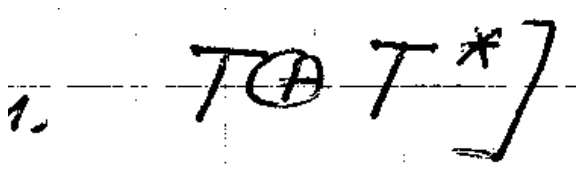
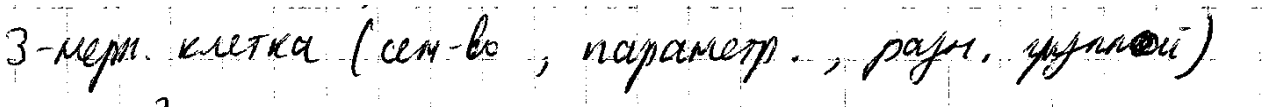


Diagram and typed description. The sign between T and T^* is $\oplus$.

Original source details: GELFAND-P001061

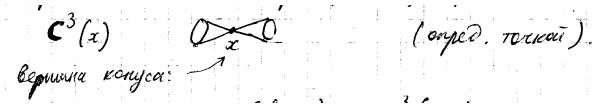
dim $G(2,T) = 4$ = complex compact Minkowski space.

3-dimensional cell (family, parametrized by a group [one uncertain abbreviation])



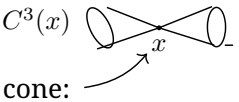
Original source details: GELFAND-P001061

Uncertain reading. The abbreviation before “group” may read «разн.» or «разл.».



Original source details: GELFAND-P001061

Diagram and typed description.

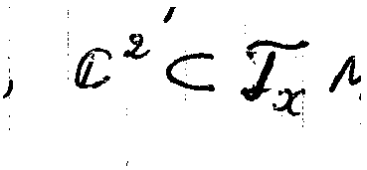


(determined by a point).

$$M^4 = \text{real part of } G(2,T) \setminus C^3(\infty) \leftarrow \infty\text{-cone}.$$

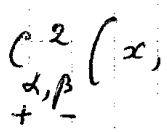
2-dimensional cells: 2 types

$$C^2_{\substack{\alpha,\beta \\ + \; -}}(x, \mathbb{C}^2 \subset T_x M) \text{ — } \pm\text{-planes}, \quad (\alpha,\beta)\text{-planes}.$$



Original source details: GELFAND-P001061

Diagram and typed description. Inside the parentheses, the symbol before $\subset$ is a double-stroked $\mathbb{C}$ with superscript 2.

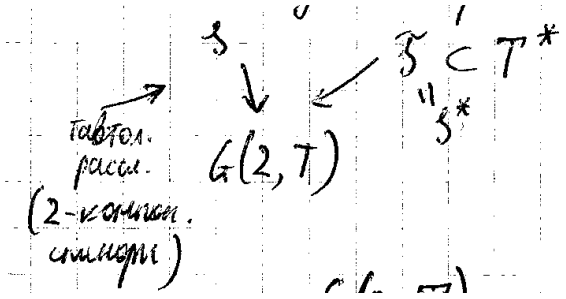


Original source details: GELFAND-P001061

Diagram and typed description. The $+$ is written below α, and the $-$ below β.

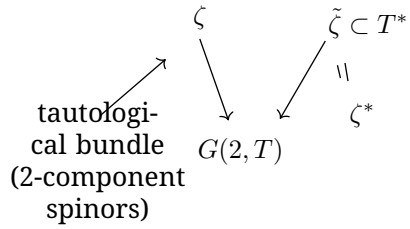
GELFAND-P001062

C^1 — null lines (light rays)



Original source details: GELFAND-P001062

Diagram and typed description.



The bundle symbol is provisionally transcribed as ζ. The meaning of the two strokes below the right-hand label is uncertain.

Problem. $G(2,T)$

Describe the superhomogeneous complex manifolds M, for which $M_{\text{red}} = G(2,T)$.

Condition: M — a flag manifold for simple Lie supergroups of classical type.

List (beginning).

a) $F(2|0,2|N,\mathbb{C}^{4|N}) = M$



Uncertain reading. A mark resembling 2 is crossed out after $2 \mid N$.

Original source details: GELFAND-P001062

(flags of dimensions $2|0, 2|N$ in $\mathbb{C}^{4|N}$).

$$N = 0,1,2,3,\dots \qquad \mathbb{C}^{4|N} = (\mathbb{C}^4)_0 \oplus (\mathbb{C}^N)_1.$$

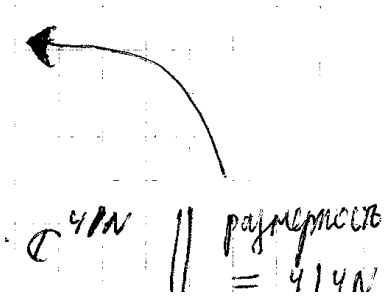


Diagram and typed description.

$\mathbb{C}^{4|N}$ — $\|$ dimension $= 4|4N$

The note "dimension $= 4 \mid 4N$" is to the right of the inclusions. The arrow points up and to the left.

Original source details: GELFAND-P001062

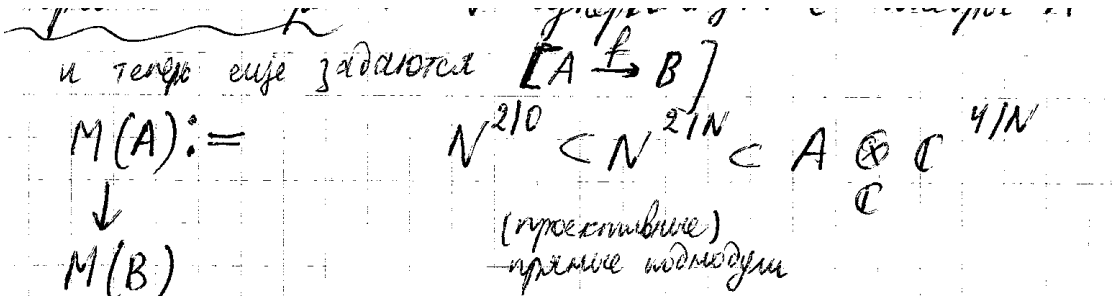
Naïve definition:

$$\text{points } x \in F(2|0,2|N,\mathbb{C}^{4|N}) = \underline{\mathbb{C}^{2|0} \subset \mathbb{C}^{2|N}} \subset \mathbb{C}^{4|N}.$$

But there are too few coordinates.

Correct definition: $\forall$ supercommutative $\mathbb{C}$-algebra A we specify $M(A)$ and now we also specify $[A \xrightarrow{f} B]$

$$\begin{array}{rl} M(A) := & N^{2|0} \subset N^{2|N} \subset A \otimes_{\mathbb{C}} \mathbb{C}^{4|N} \\ \downarrow & \text{(projective)} \\ M(B) & \text{direct submodules} \end{array}$$



Original source details: GELFAND-P001062

Diagram and typed description. The arrow goes from $M(A)$ to $M(B)$. The submodule description is written below the inclusions.

Coordinates: $G(1|0, \mathbb{C}^{m|n}) := P^{m-1|n} \supset P$.

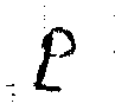


Diagram and typed description. The symbol to the right of $\supset$ appears to be P.

Original source details: GELFAND-P001062

Homogeneous coordinates: $(x_1, \dots, x_m; \xi_1, \dots, \xi_n)$.

$$\text{Cover } P^{m-1,n} = \bigcup_{i=1}^{m} U_i, \quad U_i : \left(\frac{x_1}{x_i}, \dots, \frac{x_m}{x_i}, \frac{\xi_1}{x_i}, \dots, \frac{\xi_m}{x_i} \right).$$

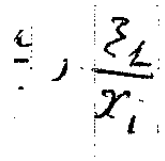


Diagram and typed description. The denominator of the first odd coordinate appears to be x_i.

Original source details: GELFAND-P001062

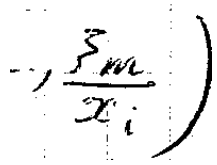


Diagram and typed description. The final fraction reads ξ_m/x_i. The list of homogeneous coordinates ends in ξ_n.

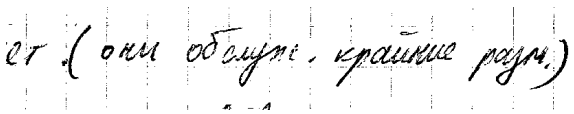
Original source details: GELFAND-P001062

Exercise: relate this to the preceding discussion (big cells).

We can have: $\xi_i = 0$ — a cell of codimension $0|1$.

It is unclear in which cohomology they lie?

Differential and integral forms are also insufficient. (they serve the extreme dimensions)



Uncertain reading. The abbreviation appears to be «обслуж.», for “serve.”

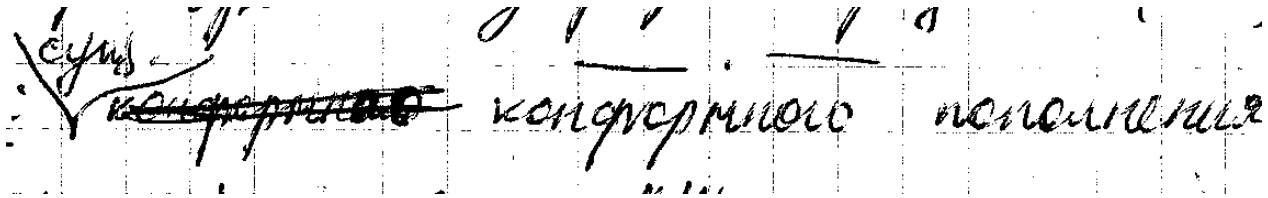
Original source details: GELFAND-P001062

Schwarz’s objects are intermediate but have strange properties.

Real points of the big cell in $F(2|0, 2|N; \mathbb{C}^{4|N}) =$

$$= N\text{-extended physicists' superspace} : (x^m; \theta_{\alpha i}, \theta^{\beta i}).$$

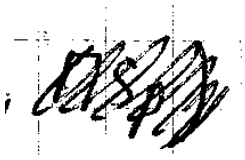
Requirement: [inserted existence of] ~~conformal~~ conformal completion.



Original source details: GELFAND-P001062

Diagram and typed description. “Existence” is written with an insertion mark. The earlier “conformal” is crossed out.

b) $FI(2|0, 2|2, \mathbb{C}^{4|4}, \Pi\mathrm{Sp})$ dimension $= 4|8$.



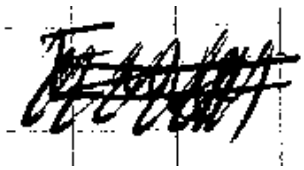
Uncertain reading. An entry before $\mathbb{C}^{4|4}$ is crossed out. It may read OSp.

Original source details: GELFAND-P001062

GELFAND-P001063

[source number: 49]

I — isotropic, ΠSp — the type of form with respect to which the flags are isotropic.
(odd symmetric form)
There are also series and isolated manifolds.



Uncertain reading. A word before “There are also” is crossed out. It may read «Бывают» (“there are”) or «Будут» (“there will be”).

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2) $M = G(2, T)$, $P(T) = G(1, T)$.

GELFAND-P001064

22/XI-82 Gelfand Seminar.
A. M. Polyakov.
(Alexander Markovich).
The string problem

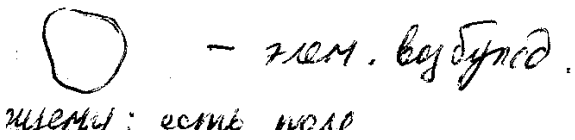
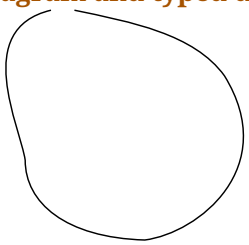


Diagram and typed description.



— elem. excitation.

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Reduces to the following: there is a field

$$\varphi = \varphi(\xi_1, \xi_2) \qquad (\sim \text{ surface in isothermal coordinates})$$

$$g_{ab}(\xi) = e^{\varphi(\xi)}\delta_{ab}. \tag{1}$$

$$\mathcal{L} = \frac{1}{\gamma}\left\{\frac{1}{2}(\partial_a\varphi)^2 + e^{\varphi}\right\}, \qquad S = \int \mathcal{L}\, d^2\xi, \tag{2}$$

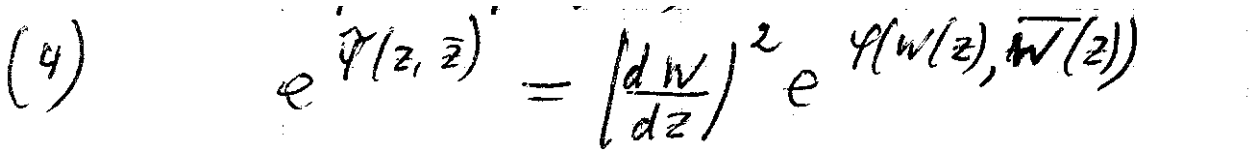
$$\frac{1}{\gamma} = \frac{26 - D}{48\pi} \tag{3}$$

(D — dimension of the space into which the two-dimensional surface under consideration is embedded)

Why does it have the form (2) (apart from γ)?

Fix the metric (1). The remaining freedom: conformal transformations.

$$e^{\widetilde{\varphi}(z,\bar{z})} = \left|\frac{dw}{dz}\right|^2 e^{\varphi(w(z),\overline{w}(z))}. \tag{4}$$



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Diagram and typed description. In the exponent on the right, the second w has a bar. The final z has none.

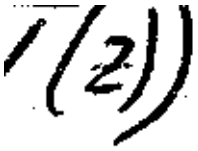
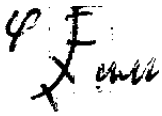


Diagram and typed description. At the right of the formula is "(2)" with an additional stroke.

Original source details: GELFAND-P001064

Lagrangian (2) — the only local Lagrangian (depending only on the value of the function and its derivatives at the point), invariant under transformation (4).

$\mathcal{L}$ is called the Liouville Lagrangian (the corresponding equation $\Delta\varphi = e^{\varphi}$ or $\partial^2\varphi = e^{\varphi}$).



Uncertain reading. A bracket-like mark after the first e^{φ} is crossed out.

Original source details: GELFAND-P001064

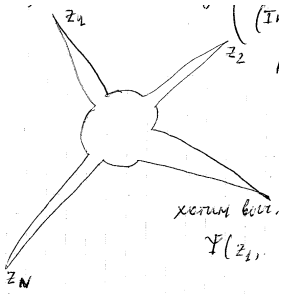
$$\partial^2\varphi \equiv \Delta\varphi = e^{\varphi}. \tag{5}$$

Can be solved explicitly. ($\sim$ curvature is constant)

Classical solution of equation (5):

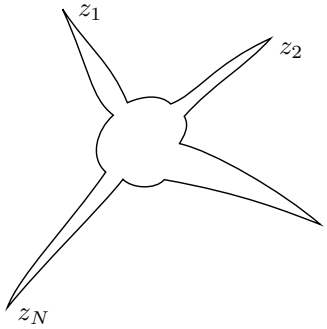
$$\varphi = \log\left(\frac{|\partial_z w|^2}{(\operatorname{Im} w)^2}\right), \qquad w(z) \text{ — any holomorphic function.} \tag{6}$$

Consider a sphere with punctures that are at ∞ in the sense of the metric.



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Diagram and typed description.



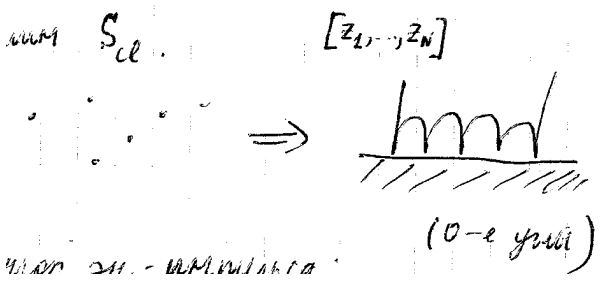
$$e^{\varphi} \underset{z\to z_i}{\longrightarrow} \frac{2}{|z-z_i|^2 \log^2|z-z_i|}.$$

we want to calculate the quantum quantity

$$\Psi(z_1,\dots,z_N) = \int \mathcal{D}\varphi(z) e^{-S(\varphi)} \underset{\gamma\to 0}{\simeq} e^{-S_{\text{cl}}(z_1,\dots,z_N)}. \tag{7}$$

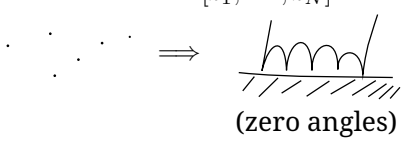
Let us calculate S_{cl}.

(8)



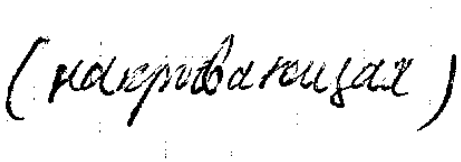
Original source details: GELFAND-P001065

Diagram and typed description.



(zero angles)

The caption below the drawing appears to read “zero angles.”



Original source details: GELFAND-P001065

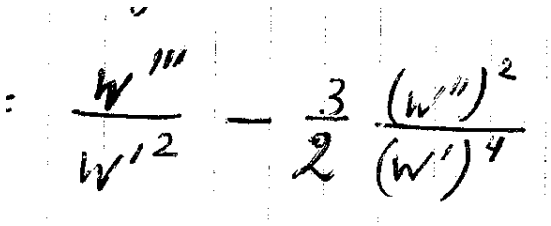
Diagram and typed description. (covering surface)

Let us calculate the energy–momentum tensor:

$$T = \tfrac{1}{2}(\partial_z \varphi)^2 - \partial_z^2 \varphi, \qquad T = T_{11} - T_{22} + 2iT_{12}, \qquad T_{11} + T_{22} = 0. \tag{9}$$

(T_{ij} is recovered from T, since T_{ij} — real)

$$T(z|z_1, \ldots, z_N) = \underset{\text{(Schwarzian derivative)}}{\mathcal{D}}[w(z)] = \frac{w'''}{w'^2} - \frac{3}{2}\frac{(w'')^2}{(w')^4}. \tag{10}$$



Original source details: GELFAND-P001065

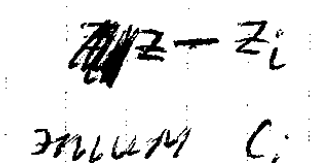
Diagram and typed description. The denominators read w'^2 and $(w')^4$.

(invariant under fractional linear transformations)

$$\widetilde{w}(z) = \frac{\alpha w + \beta}{\gamma w + \delta}.$$

Hence T — is single-valued and has only poles:

$$T(z|z_1, \ldots, z_N) = \frac{1}{4}\sum_1^N \frac{1}{(z - z_i)^2} + \sum_1^N \frac{c_i(z_1, \ldots, z_N)}{z - z_i}. \tag{11}$$

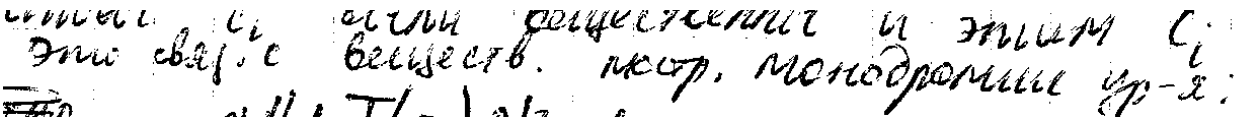


Original source details: GELFAND-P001065

Uncertain reading. An entry before $z - z_i$ in the final denominator is heavily crossed out.

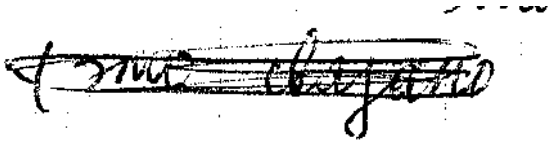
It is important that c_i be real, and this determines c_i .

(c_i — accessory coefficients)



Original source details: GELFAND-P001065

Diagram and typed description. In small handwriting above (12), “This is connected with the real monodromy coefficients of the equation.”



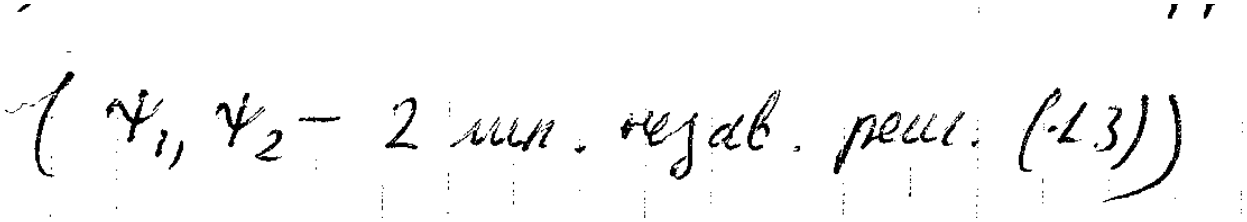
Original source details: GELFAND-P001065

Uncertain reading. A phrase before (12) is crossed out. It may read “this is connected.”

$$\psi'' + T(z)\psi = 0. \tag{12}$$

$$w(z) = \frac{\psi_1(z)}{\psi_2(z)}. \tag{13}$$

(ψ_1, ψ_2 — 2 linearly independent solutions of (13))



Original source details: GELFAND-P001065

Diagram and typed description. The reference after "solutions" is "(13)."

Condition on variation of c: the monodromy changes but remains real,

(this is the problem)

(in $\mathrm{SL}(2,\mathbb{R})$, as opposed to $\mathrm{SL}(2,\mathbb{C})$)

GELFAND-P001066

[source number: 55]

Relation:

$$c_i(z_1, \ldots, z_N) = \frac{\partial S_{\mathrm{cl}}}{\partial z_i}. \tag{14}$$



Diagram and typed description. ___ · ___

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The problem is solved via the quantum problem, then the semiclassical limit as $\gamma \to 0$!

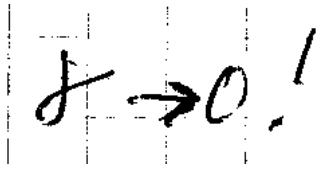
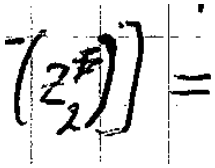


Diagram and typed description. The parameter in the semiclassical limit is transcribed as γ.

Original source details: GELFAND-P001066

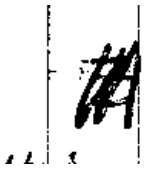
$$[T(z_1), T(z_2)] = \big(T(z_1) + T(z_2)\big)\delta'(z_1 - z_2) + \frac{c}{12}\delta'''(z_1 - z_2). \tag{15}$$



Uncertain reading. A small superscript over z_2 on the left of the first (15) is crossed out. It may be a digit.

Original source details: GELFAND-P001066

(the dependence on $z_1, \dots, z_N$ is left unwritten)



Uncertain reading. A mark before the parenthesis is heavily crossed out.

Original source details: GELFAND-P001066

or:

$$[T(z), T(w)] = \big(T(z) + T(w)\big)\delta'(z-w) + \frac{c}{12}\delta'''(z-w) \tag{15}$$

(in the classical case. Poisson brackets)

Let us calculate:

Lagrangian:

$$\mathcal{L} = \int \big(\partial_z\varphi\,\partial_{\bar z}\varphi - e^{\varphi}\big)$$

$$p = \frac{\partial\mathcal{L}}{\partial(\partial_{\bar z}\varphi)} = \partial_{\bar z}\varphi \tag{16}$$

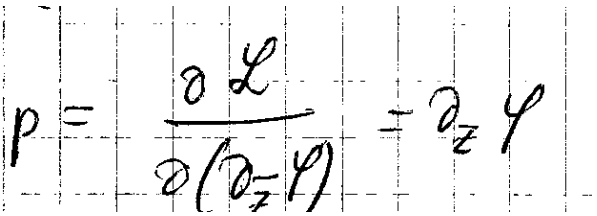


Diagram and typed description. On the right of (16), a separate bar is visible over the subscript z.

Original source details: GELFAND-P001066

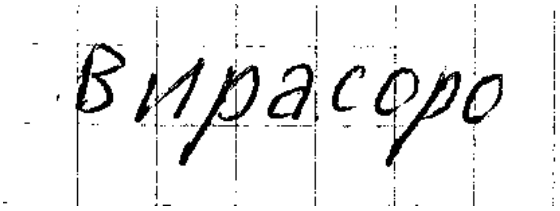
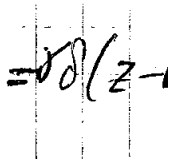


Diagram and typed description. [In the right margin] Virasoro

Original source details: GELFAND-P001066

whence

$$\{\varphi(z), \partial_w\varphi(w)\} = \gamma\delta(z-w). \tag{17}$$



Uncertain reading. A separate mark before δ appears to be γ.

Original source details: GELFAND-P001066

$\{T(z), T(w)\}$ and so on.

Expand in a Laurent series:

$$T(z) = \sum_n (L_n z^n + L_{-n} z^{-n}) \tag{18}$$

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12} n(n^2-1), \quad c = 26 - D. \tag{19}$$

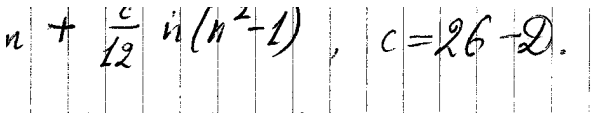


Diagram and typed description. A comma follows $n(n^2-1)$ in (19).

Original source details: GELFAND-P001066

(this could have been predicted from the Gelfand–Fuchs result, from the form of the central extension)

$$\widetilde{z} = z + \varepsilon(z) \qquad \text{(variation)}$$

$$\delta\varphi = \mathrm{Re}\big(\varepsilon(z)\partial_z\varphi + \partial_z\varepsilon\big)$$

A number is added to the action $\rightsquigarrow$ a central extension appears).

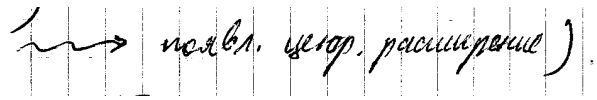


Diagram and typed description. A wavy arrow connects the two phrases.

Original source details: GELFAND-P001066

Let us try to find Ψ purely algebraically:

$$\text{Let} \qquad T_\varepsilon = \int \varepsilon(z) T(z)\, d^2 z \tag{20}$$

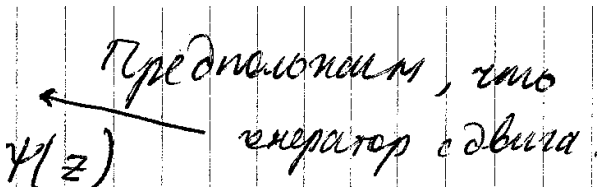


Diagram and typed description. $\longleftarrow$ Suppose it is the generator of the shift.

Original source details: GELFAND-P001066

$$\delta_\varepsilon \psi = [T_\varepsilon, \psi(z)] = \varepsilon(z)\frac{d\psi}{dz} + \Delta \cdot \varepsilon'(z)\psi(z) \tag{21}$$

(variation of the field ψ)

$$[L_n, \psi(z)] = z^{n+1}\frac{d\psi}{dz} + (n+1)\Delta\psi(z). \tag{22}$$

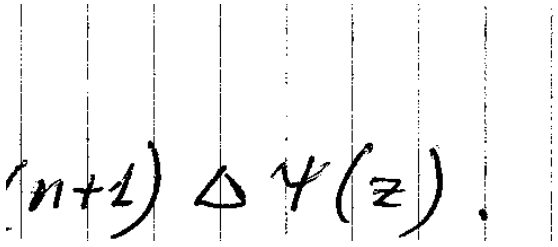


Diagram and typed description. The second term of (22) reads $(n+1)\Delta\psi(z)$.

Original source details: GELFAND-P001066

$\psi(z)$ — an operator in the same space on which the representation is defined.

GELFAND-P001067

$\psi(z)$ is the operator creating a horn on the sphere.

$$\Psi(z)|v\rangle.$$

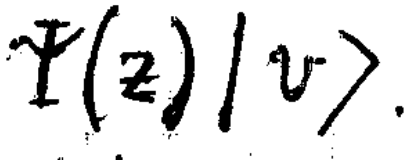


Diagram and typed description. Uppercase Ψ in the ket line.

Original source details: GELFAND-P001067

Consider the functions:

$$\langle v|\psi(z_1)\psi(z_2)\psi(z_3)\psi(z_4)|v\rangle \tag{23}$$

$$\Psi(x) = \langle v|\psi(\infty)\psi(1)\psi(x)\psi(0)|v\rangle$$

$$|0\rangle = \psi(0)|v\rangle, \qquad \langle v|\psi(\infty) = \langle 0|$$

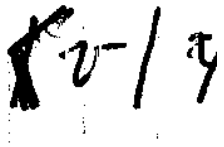


Diagram and typed description. The opening angle bracket of the bra is retraced.

Original source details: GELFAND-P001067

$$\Psi(x) = \langle 0|\psi(1)\psi(x)|0\rangle$$

$$\begin{cases} L_n|0\rangle = 0, \quad n > 0, \\ L_0|0\rangle = \Delta|0\rangle \end{cases} \tag{24}$$

Verma modules arise: such a representation without extraneous relations.

$\psi(x)$ (possibly $\Psi(x)$), overwritten symbol? is described above and maps the Verma module to itself.



Uncertain reading. $\psi(x)$ (possibly $\Psi(x)$), overwritten symbol

Original source details: GELFAND-P001067

$$\Psi(x) = e^{\Delta \cdot S_{cl}(x)} \quad \Rightarrow \quad \text{everything is determined by } \Psi(x) \tag{25}$$

crossed-out mark, possibly § or 4? Computation of Ψ.



Uncertain reading. crossed-out mark, possibly § or 4

Original source details: GELFAND-P001067

Add a relation (to the commutation relations):

$$\langle 0|\psi(x)|0\rangle = x^{-\Delta} \tag{26}$$

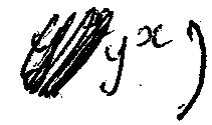
$$\langle 0|\psi(1)|0\rangle\langle 0|\psi(x)|0\rangle + \langle 0|\psi(1)|1\rangle\langle 1|\psi(x)|0\rangle + \\ + \langle 0|\psi(1)|2\rangle\langle 2|\psi(x)|0\rangle + \dots \tag{27}$$

$$I = \sum_n |n\rangle\langle n|$$

$$\langle 1|\psi(x)|0\rangle = \langle 0|L_1\psi(x)|0\rangle = \langle 0|[L_1, \psi(x)]|0\rangle = \\ = \left(x^2\frac{d}{dx} + 2\Delta x\right)\langle 0|\psi(x)|0\rangle = (\Delta \cdot x)x^{-\Delta}$$

[writing after 2Δ is crossed out, and x is inserted. The mark beneath the line may be an insertion mark or y?]

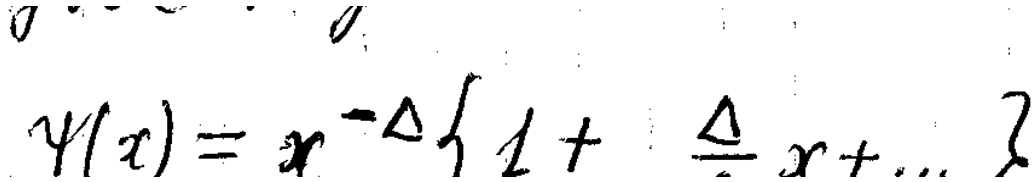
Uncertain reading. writing after 2Δ is crossed out, and x is inserted. The mark beneath the line may be an insertion mark or y



Original source details: GELFAND-P001067

and then one obtains:

$$\psi(x) = x^{-\Delta}\left\{1 + \frac{\Delta}{2}x + \dots\right\}$$



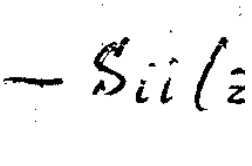
Original source details: GELFAND-P001067

Diagram and typed description. Lowercase ψ in the expansion.

$$\langle v|\psi(z_1)\dots\psi(z_N)|v\rangle = e^{-S_{ii}(z_1,\dots,z_N)}$$

[subscript of S, probably ii, possibly il?]

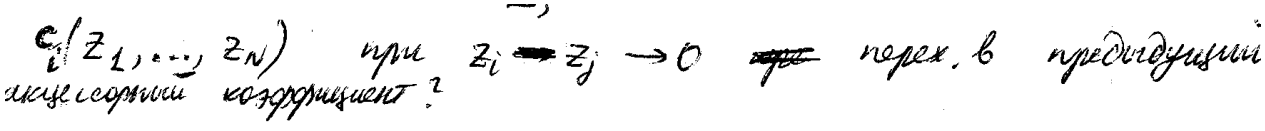
Uncertain reading. subscript of S, probably ii, possibly il



Original source details: GELFAND-P001067

$$\Delta, c \to \infty$$

Does $c_i(z_1, \dots, z_N)$ as $z_i - z_j \to 0$ [crossed-out «пр...». The mark between z_i and z_j is retraced and reads as minus?] go over to the preceding accessory coefficient?



Original source details: GELFAND-P001067

Uncertain reading. crossed-out «пр...». The mark between z_i and z_j is retraced and reads as minus

GELFAND-P001068

[source number: 56]

$S_{cl}(x)$

(3 other points $0, 1, \infty$)

~~The remaining ones~~ $\Rightarrow c_k = c_k(x)$

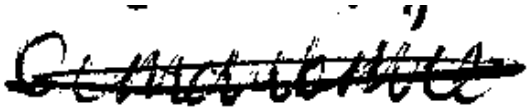


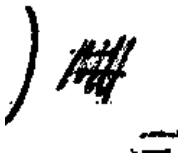
Diagram and typed description. Crossed-out word.

Original source details: GELFAND-P001068

$$c_k(x) = \frac{\partial S_{cl}(x)}{\partial x}$$

$$e^{S_{cl}(x)} = x^{-\Delta}\{1 + a_1 x + a_2 x^2 + \ldots\}.$$

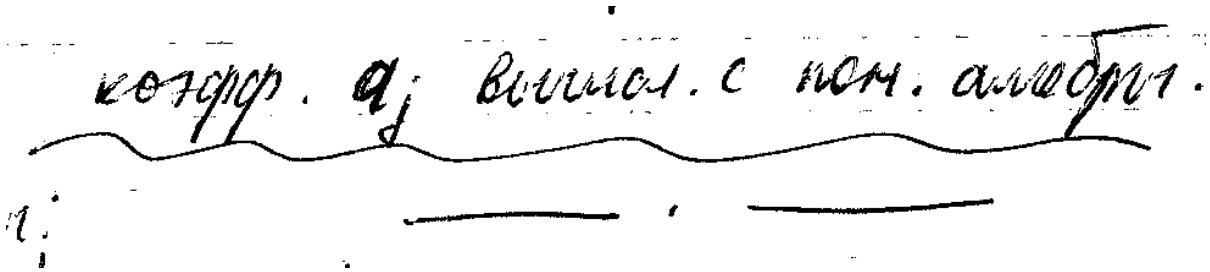
[crossed-out writing in the exponent after $S_{cl}(x)$?]



Uncertain reading. crossed-out writing in the exponent after $S_{cl}(x)$

Original source details: GELFAND-P001068

the coefficients a_j are computed by means of the algebra.



Original source details: GELFAND-P001068

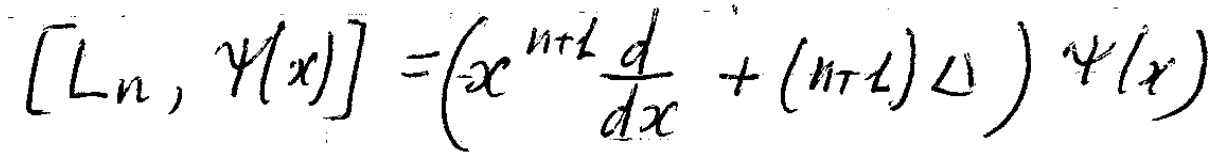
Diagram and typed description. Wavy underline and two shorter strokes underneath.

Axioms:

$$L_0|0\rangle = \Delta|0\rangle$$

$$L_n|0\rangle = 0, \quad n > 0$$

$$[L_n, \psi(x)] = \left(x^{n+1}\frac{d}{dx} + (n+1)\Delta\right)\psi(x)$$



Original source details: GELFAND-P001068

Diagram and typed description. Manuscript commutator. The factor x^n beside $(n+1)\Delta$ is absent in this line.

$$\langle 0|\psi(x)|0\rangle = x^{-\Delta}$$

$$\Psi(x) = \langle 0|\psi(1)\psi(x)|0\rangle$$

[illegible crossed-out mark after the ket?]

Uncertain reading. illegible crossed-out mark after the ket

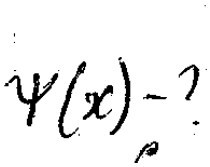


Original source details: GELFAND-P001068

$$I = \sum_n |n\rangle\langle n|, \qquad |n\rangle = L_{-n}|0\rangle.$$

[Left margin: $\psi(x)$ — ?]

Diagram and typed description. Left-margin note.



Original source details: GELFAND-P001068

Difficulty. there is no orthogonality.

It is natural from the form of T:

$$L_n = (\alpha + i\beta n)a_n + \sum_{k=1}^{n-1} a_k a_{n-k} + \sum_{k=1}^{\infty} a_k^+ a_{n+k};$$

$$L_{-n} = L_n^+;$$

$$[a_n, a_m^+] = n\delta_{n,m};$$

$$\Delta = \frac{1}{2}(\alpha^2 + \beta^2), \quad c = 1 + 12\beta^2.$$

$$a_n|0\rangle = 0.$$

There is a correspondence between the bases:

$$a_{n_1}^+ a_{n_2}^+ \ldots |0\rangle$$

[illegible crossed-out mark to the right of the first basis line?]

Uncertain reading. illegible crossed-out mark to the right of the first basis line



Original source details: GELFAND-P001068

$$L_{-n_1} L_{-n_2} \ldots |0\rangle$$

expressible in terms of each other, but the first is orthogonal!!

The commutation relations of $\psi(x)$ in the first basis are more complicated

$$L(z) = T(z) = \frac{dp(z)}{dz} + p^2(z),$$

where

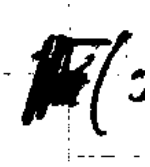
$$p(z) = \sum (a_n z^n + a_{-n} z^{-n})$$

GELFAND-P001069

Take $a_n = \dfrac{\partial}{\partial z_n}$, $a_n^+ = n z_n$.

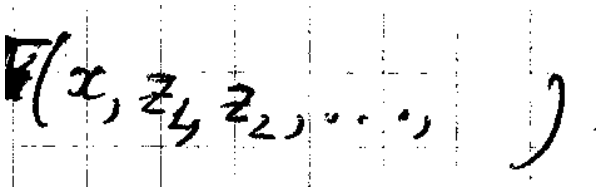
$$(\alpha + i\beta n)\frac{\partial}{\partial z_n} + \sum \frac{\partial^2}{\partial z_k \partial z_{n-k}} + \sum k z_k \frac{\partial}{\partial z_{n+k}} \Big) F(x, z_1, z_2, \dots,) =$$
$$= \left(x^{n+1} \frac{\partial}{\partial x} + (n+1)\Delta \right) F(x, z_1, , z_n).$$

[on the left F is written over another symbol. The underlying symbol is illegible?]



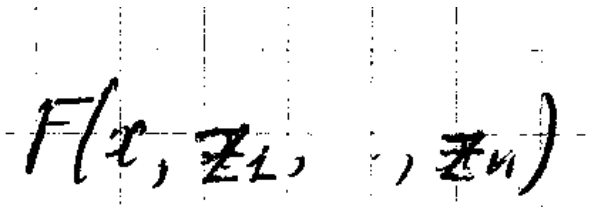
Original source details: GELFAND-P001069

Uncertain reading. on the left F is written over another symbol. The underlying symbol is illegible



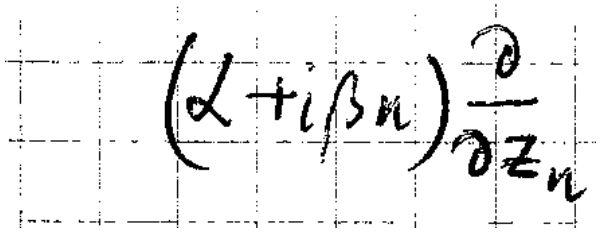
Original source details: GELFAND-P001069

Diagram and typed description. Arguments on the left. The comma touches z_1, and a gap follows the last comma.



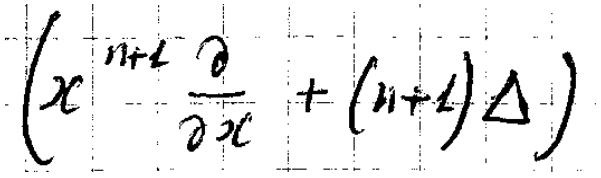
Original source details: GELFAND-P001069

Diagram and typed description. Arguments on the right, with a gap between commas.



Original source details: GELFAND-P001069

Diagram and typed description. Start of the left side. There is no separate outer opening parenthesis before the first parenthesis.



Original source details: GELFAND-P001069

Diagram and typed description. Right side. No factor x^n is written beside $(n+1)\Delta$.

GELFAND-P001070

29/XI–82 Gelfand Seminar.

Kobzarev (Igor Yuryevich)

On Strings.

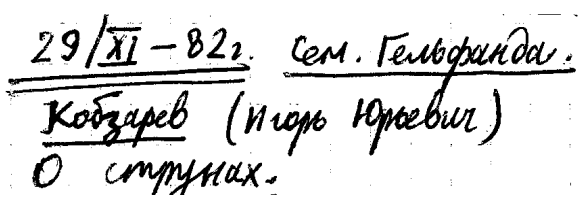


Diagram and typed description. Date, speaker's name and title.

Original source details: GELFAND-P001070

Field theory: there is a field φ, M^4 is Minkowski space, and $\mathcal{L}(\varphi)$ is the Lagrangian.
solutions and so on.

Another type of field theory
before 1970

$$\Box\psi + \mu^2\psi = 0 \tag{1}$$

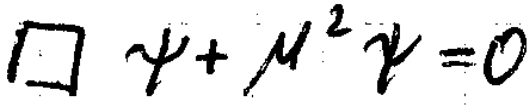


Diagram and typed description. The notation μ^2 in (1).

Original source details: GELFAND-P001070

[At right:] Interaction — perturbation theory. $\varphi_1\varphi_2\varphi_3, \ldots$
one must add $\lambda\varphi_1\varphi_2\varphi_3$ and so on.
after: of Higgs type

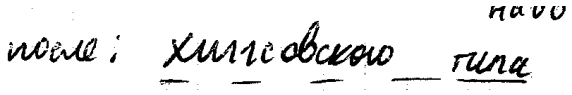


Diagram and typed description. Manuscript line "after: of Higgs type."

Original source details: GELFAND-P001070

$$\mathcal{L}(\varphi) = \frac{1}{2}(\partial_i\varphi)^2 - \lambda^2(\varphi^2 - \eta^2)^2 \tag{2}$$

Vacuum: for classical fields, the state with the least energy ||



Diagram and typed description. Double stroke to the right of the vacuum line.

Original source details: GELFAND-P001070

(corresponding Hamiltonian:

$$H = \frac{1}{2}(\partial_t\varphi)^2 + \frac{1}{2}(\bar{\nabla}\varphi)^2 + \lambda^2(\varphi^2 - \eta^2)^2 \tag{3}$$



Diagram and typed description. The digit 3 is retraced.

Original source details: GELFAND-P001070



Diagram and typed description. Bar over ∇ in (3).

Original source details: GELFAND-P001070

Degenerate vacuum: $\varphi = \eta$, $\varphi = -\eta$. (2 vacua)

$$\varphi \mapsto -\varphi \qquad \text{discrete symmetry} \tag{4}$$

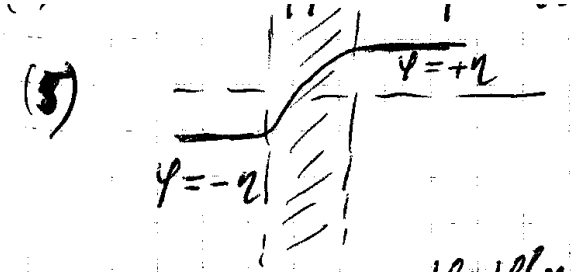
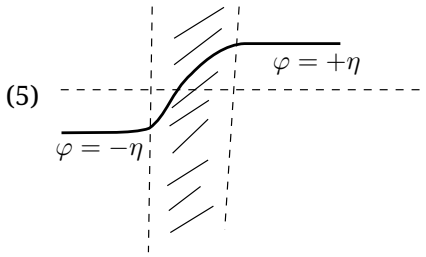


Diagram and typed description.



Original source details: GELFAND-P001070

[To the right of the figure:] Consider a wall — a solution localized on a plane.

$$\varphi = \varphi(x, y, z) \qquad \text{(independent of } t\text{)}$$

$$\Box\varphi + \varphi\lambda^2(\varphi^2 - \eta^2) = 0 \qquad \text{— Euler equation for Lagrangian (2)} \tag{6}$$

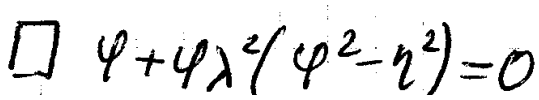


Diagram and typed description. The symbol before λ^2 is φ.

Original source details: GELFAND-P001070

(7) $\| \quad \varphi \underset{z\to\pm\infty}{=} \pm\eta$ independent of $x, y, z \Rightarrow$ it is easy to verify that such a solution exists.

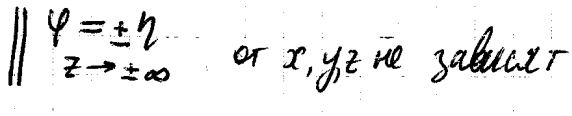


Diagram and typed description. Asymptotic condition and the manuscript's coordinate list.

Original source details: GELFAND-P001070

GELFAND-P001071

$\eta \sim m$

(8) Layer thickness [inserted below: (in (5))] $\sim \frac{1}{\eta}$.

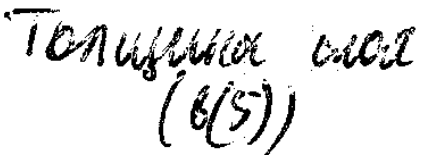


Diagram and typed description. Reference to (5) below the line on the layer thickness.

Original source details: GELFAND-P001071

[illegible crossed-out mark before “Solution”[?]] Solution (5) is found explicitly: th$(z\eta)$



Uncertain reading. illegible crossed-out mark before “Solution”

Original source details: GELFAND-P001071



Diagram and typed description. The double-underlined th$(z\eta)$.

Original source details: GELFAND-P001071

Symmetry group: $\varphi \mapsto -\varphi$.

One now usually considers a continuous group G of symmetries of the Lagrangian.

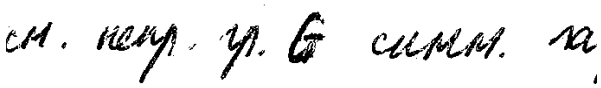


Diagram and typed description. Abbreviations in the line on the continuous group.

Original source details: GELFAND-P001071

$U(1)$-invariant theory:

$$\mathcal{L}(\varphi) = \frac{1}{2}\left|(\partial_i + iA_i)\varphi\right|^2 - (\varphi\varphi^* - \eta^2)^2 \tag{9}$$

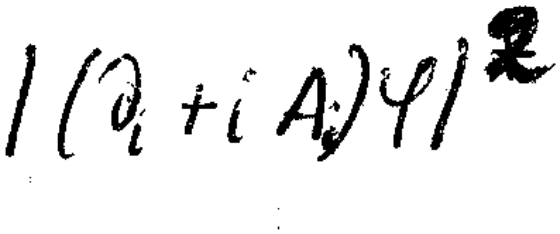


Diagram and typed description. The $+$ sign before iA_i in (9).

Original source details: GELFAND-P001071

$$\text{Vacuum: } (c_1, c_2): \quad c_1^2 + c_2^2 = \eta^2 \tag{10}$$

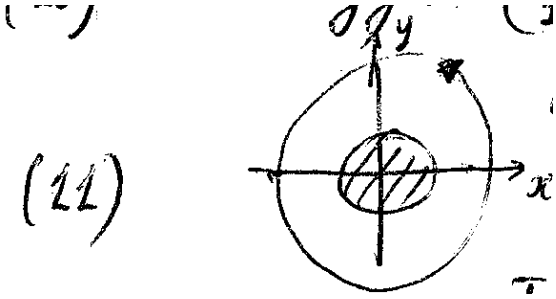
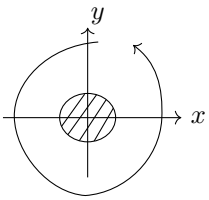


Diagram and typed description.

(11)



Original source details: GELFAND-P001071

[To the right of the figure:] A string is when the energy is concentrated along a line (one-dimensional).

On going around, the vector C turns through 2π

There is a topological invariant.

(Static solution: independent of t)

A is an additional vector field (electromagnetic).

$$\mathcal{L} = -\frac{1}{4}F_{ik}F_{ik} + \frac{1}{2}\left|(\partial_i + iA_i)\varphi\right|^2 - (\varphi\varphi^* - \eta^2)^2 \tag{12}$$

$$F_{ik} = \partial_i A_k - \partial_k A_i$$

Action of $U(1)$ on the field:

$$\begin{cases} \varphi \mapsto e^{i\alpha}\varphi \\ A_i \mapsto A_i + \nabla_i \alpha \end{cases} \tag{13}$$

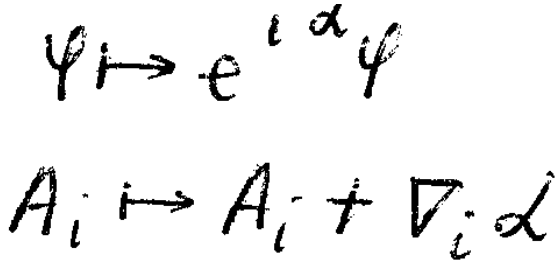
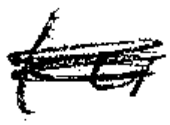


Diagram and typed description. Equation (13) has $e^{i\alpha}$ and a $+$ sign before $\nabla_i\alpha$.

Original source details: GELFAND-P001071

Consider an approximate solution [crossed-out writing, possibly a number in parentheses[?]]



Uncertain reading. crossed-out writing, possibly a number in parentheses

Original source details: GELFAND-P001071

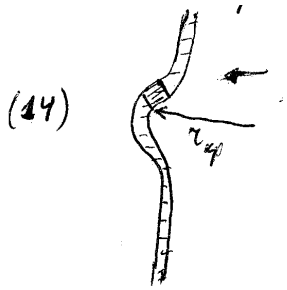
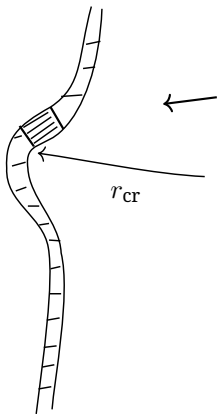


Diagram and typed description.

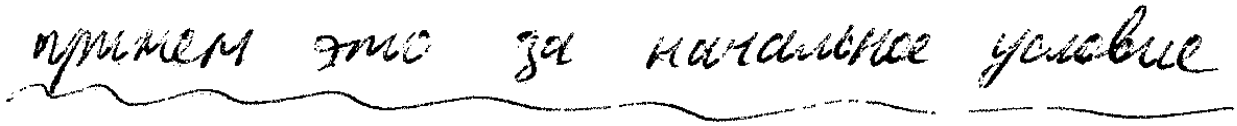
(14)



The arrow at the bend is labelled r_{cr}. The strokes inside the marked section may have arrowheads.

Original source details: GELFAND-P001071

[To the right of the figure, left arrow:] take this as the initial condition



Original source details: GELFAND-P001071

Diagram and typed description. Wavy underline and singular “initial condition.”

Approximate form of solution (11):

$$\begin{cases} \varphi_1 = \dfrac{x}{r} f(r) \\ \varphi_2 = \dfrac{y}{r} f(r) \end{cases} \tag{15}$$

(for probably “rotation,” the first letters are only partly legible[?])

Uncertain reading. probably “rotation,” the first letters are only partly legible



Original source details: GELFAND-P001071

[At right:] Construct A in the same way.

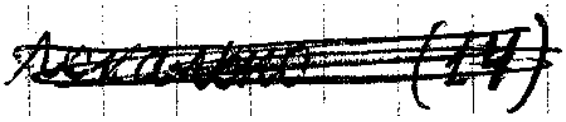
$\varepsilon(r)$, the energy density in (11), can be calculated.

GELFAND-P001072

[source number: 67]

[Crossed out: possibly “Solution (14)”[?]]

Uncertain reading. possibly “Solution (14)”



Original source details: GELFAND-P001072

They must approach the vacuum: $\varphi_1^2 + \varphi_2^2 \to \eta^2$ as $r \to \infty$.

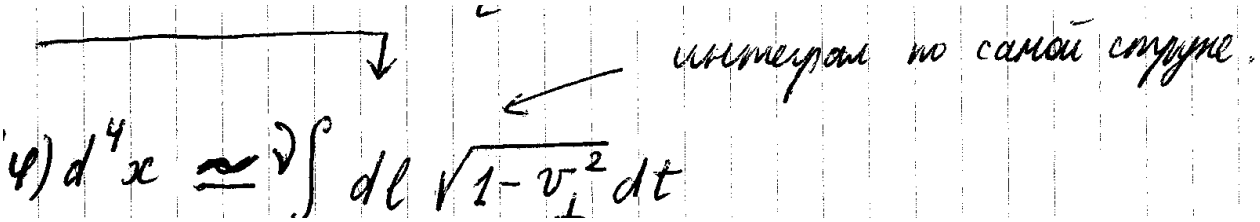
Locally (14) resembles (11).

(16) Let the radius of curvature satisfy $r_{\text{cr}} \gg \dfrac{1}{\eta}$

Effective action: [arrow rightward, then down to the right side of (17)]

$$S = \int \mathcal{L}(\varphi)\, d^4x \simeq \nu \int dl \sqrt{1 - v_\perp^2}\, dt \tag{17}$$

[Above right, arrow down-left toward the integral:] integral along the string itself.



Original source details: GELFAND-P001072

Diagram and typed description. Two arrows to the right side of (17).

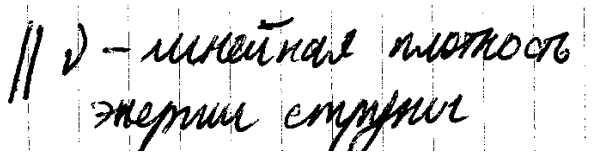


Original source details: GELFAND-P001072

Diagram and typed description. The coefficient ν before the integral in (17).

$v_\perp$ is the transverse velocity of the string

|| ν is the linear energy density of the string.

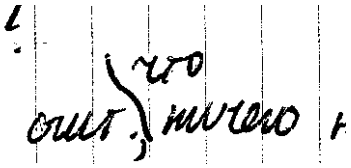


Original source details: GELFAND-P001072

Diagram and typed description. Double stroke and explanation of ν.

Evolution — ?

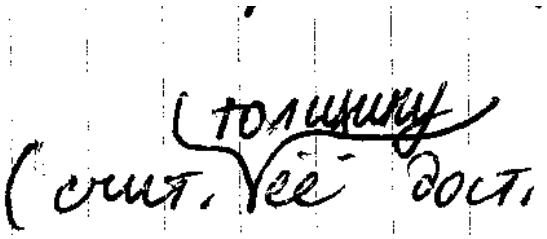
Inside the string we assume [inserted above: that] nothing happens



Original source details: GELFAND-P001072

Diagram and typed description. "That" is inserted above the line.

(we consider its [inserted above: thickness] sufficiently small — — compared with the radii of curvature)



Original source details: GELFAND-P001072

Diagram and typed description. Insertion of "thickness" before "its."

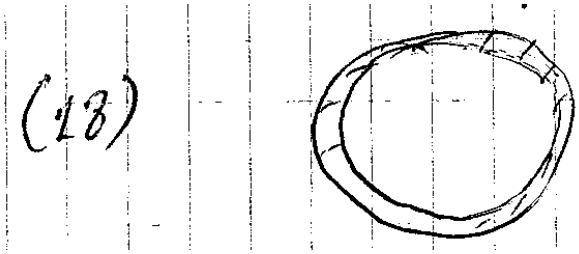
We evaluate the integral in each transverse section to obtain ν

$\sqrt{1-v_\perp^2}$ comes from d^4x (by a change of variables)

Or use relativistic invariance: ds (the interval), written in any coordinate system.

(17) is the formula of Nambu.

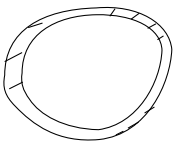
One can consider a ring-shaped string with Lagrangian given by the right side of (17)



Original source details: GELFAND-P001072

Diagram and typed description.

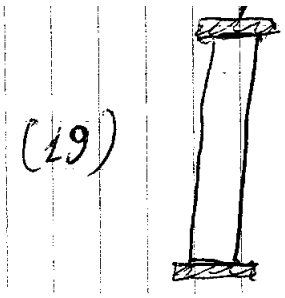
(18)



it will contract, quickly reach the speed of light, and shrink to a small lump.

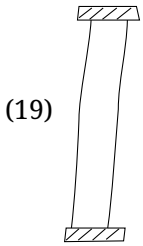
Variant.

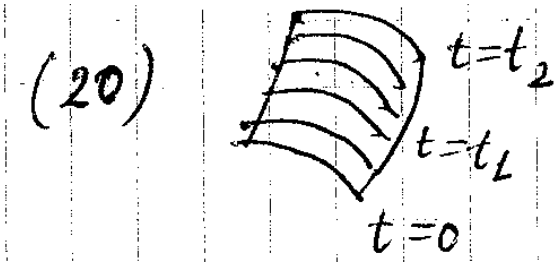
break the ends (put a monopole at each):



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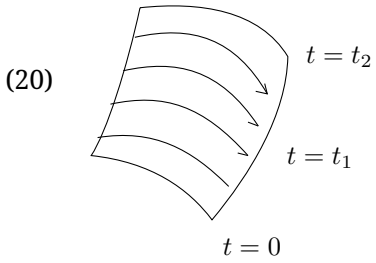
Diagram and typed description.





Original source details: GELFAND-P001072

Diagram and typed description.



Short strokes are visible at the ends of three arcs. It is uncertain whether they are arrowheads.

[To the right of the figure:] As it moves, the string sweeps out a sheet.

(21) The area of the swept surface is the Lagrangian (17_{right})



Original source details: GELFAND-P001072

Diagram and typed description. Wavy underline and reference 17_{right}.

GELFAND-P001073

$\dot{x} = \dfrac{\partial x}{\partial t}$, $x' = \dfrac{\partial x}{\partial s}$, $x(t, s)$ and calculate.

A min area arises.



Original source details: GELFAND-P001073

Diagram and typed description. The Russian word after min ends in «ь».

Strings and chromodynamics.

Quantum chromodynamics:

$$\mathcal{L}(\psi, A) = -\frac{1}{4}F^a_{ik}F^a_{ik} + \bar{\psi}(i\hat{\partial} - m)\psi \tag{22}$$

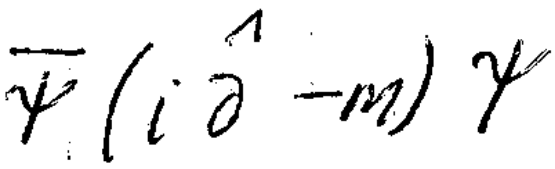


Diagram and typed description. Lowercase ψ on the right side of (22).

Original source details: GELFAND-P001073

$a = 1 \div 8$; $i, k = 0, 1, 2, 3$.

$$\hat{\partial} = \gamma^i(\partial_i + iI^a A^a_i) \tag{23}$$

[Beneath γ^i:] Dirac matrices

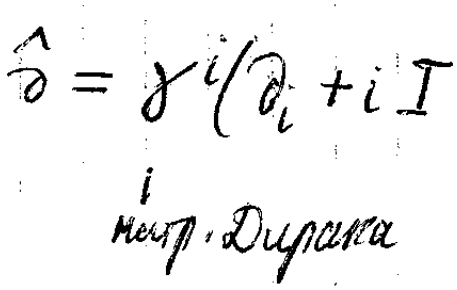


Diagram and typed description. A short dotted stroke appears above the note beneath γ^i.

Original source details: GELFAND-P001073

$$F^a_{ik} = \partial_i A^a_k - \partial_k A^a_i + g f_{abc} A^b_i A^c_k, \tag{24}$$

g is the coupling constant, and f_{abc} are the structure constants of the group.

Acts: the group $SU(3)$ (adjoint representation) — hence 8.

— acts on A

[Crossed out: illegible mark before I^a, possibly a?]



Uncertain reading. illegible mark before I^a, possibly a

Original source details: GELFAND-P001073

I^a is a generator of $SU(3)$, $I^a = I^a_{\alpha\beta}$.

in $\bar{\psi}$ and ψ there is also a color index (quarks are triplets) — — contraction is over that index.



Diagram and typed description. The note “quarks are triplets.”

Original source details: GELFAND-P001073

In fact several types of quarks are needed: at least 2 types

$$\mathcal{L}(\psi, A) = -\frac{1}{4}F^a_{ik}F^a_{ik} + \bar{u}(i\bar{\partial} - m)u + \bar{d}(i\hat{\partial} - m)d \tag{25}$$

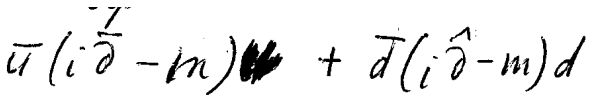


Diagram and typed description. There is a horizontal bar over ∂ in the first quark term, and a hat in the second.

Original source details: GELFAND-P001073

[heavily overwritten symbol after the first parenthesis, provisionally u?]



Uncertain reading. heavily overwritten symbol after the first parenthesis, provisionally u

Original source details: GELFAND-P001073

(u-quarks and d-quarks)

$$\hat{\partial}_i = \nabla_i + iA^a_i I^a, \qquad I^a = I^a_{\alpha\beta} \tag{26}$$

A is a Yang–Mills field, or a gluon field (it glues quarks together)

(gluon field)

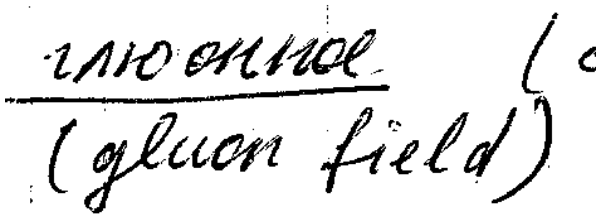


Diagram and typed description. English “gluon field” beneath the underlined word.

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